GENERAL APTITUDE

Previous Year Questions • 2021–2026

GATE — ALL BRANCHES

Himanshu Agarwal & Anish Saha

PrepFusion™

Where Concept Meets Clarity!

Preface

Welcome to PrepFusion™!

This book is a complete compilation of the questions GATE has actually asked in General Aptitude (all GATE branches), gathered from the original papers and arranged year by year and session by session. Nothing is paraphrased or reconstructed: every question appears as it was set, tagged with its year, session and marks.

Each question carries an identifier that maps one-to-one onto its worked solution in the companion solutions volume, so you can move between practice and explanation without a lookup table. Answer keys close every section, and answers revised by the examining authority (MTA) are marked where they occur.

Used end to end, the book shows you what the paper rewards — which topics repeat, how the framing has shifted over the years, and where your preparation still has gaps.

Meet your Authors

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- AIR 27 — GATE (ECE)
- AIR 45 — GATE (IN)
- 5+ years of teaching experience

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- AIR 19 — GATE (IN)
- AIR 66 — GATE (EE)
- 3+ years in teaching & the VLSI industry

> *"We built this book the way every aspirant deserves one to be built — organized strictly by year and by session, so that every hour of practice maps directly onto how GATE actually tests you."*
>
> — Himanshu Agarwal & Anish Saha

How to Read the Question Numbers

Every question in this book carries a three-part identifier instead of a plain serial number:

Q24.5.3

24	GATE exam year — 2024 (two digits; 98 = 1998)
5	Session number of that year's exam
3	Question number within that session

So **Q24.5.3** is the 3rd aptitude question of Session 5, GATE 2024. Numbers are therefore not sequential through the book — they tell you the year and session at a glance. The same identifier is used in the questions book, the solutions book and the answer keys, so you can jump between them without a lookup table. In the PDF bookmarks panel, expand a session to see every question ID and click one to go straight to it.

What the Bubbles Mean

Every question ends with three empty circles for you to shade in yourself:

Easy Medium Hard

The companion solutions volume marks the same question with *our* recommended difficulty, using the identical bubbles. If you think a rating should be different, tell us directly — report it under the **Study Hub** section at **pyq.prepfusion.in**.

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YEAR

2026

8 Sessions · 80 Questions

Sessions Summary

Session	Participating Branches	Total Qs
Session 1	AG, ES, GG, IN, MA, MN, TF, XE, XL	10
Session 2	AE, BT, CH, CY, GE, PH, XH	10
Session 3	CS-1, ST	10
Session 4	CS-2, EY, NM, PE	10
Session 5	CE-1, EE, PI	10
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Year Snapshot

Total Questions: 80 **MCQ:** 80 **MSQ:** 0 **NAT:** 0

Session : 1 | Forenoon | 07/02/26

Q26.1.1 MCQ 2026 1M

Suresh said, "I did it yesterday."
Which one of the following options is the correct form of this sentence in indirect speech?

A. Suresh said that I did it yesterday.

B. Suresh says I did it yesterday.

C. Suresh says that he did it the day before.

D. Suresh said that he had done it the day before.

Q26.1.2 MCQ 2026 1M

To continue the sequence of tiles shown, the tile indicated by the question mark should be

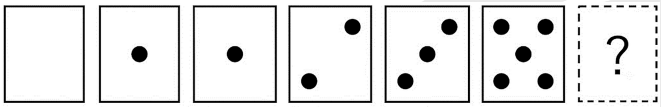
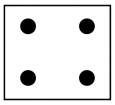
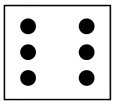
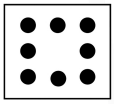
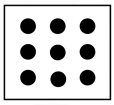


Fig. Q26.1.2

A.  **B.**  **C.**  **D.** 

Q26.1.3 MCQ 2026 1M

Consider an art gallery whose walkways are shown as lines in the diagram. A black dot represents a junction of two walkways. A guard may be placed at a junction to watch over the walkways that join at that junction. The minimum number of guards needed to watch all the walkways is ____.

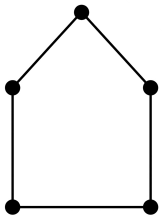


Fig. Q26.1.3

A. 2 **B.** 3 **C.** 4 **D.** 5

Q26.1.4 MCQ 2026 1M

The 2nd of June is a Thursday in a certain year. Which day of the week is the 3rd of July in that year?

A. Thursday **B.** Friday **C.** Saturday **D.** Sunday

Q26.1.5 MCQ 2026 1M

A coin with heads facing up is shown as (H) and a coin with tails facing up is shown as (T).
Six coins are placed in the Starting Arrangement, as shown in the figure below. A "step" is defined as interchanging a pair of adjacent coins without flipping them. The minimum number of steps needed to go from the Starting Arrangement to the Final Arrangement, as shown in the figure, is

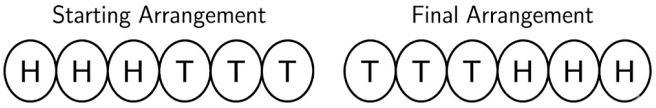


Fig. Q26.1.5

A. 3 **B.** 6 **C.** 9 **D.** 12

Q26.1.6 MCQ 2026 2M

Exacerbate : Mitigate :: _____
Choose the option with the correct pair of words to fill the blank.

A. Aggravate : Alleviate **B.** Alleviate : Precipitate

C. Aggravate : Precipitate **D.** Emancipate : Exonerate

Q26.1.7 MCQ 2026 2M

A paper shown in Panel I is folded along the dashed lines (- - -) to construct a cube. The shaded regions shown in Panel I appear on the outer surface of the cube. Referring to cubes shown in Panel II, which one of the options is correct?

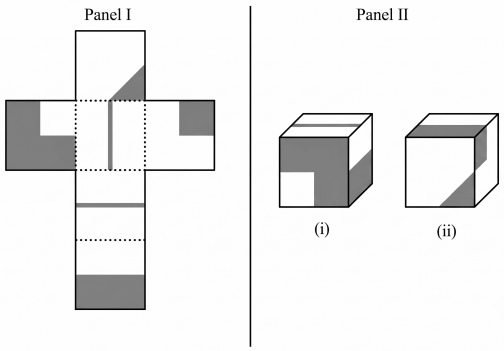


Fig. Q26.1.7

A. Only (i) can correspond to the unfolded cube in Panel I.

B. Only (ii) can correspond to the unfolded cube in Panel I.

C. Both (i) and (ii) can correspond to the unfolded cube in Panel I.

D. Neither (i) nor (ii) can correspond to the unfolded cube in Panel I.

Q26.1.8 MCQ | 2026 | 2M

In a population, patients who have high cholesterol also have high blood-pressure (BP). Some patients with high BP also have diabetes. There are no patients who have both high cholesterol and diabetes. Furthermore,

1. the total number of patients with at least one of these conditions is 75,
2. the number of patients with high cholesterol is 10,
3. the number of patients with high BP is 45, and
4. the number of patients with only high BP and no other conditions is 20.

Then the number of patients who have both diabetes and high BP is _____.

A. 0 **B.** 15 **C.** 20 **D.** 10

Q26.1.9 MCQ | 2026 | 2M

Four people P, Q, R, and S, of different ages, make the following observations.

P – I am younger than S.

Q – I am neither the youngest nor the oldest.

R – P is older than me.

Based on these observations, the youngest person is _____.

A. P **B.** Q **C.** R **D.** S

Q26.1.10 MCQ | 2026 | 2M

Circles C_1, C_2, and C_3, with centers O_1, O_2, and O_3, and radii r_1, r_2, and r_3, respectively, touch each other as shown in the following figure.

Given $r_1 = 2$ cm, $r_2 = 1$ cm and the angle $\angle O_1O_3O_2$ is $90°$, r_3 = _____ cm.

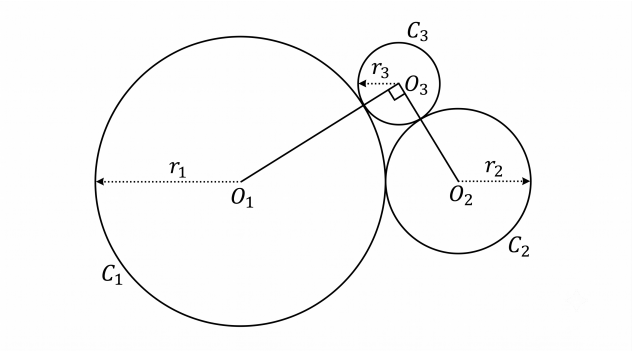


Fig. Q26.1.10

A. $\frac{1}{2}\left(-3+\sqrt{17}\right)$ **B.** $\frac{1}{2}\left(3+\sqrt{17}\right)$

C. $\frac{1}{2}\left(-2+\sqrt{17}\right)$ **D.** $\frac{1}{2}\left(-3+2\sqrt{17}\right)$

Session : 2 | Afternoon | 07/02/26

Q26.2.1 MCQ | 2026 | 1M

"He often _____ the numbers. False claims are not going to help. Honesty _____ trust", said the manager.

Choose the option with the correct order of words to fill the blanks.

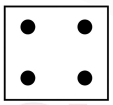
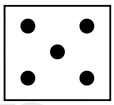
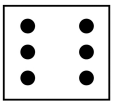
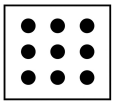
A. exaggerates; engenders **B.** excels; encourages

C. aggravates; alleviates **D.** diminishes; eliminates

Q26.2.2 MCQ | 2026 | 1M

In the sequence of tiles shown below, the missing tile indicated by the question mark should be

Fig. Q26.2.2

A.  **B.**  **C.**  **D.** 

Q26.2.3 MCQ | 2026 | 1M

A school has 100 students distributed among ten standards, numbered 1 to 10.

Based on this, which one of the following statements is always correct?

A. There are at least 10 students who belong to the same standard.

B. There is at least one student in each standard.

C. There are at most 10 students in standard 10.

D. The total number of students from standards 1 to 5 is at least 50.

Q26.2.4 MCQ 2026 1M

How many 3-digit numbers can be formed using three distinct single digit prime numbers?

A. 64 **B.** 24 **C.** 12 **D.** 4

Q26.2.5 MCQ 2026 1M

In a group of students, 10 students like Mathematics, 12 students like English, 4 students like both Mathematics and English, and 6 students like neither Mathematics nor English. The number of students in the group is _____.

A. 18 **B.** 20 **C.** 24 **D.** 32

Q26.2.6 MCQ 2026 2M

Charity : P :: Retaliation : Q

Choose the appropriate pair of words P and Q that fit the analogy.

A. P = Parsimonious; Q = Vengeful

B. P = Altruistic; Q = Amicable

C. P = Resentful; Q = Spiteful

D. P = Magnanimous; Q = Vindictive

Q26.2.7 MCQ 2026 2M

A paper shown in Panel I is folded along the dashed lines (- - - - -) to construct a cube. The shaded regions shown in Panel I appear on the outer surface of the cube. Referring to cubes shown in Panel II, which one of the options is correct?

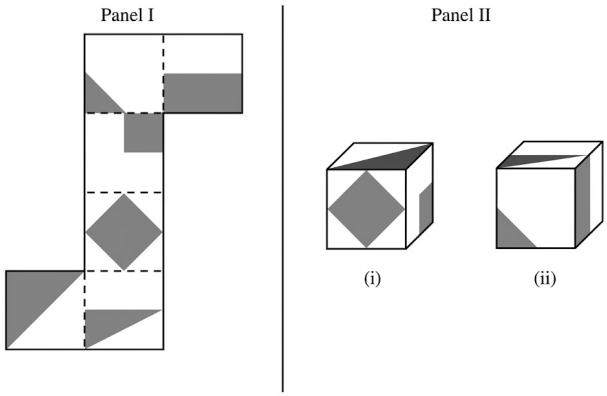


Fig. Q26.2.7

A. Only (i) can correspond to the unfolded cube in Panel I.

B. Only (ii) can correspond to the unfolded cube in Panel I.

C. Both (i) and (ii) can correspond to the unfolded cube in Panel I.

D. Neither (i) nor (ii) can correspond to the unfolded cube in Panel I.

Q26.2.8 MCQ 2026 2M

Consider the cube shown below with its 8 corners labelled a, b, c, d, e, f, g, and h. The figure is representative. All corners are to be colored such that any two corners that are connected by an edge must be of different colors. The minimum number of colors required to achieve this is ___.

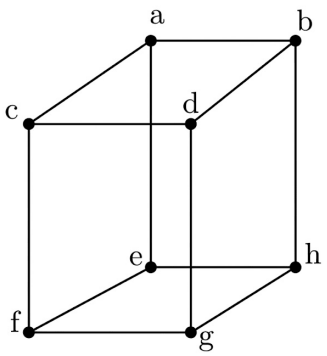


Fig. Q26.2.8

A. 8 **B.** 4 **C.** 3 **D.** 2

Q26.2.9 MCQ 2026 2M

Four hills H1, H2, H3, and H4 are present in an area. The following observations are made about them:

(i) Neither H2 nor H3 is the easternmost hill.
(ii) Neither H2 nor H3 is the westernmost hill.
(iii) Neither the easternmost hill nor the westernmost hill is the southernmost hill.
(iv) Two hills are located to the west of H2.
(v) The southernmost hill has at least two hills to its east.

The southernmost hill is _____.

A. H1 **B.** H2 **C.** H3 **D.** H4

Q26.2.10 MCQ 2026 2M

As shown in the figure, circle C_1 with center O_1 and radius r_1 touches the square $VWXY$ at points P and Q while circle C_2 with center O_2 and radius r_2 touches the square $VWXY$ at points R and S. The two circles touch each other at T. Given $r_1 = 1$ cm and $\overline{VY} = \overline{VW} = 4$ cm, r_2 = _____ cm.

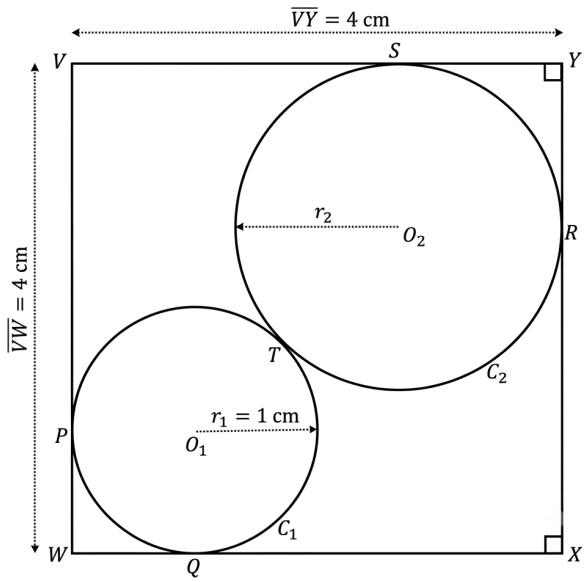


Fig. Q26.2.10

A. $4 - 3\sqrt{2}$ **B.** $1 + 2\sqrt{2}$ **C.** $7 - 4\sqrt{2}$ **D.** $5 + 3\sqrt{2}$

Session : 3 | Forenoon | 08/02/26

Q26.3.1 MCQ 2026 1M

The antonym of the word protagonist is_____.

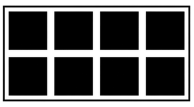
A. agnostic **B.** antagonist **C.** arsonist **D.** anarchist

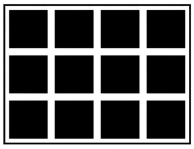
Q26.3.2 MCQ 2026 1M

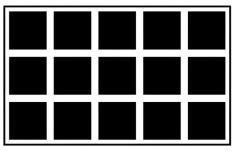
The figure shows two 4-tile patterns.

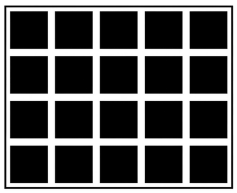
Fig. Q26.3.2

Either one or both of the patterns can be used any number of times and in any orientation to construct a new pattern. Which one of the options below **cannot** be constructed by using only these two 4-tile patterns assuming there are no overlaps among them?

A. 

B. 

C. 

D. 

Q26.3.3 MCQ 2026 1M

Consider a knock-out women's badminton singles tournament where there are no ties. The loser in each game is eliminated from the tournament. Every player plays until she is defeated or remains the last undefeated player. The last undefeated player is declared the winner of the tournament. If there are 64 players in the beginning of the tournament, how many games should be played in total to declare the winner of the tournament?

A. 127 **B.** 64 **C.** 63 **D.** 32

Q26.3.4 MCQ 2026 1M

A student needs to enroll for a minimum of 60 credits. A student cannot enroll for more than 70 credits. The credits are divided amongst project and three distinct sets of courses namely, core courses, specialization courses, and elective courses. It is compulsory for a student to enroll for exactly 15 credits of core courses and exactly 20 credits of project. In addition, a student has to enroll for a minimum of 10 credits of specialization courses. The maximum credits of elective courses that a student can enroll for is_____.

A. 10 **B.** 15 **C.** 20 **D.** 25

Q26.3.5 MCQ 2026 1M

'When the teacher is in the room, all students stand silently.' If the above statement is true, which one of the following statements is **not necessarily** true?

A. If any student is not standing silently, then the teacher is not in the room.

B. When the teacher is in the room, all students are silent.

C. If all students are standing, then the teacher is in the room.

D. When the teacher is in the room, all students are standing.

Q26.3.6 MCQ 2026 2M

Combinatorics deals with problems involving counting. For example, "How many distinct arrangements of N distinct objects in M spaces on a circle are possible?" is a typical problem in combinatorics. This kind of counting is sometimes used in the modeling of several physical phenomena. Often, in such models, the different combinatorial possibilities are assigned probability values. Assigning probabilities enables the computation of the average values of physical quantities.

Consider the following statements:

P: Combinatorics is always invoked in the modeling of physical phenomena.

Q: Modeling some physical phenomena involves assigning probabilities to combinatorial possibilities in order to compute average values of physical quantities.

Based on the passage above, what can be inferred about statements P and Q?

A. P is False and Q is False **B.** P is False and Q is True

C. P is True and Q is False **D.** P is True and Q is True

Q26.3.7 MCQ 2026 2M

In Panel I of the figure below, the front view and top view of a structure are shown. Which one of the 3D structures shown in Panel II possesses the views shown in Panel I?

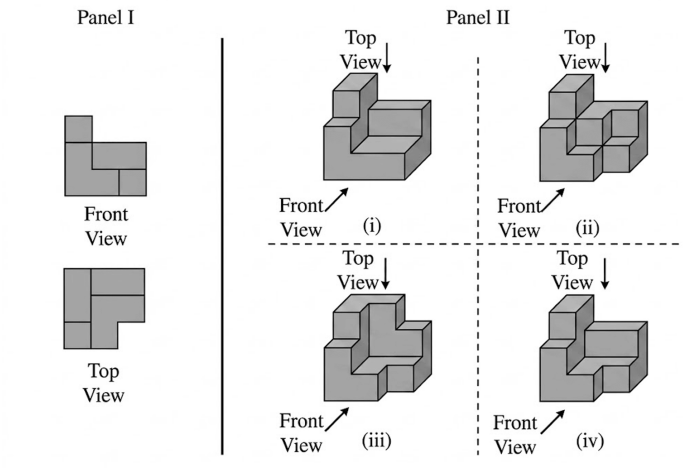


Fig. Q26.3.7

A. (i) **B.** (ii) **C.** (iii) **D.** (iv)

Q26.3.8 MCQ 2026 2M

For positive real numbers S and K, the function $H_K(S)$ is defined as: $H_K(S) = \max(S - K, 0)$. The max function is defined as:

$$\max(a, b) = \begin{cases} a, & \text{when } a > b \\ b, & \text{when } a \leq b \end{cases}$$

The graph below shows the plot of a function $N(S)$ versus S.

$N(S)$ can be expressed as _____.

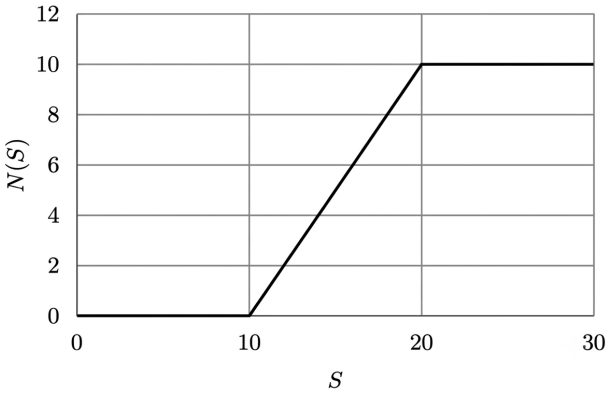


Fig. Q26.3.8

A. $H_{10}(S) - H_{20}(S)$ **B.** $H_{10}(S) - 2H_{20}(S)$

C. $-H_{10}(S) + H_{20}(S)$ **D.** $H_{15}(S) - H_{20}(S)$

Q26.3.9 MCQ 2026 2M

In the 2020 summer Olympics' Javelin throw finals, Neeraj Chopra exhibited a spectacular performance to win the gold medal. The silver medal was won by Jakub Vadlejch and the bronze medal was won by Vitezlav Vesely. There were six rounds of throws with each athlete having one throw per round. The best of all the throws of each athlete is considered for the medal. Following were the observations about the throws:

i. The first and second rounds were dominated by Neeraj Chopra with a gold medal performance in his second throw, while the other two athletes did not have any medal winning throws in these rounds.

ii. The throws in the last round by both Jakub Vadlejch and Vitezlav Vesely were fouls and were not considered for scoring.

iii. After four rounds, Vitezlav Vesely was in the second position and could not improve upon his best throw in the succeeding rounds.

iv. In the fourth round, the throw by Jakub Vadlejch was the best in that round.

In which round did Vitezlav Vesely have his best throw?

A. Third **B.** Fourth **C.** Fifth **D.** Sixth

Q26.3.10 MCQ 2026 2M

An unbiased six-faced dice whose faces are marked with numbers 1, 2, 3, 4, 5, and 6 is rolled twice in succession and the number on the top face is recorded each time. The probability that the number appearing in the second roll is an integer multiple of the number appearing in the first roll is _____.

A. $\frac{1}{6}$ **B.** $\frac{5}{18}$ **C.** $\frac{7}{18}$ **D.** $\frac{5}{6}$

Session : 4 | Afternoon | 08/02/26

Q26.4.1 MCQ 2026 1M

Expedite, Hasten, Hurry, _____
Fill the blank by choosing a word with a meaning similar to that of the words given above.

A. Accelerate **B.** Retard **C.** Provide **D.** Disable

Q26.4.2 MCQ 2026 1M

A black square PQRS has been cut into two parts. One part of it is shown in Panel I. Which one of the shapes in Panel II is the other part?

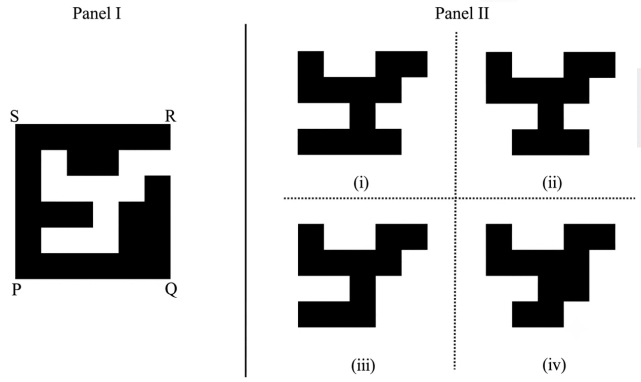


Fig. Q26.4.2

A. (i) **B.** (ii) **C.** (iii) **D.** (iv)

Q26.4.3 MCQ 2026 1M

A day can only be cloudy or sunny. The probability of a day being cloudy is 0.5, independent of the condition on other days. What is the probability that in any given four days, there will be three cloudy days and one sunny day?

A. 1/4 **B.** 3/4 **C.** 2/3 **D.** 3/8

Q26.4.4 MCQ 2026 1M

The values of Stock A and Stock B on a particular day are Rs. 50 and Rs. 80, respectively. An investor invests Rs. 100 in Stock A and Rs. 80 in Stock B. He sells all the stocks the next day when the value of Stock A is Rs. 55 and Stock B is Rs. 70. The profit made by the investor is Rs. ___.

A. 0 **B.** 5 **C.** 10 **D.** 20

Q26.4.5 MCQ 2026 1M

'When it is raining, peacocks dance.'
Based only on this sentence, which one of the following options is necessarily true?

A. Peacocks dance only when it is raining.

B. When peacocks dance, it is raining.

C. When peacocks are not dancing, it is not raining.

D. When it is not raining, peacocks do not dance.

Q26.4.6 MCQ 2026 2M

Water : P :: Food : Q
Choose the P and Q combination from the options below to form a meaningful analogy.

A. P = Thirst; Q = Hunger **B.** P = Drink; Q = Hunger

C. P = Thirst; Q = Satiated **D.** P = Wet; Q = Critic

Q26.4.7 MCQ 2026 2M

Two tiles are missing in Panel I. Which one of the options in Panel II is the appropriate choice for the missing tiles?

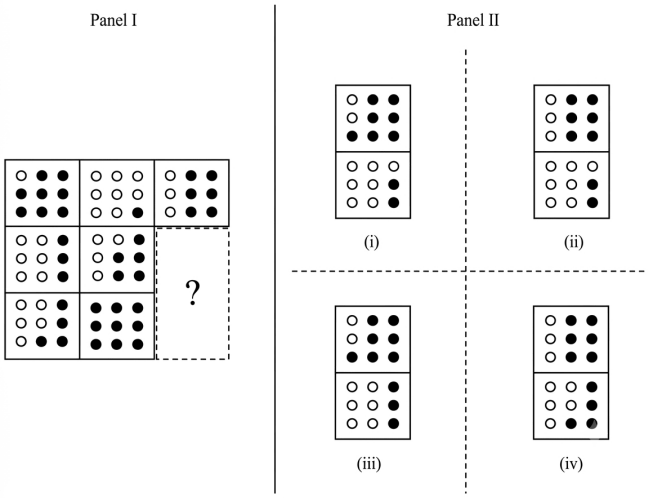


Fig. Q26.4.7

A. (i) **B.** (ii) **C.** (iii) **D.** (iv)

Q26.4.8 MCQ 2026 2M

Figures (i) and (ii) represent intercity highway systems. The black dots represent cities and the line segments between them represent intercity highways. A salesperson needs to make a trip. She needs to start from a city, visit each of the remaining cities exactly once, and finally return to the same city from which she started.
Which one of the following options is then true?

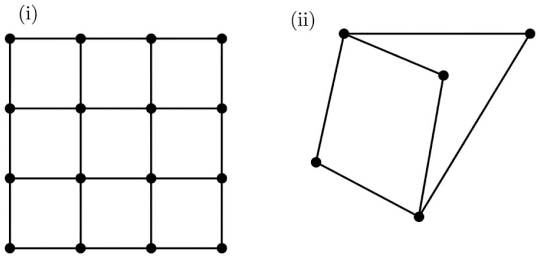


Fig. Q26.4.8

A. Such a trip is possible for (i), but not for (ii).

B. Such a trip is possible for (ii), but not for (i).

C. Such a trip is possible for both (i) and (ii).

D. Such a trip is possible neither for (i) nor for (ii).

Q26.4.9 MCQ 2026 2M

The figure in Panel I below is a grid of cells with four rows and four columns. The numbers on the top and on the left represent the number of cells that are to be shaded in that column and row, respectively. Which one of the options shown in Panel II below represents the grid shaded correctly?

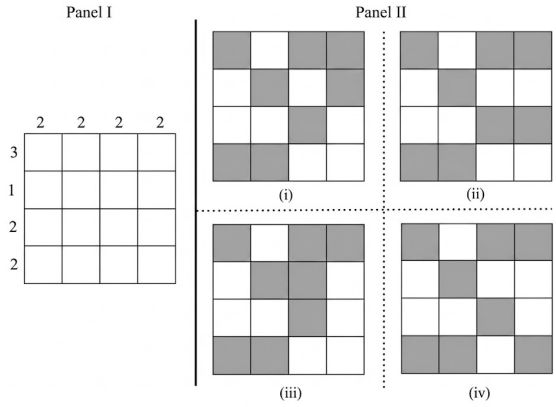


Fig. Q26.4.9

A. (i) **B.** (ii) **C.** (iii) **D.** (iv)

Q26.4.10 MCQ 2026 2M

An unbiased six-faced dice whose faces are marked with numbers 1, 2, 3, 4, 5, and 6 is rolled twice in succession and the number on the top face is recorded each time. The probability that the sum of the two recorded numbers is a prime number is _____

A. $\frac{3}{36}$ **B.** $\frac{13}{36}$ **C.** $\frac{15}{36}$ **D.** $\frac{19}{36}$

Session : 5 | Forenoon | 14/02/26

Q26.5.1 MCQ 2026 1M

'The shopkeeper sells lemons.'
In this sentence, the word 'lemons' is the _____.

A. object **B.** subject **C.** predicate **D.** verb

Q26.5.2 MCQ 2026 1M

The figure below is supposed to show three non-overlapping shapes – one oval and two triangles. Which one of the following figures P, Q, R, or S fits the missing portion indicated by '?' and completes the oval and the two triangles?

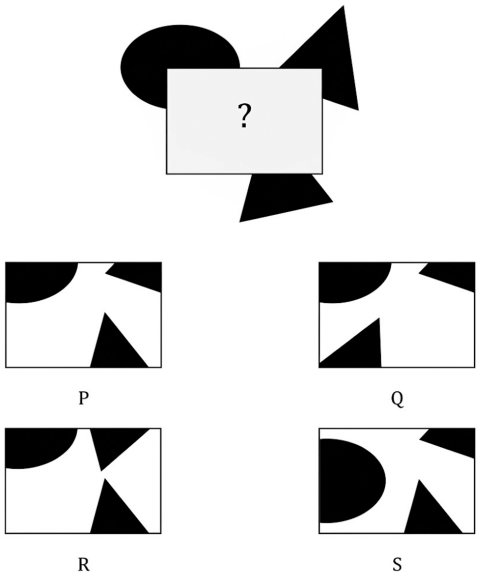


Fig. Q26.5.2

A. P **B.** Q **C.** R **D.** S

Q26.5.3 MCQ 2026 1M

At how many points will the curves $y = x^2$ and $y = -x^2 - 2x - 1$ intersect in the real (x, y) plane?

A. 0 **B.** 1 **C.** 2 **D.** 3

Q26.5.4 MCQ 2026 1M

'If Anish had scored hundred runs in today's match, he would have been made the captain of his team. He would have then become the youngest captain in his team's history. Unfortunately, he got out without scoring any runs. Hence, there won't be any change in the captaincy for now.'
Based on the paragraph above, which one of the following statements is true?

A. Anish made hundred runs but was denied captaincy.

B. Anish was the captain of his team before the game today.

C. The current captain is older than Anish.

D. Anish is the youngest player in his team.

Q26.5.5 MCQ 2026 1M

Which one of the following figures P, Q, R, or S, correctly shows the 45° clockwise-rotated version of figure (I)?

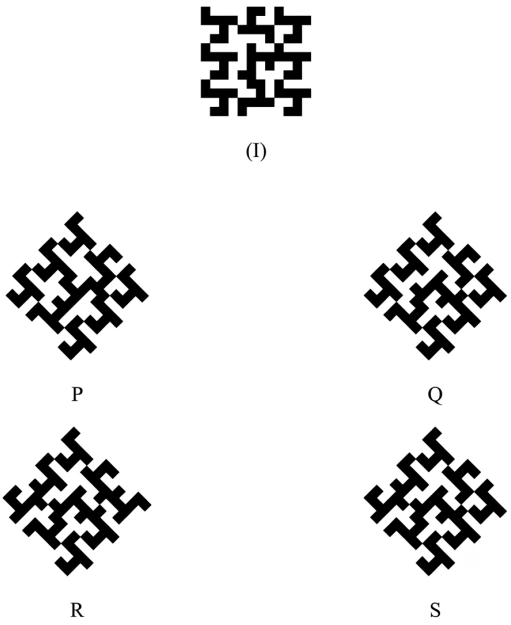


Fig. Q26.5.5

A. P **B.** Q **C.** R **D.** S

Q26.5.6 MCQ 2026 2M

Match the words in Column I with their synonyms in Column II.

Column I	Column II
(i) Lonely	(p) Verbatim
(ii) Literal	(q) Solitary
(iii) Lousy	(r) Deadly
(iv) Lethal	(s) Terrible

A. (i)-(q); (ii)-(p); (iii)-(s); (iv)-(r)

B. (i)-(q); (ii)-(s); (iii)-(r); (iv)-(p)

C. (i)-(s); (ii)-(p); (iii)-(q); (iv)-(r)

D. (i)-(r); (ii)-(s); (iii)-(p); (iv)-(q)

Q26.5.7 MCQ 2026 2M

In the given figure, $\overline{PQ}$ is the diameter of a circle with center O. Two points R and S are chosen on the circle such that $\angle ROS = 80°$. When $\overline{PR}$ and $\overline{QS}$ are extended, they meet at T. The value of $\angle RTS$ is _____.

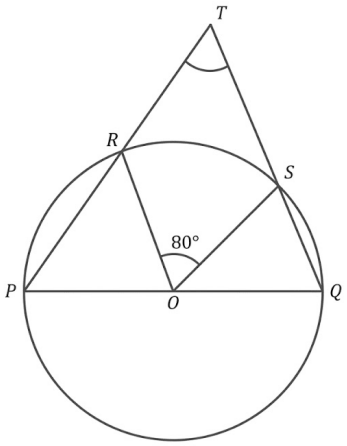


Fig. Q26.5.7

A. 40° **B.** 50° **C.** 60° **D.** 80°

Q26.5.8 MCQ 2026 2M

Based on the relationship between each polygon and the number inside it, the value of 'X' is _____

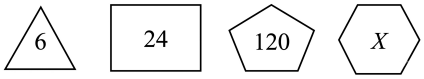


Fig. Q26.5.8

A. 720 **B.** 596 **C.** 24 **D.** 240

Q26.5.9 MCQ 2026 2M

Consider a linear arrangement of seven bulbs, each of which can be in the ON or OFF states. The initial configuration of the bulbs is shown in the figure. In every Step, the states of the bulbs are changed based on the following rules:

- Any OFF bulb with exactly one ON neighbor at the end of the previous Step is turned ON.

- Any ON bulb with both neighbors ON at the end of the previous Step is turned OFF.
- The state of any bulb not meeting the conditions above is left unchanged.

The state of bulbs at the end of Step 1 and Step 2 are also shown in the figure.
The number of bulbs which are ON at the end of Step 8 is

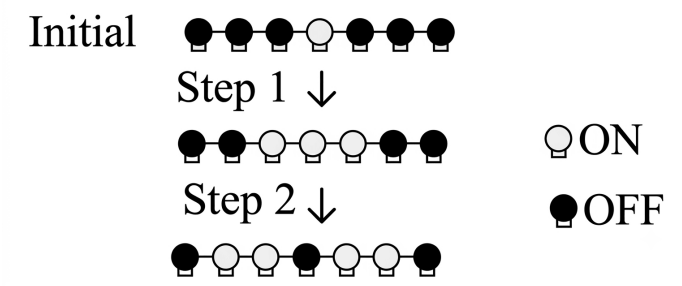


Fig. Q26.5.9

A. 5 **B.** 4 **C.** 3 **D.** 0

Q26.5.10 MCQ | 2026 | 2M

P and Q are two positive integers such that $P^2 = Q^2 + 13$. The product of the numbers P and Q is _____

A. 13 **B.** 26 **C.** 39 **D.** 42

Session : 6 | Afternoon | 14/02/26

Q26.6.1 MCQ | 2026 | 1M

'The team _____ more than 300 runs in 20 overs _____ rains. However, some players needed to improve their batting skills.'
Choose the option with the correct sequence of words to fill the blanks.

A. score; despite **B.** scoring; instead of

C. scored; despite **D.** scoring; in spite of

Q26.6.2 MCQ | 2026 | 1M

If a positive real x satisfies the following equation

$$\log_2 x + \log_{\sqrt{2}} x = 48,$$

then the value of x is_____

A. 2^{16} **B.** 4^{16} **C.** 2^{14} **D.** 4^{14}

Q26.6.3 MCQ | 2026 | 1M

The next figure (indicated by '?') in the sequence is

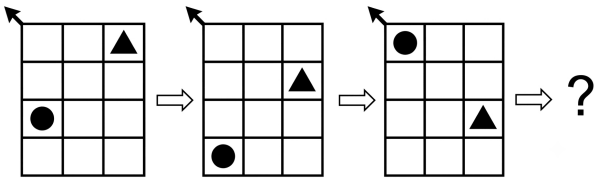
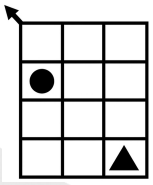
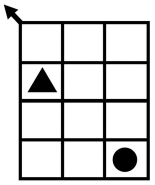
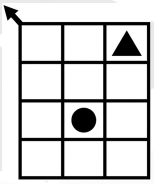
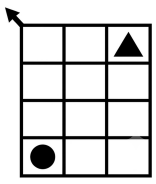


Fig. Q26.6.3

A.


B.


C.


D.


Q26.6.4 MCQ | 2026 | 1M

'All the mangoes in the basket are good.'
If the above statement is **false**, then which one of the following statements is necessarily true?

A. All the mangoes in the basket are not good.

B. No mango in the basket is good.

C. In the basket, some of the mangoes are good and some are not good.

D. There exists at least one mango in the basket that is not good.

Q26.6.5 MCQ | 2026 | 1M

Consider the following statements about four numbers:

(S1) The average of the four numbers is 25

(S2) Each number is at most 40

(S3) Each number is at least 20

Choose the option that is necessarily correct.

A. (S1) and (S2) together imply (S3)

B. (S2) and (S3) together imply (S1)

C. (S1) and (S3) together imply (S2)

D. (S1) implies (S3)

Q26.6.6 MCQ 2026 2M

'People are crowding around ____ pit into which ____ elephant has fallen. I have never seen an elephant looking more bewildered ____ miserable. Here it is in a most undignified position, thrust into a pit and made to look up ____ a vast, curiosity-stricken crowd.'

Choose the option with the correct sequence of words to fill the blanks.

A. an; a; at; and **B.** a; an; and; at

C. and; a; an; at **D.** at; a; an; and

Q26.6.7 MCQ 2026 2M

The table lists the unit selling price of five products P, Q, R, S, and T. On a particular day, 250 items were sold with the average selling price of Rs. 60.

The following observations were made:

(i) The quantity of S sold was twice that of T.
(ii) The quantity of R sold was thrice that of T.
(iii) The quantity of Q sold was four times that of T.

Product	P	Q	R	S	T
Unit selling price (Rs.)	100	50	40	60	60

What is the quantity of product P sold on that day?

A. 40 **B.** 50 **C.** 60 **D.** 70

Q26.6.8 MCQ 2026 2M

Consider a string P of length l that is laid out as a straight-line segment. Another string K is laid out as a semicircular arc with string P as its diameter, as represented in Figure (i). When both the strings are shortened by a length x they can be re-arranged such that the shortened string K forms a full circle with the shortened string P as its diameter, as represented in Figure (ii). The value of x/l is _____

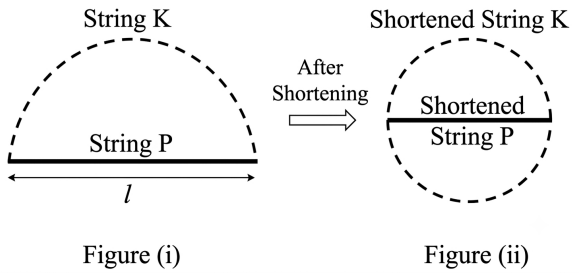


Fig. Q26.6.8

A. π **B.** $\frac{\pi - 1}{2\pi}$ **C.** $\frac{\pi}{2(\pi - 1)}$ **D.** $\frac{\pi}{\pi - 1}$

Q26.6.9 MCQ 2026 2M

The Roman senator Meritorius, his brother, his son, and his daughter have varying oratory skill levels. They are seated in rows and columns as shown in the figure with exactly one person sitting in each box. It is known that:

(i) Meritorius' daughter and his brother are seated in the same column.
(ii) His son is seated diagonally across the sibling of the worst orator.
(iii) The best and worst orators are seated in the same row.

Who is the best orator?

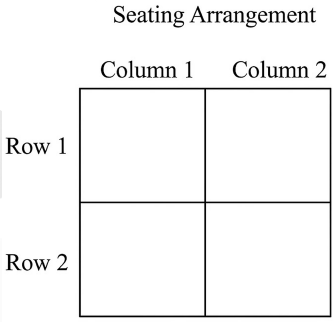


Fig. Q26.6.9

A. Meritorius **B.** Meritorius' brother

C. Meritorius' son **D.** Meritorius' daughter

Q26.6.10 MCQ 2026 2M

Which one of the patterns labelled P, Q, R, and S is used to generate the following figure?

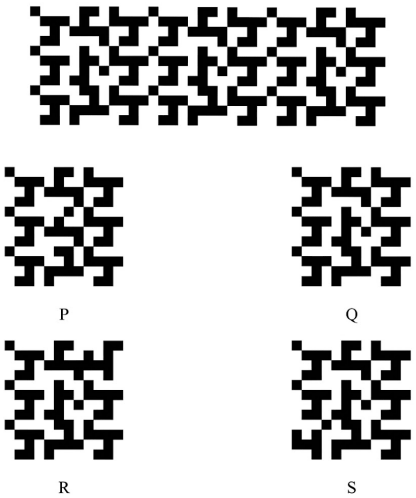


Fig. Q26.6.10

A. P **B.** Q **C.** R **D.** S

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Q26.7.1 MCQ 2026 1M

Among the following options, the antonym of the word 'nocturnal' is _____.

A. normal **B.** diurnal **C.** abnormal **D.** exceptional

Q26.7.2 MCQ 2026 1M

The statements (S1), (S2), and (S3) pertain to the scores obtained by students in an exam. The maximum possible marks in the exam is 150.

(S1) The highest score is 100.
(S2) The fourth highest score is 76.
(S3) There are at least four students whose scores are within 25 of each other.

Which one of the following options is necessarily correct?

A. (S1) and (S2) together imply (S3)

B. (S1) and (S3) together imply (S2)

C. (S2) and (S3) together imply (S1)

D. (S1) implies (S3)

Q26.7.3 MCQ 2026 1M

The figure below has exactly three intersecting line segments with a rectangular portion WXYZ missing. Which one of the following options P, Q, R, and S is the missing portion WXYZ?

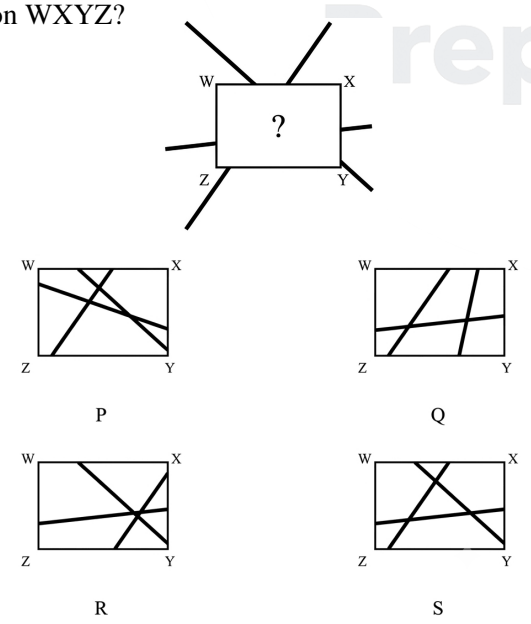

Fig. Q26.7.3

A. P **B.** Q **C.** R **D.** S

Q26.7.4 MCQ 2026 1M

Real numbers y, p, and n (all greater than 1) satisfy

$$(\log_{p^{1/n}} y)(\log_{y^{1/n}} p) = 16.$$

where the logarithms are taken to the bases $p^{1/n}$ and $y^{1/n}$. The value of n is _____

A. 2 **B.** 4 **C.** 8 **D.** 16

Q26.7.5 MCQ 2026 1M

The following observation is made about the scores obtained by 100 students in an exam:
'For each student, there exists another student in the class such that their scores are at most ten marks away.'
If the above statement is **false**, which one of the following statements is necessarily true?

A. For each student, the scores of all the other students are more than 10 marks away.

B. There exists at least one student in the class for whom the scores of all the other students are more than 10 marks away.

C. There is exactly one student in the class for whom the scores of some students are more than 10 marks away.

D. For each student, the score of exactly one other student is more than 10 marks away.

Q26.7.6 MCQ 2026 2M

Each one of the following clues contains a keyword that is partially filled.
Clue 1: Synonym of recognize (8 letters): _D_NT_FY
Clue 2: A story long enough to fill a book (5 letters): _ _ _ EL
Clue 3: Two of something (6 letters): _ _ _ PLE
Clue 4: A fraction of something, split equally into two parts (4 letters): _ _ _ F
The first letter of each of the keywords can be rearranged to form a four-letter word.
Which one of the options below is a possible choice for the four-letter word?

A. CHIN **B.** COIN **C.** ITCH **D.** NOSE

Q26.7.7 MCQ 2026 2M

Three children P, Q, R and two grown-ups X, Y play a badminton doubles tournament. X and Y are parents to two of the children playing. The child of X is not the same as the child of Y. Exactly one of the children does not have a parent playing in the tournament. The following rules are followed:

(i) A parent and his/her child cannot be on the same team.

(ii) A match can feature at most one parent and his/her child, that is, a maximum of one parent-child pair can play in a match.

The following matches were played:

	TEAM 1	TEAM 2
MATCH 1	P and X	Q and R
MATCH 2	P and R	X and Y
MATCH 3	R and X	Q and Y

Which one of the following options is correct?

A. P does not have any parent playing

B. Q does not have any parent playing

C. R does not have any parent playing

D. X does not have a child playing

Q26.7.8 MCQ 2026 2M

Let P_k represent the perimeter of a square with sides of length k. The value of the expression $P_1+P_2+P_3+\cdots+P_{10}$ is ____

A. 55 **B.** 110 **C.** 220 **D.** 440

Q26.7.9 MCQ 2026 2M

The city of Atlantis was crafted by the God of the seas, Poseidon. It was made of alternating concentric circular rings of land (shaded) and water (not shaded) as represented in the figure (not to scale). The radius of Inner Island was 2.5 stades (a unit of length used in ancient Greece). The water surrounding Inner Island was one stade wide (length AB). This was surrounded by two pairs of alternating rings of land and water. The first pair of land and water was two stades wide each (lengths BC and CD), and the outer pair is three stades wide each (lengths DE and EF).

The ratio of the surface area of the land to that of the water in the city of Atlantis is _____(round off to two decimal places).

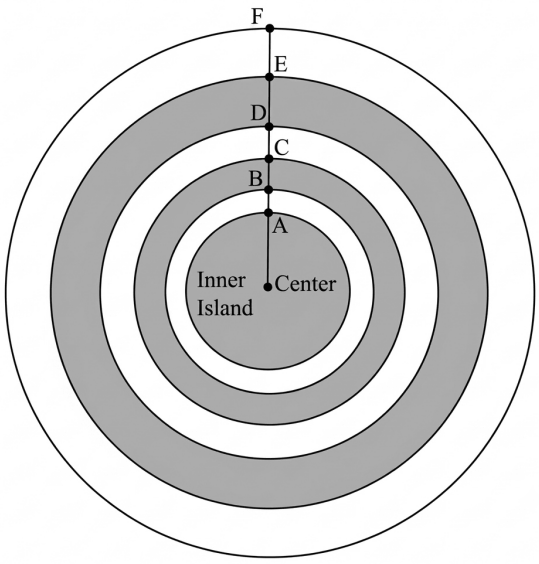

Fig. Q26.7.9

A. 0.45 **B.** 0.60 **C.** 0.75 **D.** 0.90

Q26.7.10 MCQ 2026 2M

If

Ref Fig 12 34 → ꟻ⅁ 34

Then

GATE - 2026 Aptitude Test → ?

P: ɹǝsʇ ∀ʇʇıʇnpǝ / ƆΛTƎ - ƧOƧƧ

Q: ∀ʇʇıʇnpǝ ɹǝsʇ / ƧOƧƧ - ƆΛTƎ

R: ∀ʇʇıʇnpǝ ɹǝsʇ / ƆΛTƎ - ƧOƧƧ

S: ɹǝsʇ ∀ʇʇıʇnpǝ / ƧOƧƧ - ƆΛTƎ

A. P **B.** Q **C.** R **D.** S

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Q26.8.1 MCQ 2026 1M

Verbosity: Brevity :: Insolence: _____
Choose the word that best fills the blank.

A. Innocence **B.** Respect **C.** Solace **D.** Wealth

Q26.8.2 MCQ 2026 1M

The product of the digits of a three-digit number is 70. The sum of the digits of this three-digit number is _____

A. 12 **B.** 14 **C.** 16 **D.** 18

Q26.8.3 MCQ 2026 1M

The four pieces of a puzzle are shown in the figure below.

Fig. Q26.8.3

Which one of the figures labelled as P, Q, R, and S can be constructed by using each of the four pieces only once without overlaps?

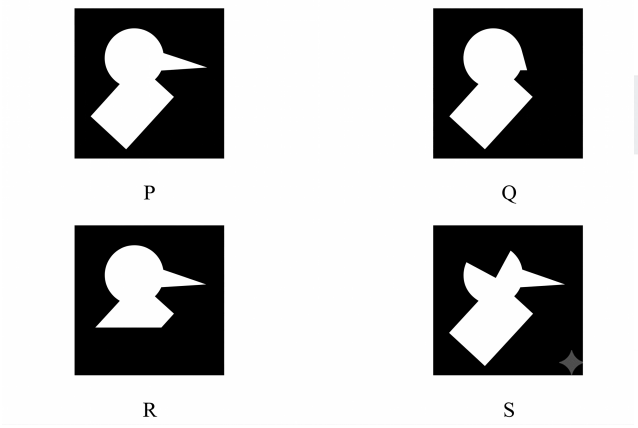


Fig. Q26.8.3

A. P **B.** Q **C.** R **D.** S

Q26.8.4 MCQ 2026 1M

Consider two distinct positive real numbers m, n, with $m > n$. Let $x = n^{\log_{10}(m)}$ and $y = m^{\log_{10}(n)}$. The relation between x and y is _____.

A. $x > y$ **B.** $x < y$ **C.** $x = y$ **D.** $x = \log_{10}(y)$

Q26.8.5 MCQ 2026 1M

'If his latest movie had been a commercial success, the actor would have made enough money to sponsor his next movie.'
Based only on the above sentence, which one of the following statements is true?

A. The actor will certainly sponsor his next movie.

B. His latest movie was a commercial success.

C. The actor made enough money from his latest movie.

D. His latest movie was not commercially successful.

Q26.8.6 MCQ 2026 2M

'My friend and I parted ____ the door ____ the cabin that I had rented ____ the night.'
Choose the option with the correct sequence of words to fill the blanks.

A. at; of; for **B.** for; at; of **C.** of; for; in **D.** in; of; for

Q26.8.7 MCQ 2026 2M

Five integers are picked from 0 to 20, with possible repetitions, such that their mean is 12, median is 18, and they have a single mode of 20.
Ignoring permutations, the number of ways to pick these five integers is _____

A. 0 **B.** 1 **C.** 2 **D.** 3

Q26.8.8 MCQ 2026 2M

Rishi and Swathi are students of Class 5. Pavan and Tanvi are students of Class 4. Rishi and Pavan are boys. Swathi and Tanvi are girls. The four students played a total of three games of chess. The games were played one after another. A player who lost a game did not participate in any more games. It was observed that:

(i) the first game was the only game where two students of the same class played against each other,
(ii) the students of Class 5 won more games than the students of Class 4, and
(iii) the boys won two games and the girls won one game.

The student who did not lose any game is _____.

A. Pavan **B.** Rishi **C.** Swathi **D.** Tanvi

Q26.8.9 MCQ 2026 2M

P, Q, R, S, X, and Y are distinct single-digit whole numbers taking values from 0 to 9.
PQ is a two-digit number with Q being in the units place and P in the tens place. Similarly, RS is a two-digit number. It is known that PQ and RS are consecutive numbers and $(PQ)^2 + (RS)^2 = XYP$, with XYP being a three-digit number.
The value of Y is _____

A. 4 **B.** 5 **C.** 6 **D.** 7

Q26.8.10 MCQ 2026 2M

In the given figure, P, Q, and R are three points on a circle of radius 10 cm with O as its center, $\overline{PQ} = \overline{RQ}$, and $\angle PQR = 45°$. The figure is representative.
The area of the shaded region $PQRO$ is _____ cm^2.

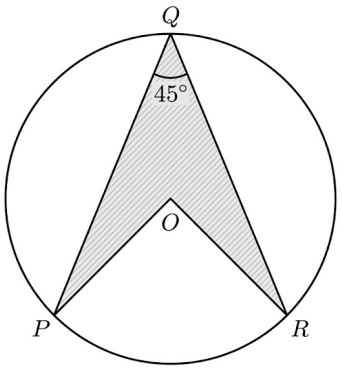


Fig. Q26.8.10

A. 50 **B.** $25\sqrt{2}$ **C.** $50\sqrt{2}$ **D.** 100

Answer Key

Session : 1					
Q26.1.1 D	Q26.1.2 C	Q26.1.3 B	Q26.1.4 D	Q26.1.5 C	Q26.1.6 A
Q26.1.7 B	Q26.1.8 B	Q26.1.9 C	Q26.1.10 A		
Session : 2					
Q26.2.1 A	Q26.2.2 B	Q26.2.3 A	Q26.2.4 B	Q26.2.5 C	Q26.2.6 D
Q26.2.7 A	Q26.2.8 D	Q26.2.9 C	Q26.2.10 C		
Session : 3					
Q26.3.1 B	Q26.3.2 C	Q26.3.3 C	Q26.3.4 D	Q26.3.5 C	Q26.3.6 B
Q26.3.7 D	Q26.3.8 A	Q26.3.9 A	Q26.3.10 C		
Session : 4					
Q26.4.1 A	Q26.4.2 C	Q26.4.3 A	Q26.4.4 A	Q26.4.5 C	Q26.4.6 A
Q26.4.7 A	Q26.4.8 A	Q26.4.9 B	Q26.4.10 C		
Session : 5					
Q26.5.1 A	Q26.5.2 A	Q26.5.3 A	Q26.5.4 C	Q26.5.5 B	Q26.5.6 A
Q26.5.7 B	Q26.5.8 A	Q26.5.9 B	Q26.5.10 D		
Session : 6					
Q26.6.1 C	Q26.6.2 A	Q26.6.3 A	Q26.6.4 D	Q26.6.5 C	Q26.6.6 B
Q26.6.7 B	Q26.6.8 C	Q26.6.9 B	Q26.6.10 B		
Session : 7					
Q26.7.1 B	Q26.7.2 A	Q26.7.3 D	Q26.7.4 B	Q26.7.5 B	Q26.7.6 A
Q26.7.7 C	Q26.7.8 C	Q26.7.9 C	Q26.7.10 C		
Session : 8					
Q26.8.1 B	Q26.8.2 B	Q26.8.3 A	Q26.8.4 C	Q26.8.5 D	Q26.8.6 A
Q26.8.7 B	Q26.8.8 D	Q26.8.9 C	Q26.8.10 C		

YEAR

2025

8 Sessions · 80 Questions

Sessions Summary

Session	Participating Branches	Total Qs
Session 1	CS1, AG, MA	10
Session 2	CS2, NM, MT, TF, IN	10
Session 3	ME, PE, AR	10
Session 4	EE	10
Session 5	CY, AE, DA, ES, PI	10
Session 6	EC, GE, XH, BM, EY	10
Session 7	CE1, GG, CH, PH, BT	10
Session 8	CE2, ST, XE, XL, MN	10

PrepFusion

Year Snapshot

Total Questions: 80 **MCQ:** 80 **MSQ:** 0 **NAT:** 0

Session : 1 | Forenoon | 01/02/25

Q25.1.1 MCQ 2025 1M

Ravi had ____ younger brother who taught at ____ university. He was widely regarded as ____ honorable man.
Select the option with the correct sequence of articles to fill in the blanks.

A. a; a; an **B.** the; an; a **C.** a; an; a **D.** an; an; a

Q25.1.2 MCQ 2025 1M

The CEO's decision to downsize the workforce was considered myopic because it sacrificed long-term stability to accommodate short-term gains.
Select the most appropriate option that can replace the word "myopic" without changing the meaning of the sentence.

A. visionary **B.** shortsighted

C. progressive **D.** innovative

Q25.1.3 MCQ 2025 1M

The average marks obtained by a class in an examination were calculated as 30.8. However, while checking the marks entered, the teacher found that the marks of one student were entered incorrectly as 24 instead of 42. After correcting the marks, the average becomes 31.4. How many students does the class have?

A. 25 **B.** 28 **C.** 30 **D.** 32

Q25.1.4 MCQ 2025 1M

Consider the relationships among P, Q, R, S, and T:

- P is the brother of Q.
- S is the daughter of Q.
- T is the sister of S.
- R is the mother of Q.

The following statements are made based on the relationships given above.

(1) R is the grandmother of S.
(2) P is the uncle of S and T.
(3) R has only one son.
(4) Q has only one daughter.

Which one of the following options is correct?

A. Both (1) and (2) are true. **B.** Both (1) and (3) are true.

C. Only (3) is true. **D.** Only (4) is true.

Q25.1.5 MCQ 2025 1M

According to the map shown in the figure, which one of the following statements is correct?
Note: The figure shown is representative.

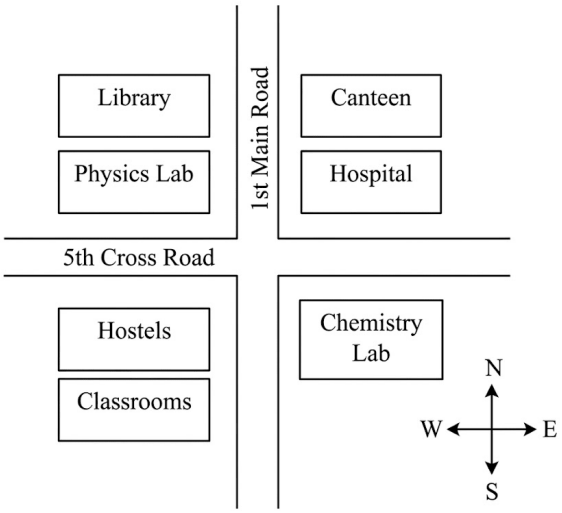


Fig. Q25.1.5

A. The library is located to the northwest of the canteen.

B. The hospital is located to the east of the chemistry lab.

C. The chemistry lab is to the southeast of physics lab.

D. The classrooms and canteen are next to each other.

Q25.1.6 MCQ 2025 2M

"I put the brown paper in my pocket along with the chalks, and possibly other things. I suppose every one must have reflected how primeval and how poetical are the things that one carries in one's pocket: the pocket-knife, for instance the type of all human tools, the infant of the sword. Once I planned to write a book of poems entirely about the things in my pocket. But I found it would be too long: and the age of the great epics is past."

(From G.K. Chesterton's "A Piece of Chalk")

Based only on the information provided in the above passage, which one of the following statements is true?

A. The author of the passage carries a mirror in his pocket to reflect upon things.

B. The author of the passage had decided to write a poem on epics.

C. The pocket-knife is described as the infant of the sword.

D. Epics are described as too inconvenient to write.

Q25.1.7 MCQ | 2025 | 2M

In the diagram, the lines QR and ST are parallel to each other. The shortest distance between these two lines is half the shortest distance between the point P and line QR. What is the ratio of the area of the triangle PST to the area of the trapezium SQRT?

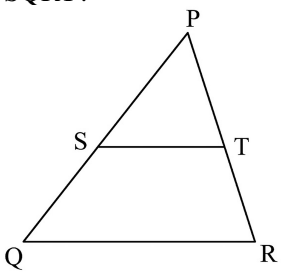


Fig. Q25.1.7

A. $\frac{1}{3}$ **B.** $\frac{1}{4}$ **C.** $\frac{2}{5}$ **D.** $\frac{1}{2}$

Q25.1.8 MCQ | 2025 | 2M

A fair six-faced dice, with the faces labelled '1', '2', '3', '4', '5', and '6', is rolled thrice. What is the probability of rolling '6' exactly once?

A. $\frac{75}{216}$ **B.** $\frac{1}{6}$ **C.** $\frac{1}{18}$ **D.** $\frac{25}{216}$

Q25.1.9 MCQ | 2025 | 2M

A square paper, shown in figure (I), is folded along the dotted lines as shown in the figures (II) and (III). Then a few cuts are made as shown in figure (IV). Which one of the following patterns will be obtained when the paper is unfolded?

Note: The figures shown are representative

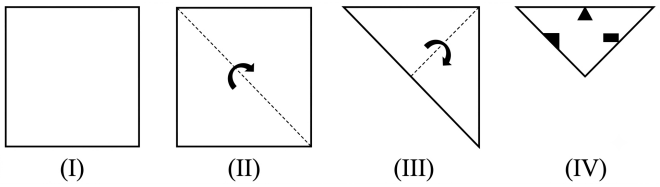
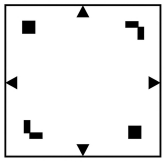
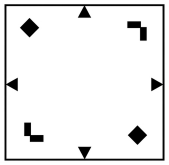


Fig. Q25.1.9

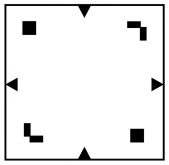
A.



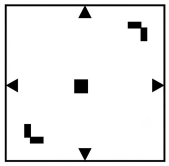
B.



C.



D.



Q25.1.10 MCQ | 2025 | 2M

A shop has 4 distinct flavors of ice-cream. One can purchase any number of scoops of any flavor. **The order in which the scoops are purchased is inconsequential**. If one wants to purchase 3 scoops of ice-cream, in how many ways can one make that purchase?

A. 4 **B.** 20 **C.** 24 **D.** 48

Session : 2 | Afternoon | 01/02/25

Q25.2.1 MCQ | 2025 | 1M

Despite his initial hesitation, Rehman's _____ to contribute to the success of the project never wavered.
Select the most appropriate option to complete the above sentence.

A. ambivalence **B.** satisfaction **C.** resolve **D.** revolve

Q25.2.2 MCQ | 2025 | 1M

Bird : Nest :: Bee : _____
Select the correct option to complete the analogy.

A. Kennel **B.** Hammock **C.** Hive **D.** Lair

Q25.2.3 MCQ | 2025 | 1M

If $Pe^{x} = Qe^{-x}$ for all real values of x, which one of the following statements is true?

A. $P = Q = 0$ **B.** $P = Q = 1$

C. $P = 1; Q = -1$ **D.** $\frac{P}{Q} = 0$

Q25.2.4 MCQ | 2025 | 1M

The paper as shown in the figure is folded to make a cube where each square corresponds to a particular face of the cube. Which one of the following options correctly represents the cube?

Note: The figures shown are representative.

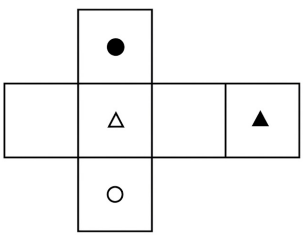
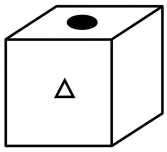
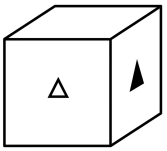
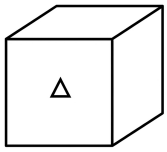
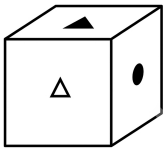


Fig. Q25.2.4

A. 

B. 

C. 

D. 

Q25.2.5 MCQ | 2025 | 1M

Let p_1 and p_2 denote two arbitrary prime numbers. Which one of the following statements is correct for all values of p_1 and p_2?

A. $p_1 + p_2$ is not a prime number.

B. $p_1 p_2$ is not a prime number.

C. $p_1 + p_2 + 1$ is a prime number.

D. $p_1 p_2 + 1$ is a prime number.

Q25.2.6 MCQ | 2025 | 2M

Based only on the conversation below, identify the logically correct inference:
"Even if I had known that you were in the hospital, I would not have gone there to see you", Ramya told Josephine.

A. Ramya knew that Josephine was in the hospital.

B. Ramya did not know that Josephine was in the hospital.

C. Ramya and Josephine were once close friends; but now, they are not.

D. Josephine was in the hospital due to an injury to her leg.

Q25.2.7 MCQ | 2025 | 2M

If IMAGE and FIELD are coded as FHBNJ and EMFJG respectively then, which one among the given options is the most appropriate code for BEACH?

A. CEADP **B.** IDBFC **C.** JGIBC **D.** IBCEC

Q25.2.8 MCQ | 2025 | 2M

Which one of the following options is correct for the given data in the table?

Iteration (i)	0	1	2	3
Input (I)	20	−4	10	15
Output (X)	20	16	26	41
Output (Y)	20	−80	−800	−12000

A. $X(i) = X(i-1) + I(i);\ Y(i) = Y(i-1)I(i);\ i > 0$

B. $X(i) = X(i-1)I(i);\ Y(i) = Y(i-1) + I(i);\ i > 0$

C. $X(i) = X(i-1)I(i);\ Y(i) = Y(i-1)I(i);\ i > 0$

D. $X(i) = X(i-1) + I(i);\ Y(i) = Y(i-1)I(i-1);\ i > 0$

Q25.2.9 MCQ | 2025 | 2M

In the given figure, PQRS is a square of side 2 cm and PLMN is a rectangle. The corner L of the rectangle is on the side QR. Side MN of the rectangle passes through the corner S of the square.
What is the area (in cm^2) of the rectangle PLMN?

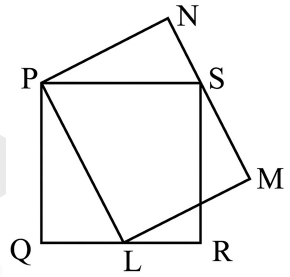


Fig. Q25.2.9

A. $2\sqrt{2}$ **B.** 2 **C.** 8 **D.** 4

Q25.2.10 MCQ | 2025 | 2M

The diagram below shows a river system consisting of 7 segments, marked P, Q, R, S, T, U, and V. It splits the land into 5 zones, marked Z1, Z2, Z3, Z4, and Z5. We need to connect these zones using the least number of bridges. Out of the following options, which one is correct?

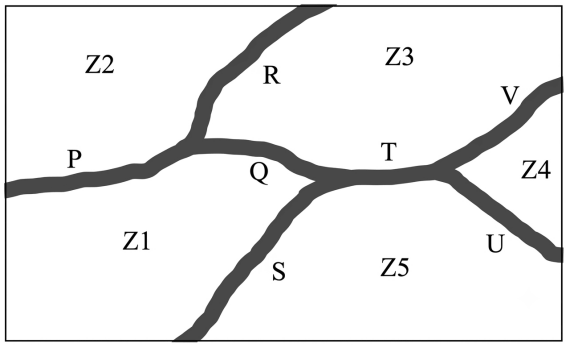


Fig. Q25.2.10

A. Bridges on P, Q, and T

B. Bridges on P, Q, S, and T

C. Bridges on Q, R, T, and V

D. Bridges on P, Q, S, U, and V

Session : 3 | Forenoon | 02/02/25

Q25.3.1 MCQ 2025 1M

Fish: Shoal :: Lion: _____

Select the correct option to complete the analogy.

A. Pride **B.** School **C.** Forest **D.** Series

Q25.3.2 MCQ 2025 1M

Identify the grammatically correct sentence:

A. It is I who am responsible for this fiasco.

B. It is myself who is responsible for this fiasco.

C. It is I who is responsible for this fiasco.

D. It is I who are responsible for this fiasco.

Q25.3.3 MCQ 2025 1M

Two cars, P and Q, start from a point X in India at 10 AM. Car P travels North with a speed of 25 km/h and car Q travels East with a speed of 30 km/h. Car P travels continuously but car Q stops for some time after travelling for one hour. If both the cars are at the same distance from X at 11:30 AM, for how long (in minutes) did car Q stop?

A. 10 **B.** 12 **C.** 15 **D.** 18

Q25.3.4 MCQ 2025 1M

The ceiling function of a real number x, denoted by $ce(x)$, is defined as the smallest integer that is greater than or equal to x. Similarly, the floor function, denoted by $fl(x)$, is defined as the largest integer that is smaller than or equal to x. Which one of the following statements is NOT correct for all possible values of x?

A. $ce(x) \geq x$ **B.** $fl(x) \leq x$

C. $ce(x) \geq fl(x)$ **D.** $fl(x) < ce(x)$

Q25.3.5 MCQ 2025 1M

P and Q play chess frequently against each other. Of these matches, P has won 80% of the matches, drawn 15% of the matches and lost 5% of the matches.

If they play 3 more matches, what is the probability of P winning exactly 2 of these 3 matches?

A. $\frac{48}{125}$ **B.** $\frac{16}{125}$ **C.** $\frac{16}{25}$ **D.** $\frac{25}{48}$

Q25.3.6 MCQ 2025 2M

Identify the option that has the most appropriate sequence such that a coherent paragraph is formed:

P. At once, without thinking much, people rushed towards the city in hordes with the sole aim of grabbing as much gold as they could.

Q. However, little did they realize about the impending hardships they would have to face on their way to the city: miles of mud, unfriendly forests, hungry beasts and inimical local lords all of which would reduce their chances of getting gold to almost zero.

R. All of them thought that easily they could lay their hands on gold and become wealthy overnight.

S. About a hundred years ago, the news that gold had been discovered in Kolar spread like wildfire and the whole State was in raptures.

A. $P \rightarrow Q \rightarrow R \rightarrow S$ **B.** $Q \rightarrow S \rightarrow R \rightarrow P$

C. $S \rightarrow Q \rightarrow P \rightarrow R$ **D.** $S \rightarrow P \rightarrow R \rightarrow Q$

Q25.3.7 MCQ 2025 2M

If HIDE and CAGE are coded as 19-23-7-11 and 5-2-17-11 respectively, then what is the code for HIGH?

A. 5-17-1-2 **B.** 17-19-13-17

C. 13-3-1-2 **D.** 19-23-17-19

Q25.3.8 MCQ 2025 2M

The given figure is reflected about the horizontal dashed line and then rotated clockwise by 90° about an axis perpendicular to the plane of the figure.

Which one of the following options correctly shows the resultant figure?

Note: The figures shown are representative.

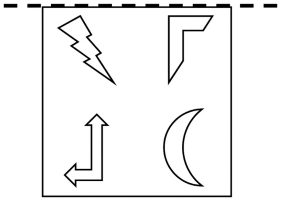
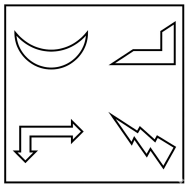
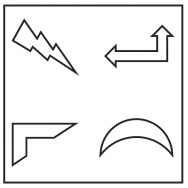
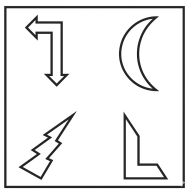
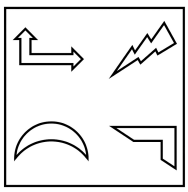


Fig. Q25.3.8

A. 

B. 

C. 

D. 

Q25.3.9 MCQ | 2025 | 2M

Which one of the following options has the correct sequence of objects arranged in the increasing number of mirror lines (lines of symmetry)?

A. Circle; Square; Equilateral triangle; Isosceles triangle

B. Isosceles triangle; Equilateral triangle; Square; Circle

C. Equilateral triangle; Isosceles triangle; Square; Circle

D. Isosceles triangle; Square; Equilateral triangle; Circle

Q25.3.10 MCQ | 2025 | 2M

A final year student appears for placement interview in two companies, S and T. Based on her interview performance, she estimates the probability of receiving job offers from companies S and T to be 0.8 and 0.6, respectively. Let p be the probability that she receives job offers from both the companies. Select the most appropriate option.

A. $0 \le p \le 0.2$ **B.** $0.4 \le p \le 0.6$

C. $0.2 \le p \le 0.4$ **D.** $0.6 \le p \le 1.0$

Session : 4 | Afternoon | 02/02/25

Q25.4.1 MCQ | 2025 | 1M

Kavya _____ go to work yesterday as she _____ feeling well.

Select the most appropriate option to complete the above sentence.

A. didn't; isn't **B.** wouldn't; wasn't

C. wasn't; wasn't **D.** couldn't; wasn't

Q25.4.2 MCQ | 2025 | 1M

Good : Evil :: Genuine : _____

Select the most appropriate option to complete the analogy.

A. Counterfeit **B.** Contraband

C. Counterfoil **D.** Counterpart

Q25.4.3 MCQ | 2025 | 1M

The relationship between two variables x and y is given by $x + py + q = 0$ and is shown in the figure. Find the values of p and q.

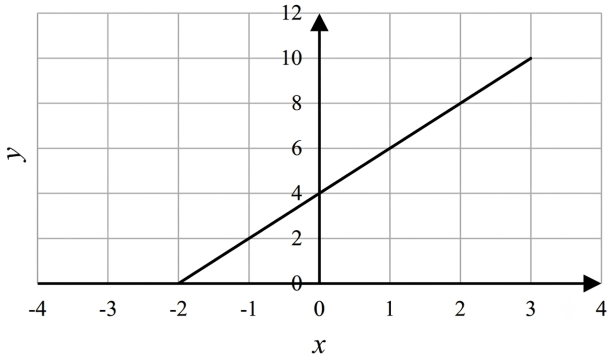


Fig. Q25.4.3

A. $p = -\frac{1}{2}; q = 2$ **B.** $p = 2; q = -2$

C. $p = \frac{1}{2}; q = 4$ **D.** $p = 2; q = 4$

Q25.4.4 MCQ | 2025 | 1M

Each row of Column-I has three items and each item is represented by a circle in Column-II. The arrangement of circles in Column-II represents the relationship among the items in Column-I. Identify the option that has the most appropriate match between Column-I and Column-II.

Column-I		Column-II	
(1)	Animals, Zebra, Giraffe	(P)	
(2)	Director, Producer, Actor	(Q)	
(3)	Word, Sentence, Novel	(R)	
(4)	Pianist, Guitarist, Instrumentalist	(S)	

Fig. Q25.4.4

A. (1)-(Q); (2)-(R); (3)-(S); (4)-(P)

B. (1)-(S); (2)-(P); (3)-(R); (4)-(Q)

C. (1)-(Q); (2)-(P); (3)-(S); (4)-(R)

D. (1)-(R); (2)-(S); (3)-(Q); (4)-(P)

Q25.4.5 MCQ | 2025 | 1M

What is the value of $\left(\frac{3^{81}}{27^4}\right)^{\frac{1}{3}}$?

A. 3^{13} **B.** 3^{96} **C.** 3^{23} **D.** 3^{69}

Q25.4.6 MCQ | 2025 | 2M

Identify the option that has the most appropriate sequence such that a coherent paragraph is formed:

P. It is because deer, like most of the animals that tigers normally prey on, run much faster! It simply means, another day of empty stomach for the big cats.

Q. Tigers spend most of their life searching for food.

R. If they trace the scent of deer, tigers follow the trail, chase the deer for a mile or two in the dark, and yet may not catch them.

S. For several nights, they relentlessly prowl through the forest, hunting for a trail that may lead to their prey.

A. $S \rightarrow P \rightarrow R \rightarrow Q$ **B.** $R \rightarrow P \rightarrow S \rightarrow Q$

C. $Q \rightarrow S \rightarrow R \rightarrow P$ **D.** $P \rightarrow Q \rightarrow S \rightarrow R$

Q25.4.7 MCQ | 2025 | 2M

In the given figure, EF and HJ are coded as 30 and 80, respectively. Which one among the given options is most appropriate for the entries marked (i) and (ii) ?

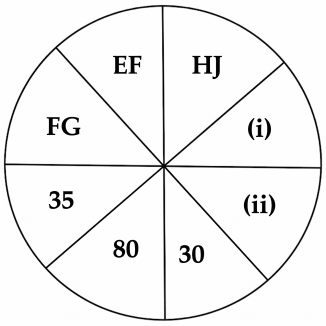


Fig. Q25.4.7

A. (i) EH; (ii) 40 **B.** (i) JK; (ii) 36

C. (i) EG; (ii) 42 **D.** (i) PS; (ii) 14

Q25.4.8 MCQ | 2025 | 2M

Scores obtained by two students P and Q in seven courses are given in the table below. Based on the information given in the table, which one of the statements is INCORRECT?

P	22	89	50	45	78	60	39
Q	35	65	60	56	81	45	50

A. Average score of P is less than the average score of Q.

B. Median score of P is same as the median score of Q.

C. Difference between the maximum and minimum scores of P is greater than the difference between the maximum and minimum scores of Q.

D. Median score and the average score of Q are same.

Q25.4.9 MCQ | 2025 | 2M

Spheres of unit diameter are centered at (l, m, n) where l, m, and n take every possible integer values.
The distance between two spheres is computed from the center of one sphere to the center of the other sphere. For a given sphere, x is the distance to its nearest sphere and y is the distance to its next nearest sphere. The value of $\frac{y}{x}$ is:

A. $2\sqrt{2}$ **B.** $\frac{1}{\sqrt{2}}$ **C.** $\sqrt{2}$ **D.** 2

Q25.4.10 MCQ | 2025 | 2M

In the triangle PQR, the lengths of PT and TR are in the ratio of 3:2.
ST is parallel to QR. Two semicircles are drawn with PS and PQ as diameters, as shown in the figure.
Which one of the following statements is true about the shaded area PQS?

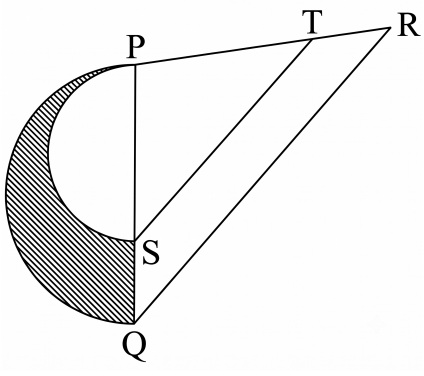


Fig. Q25.4.10

A. The shaded area is $\frac{16}{9}$ times the area of the semicircle with the diameter PS.

B. The shaded area is equal to the area of the semicircle with the diameter PS.

C. The shaded area is $\frac{14}{9}$ times the area of the semicircle with the diameter PS.

D. The shaded area is $\frac{14}{25}$ times the area of the semicircle with the diameter PQ.

Session : 5 | Forenoon | 15/02/25

Q25.5.1 MCQ 2025 1M

Courage : Bravery :: Yearning : _____

Select the most appropriate option to complete the analogy.

A. Longing **B.** Yelling **C.** Yawning **D.** Glaring

Q25.5.2 MCQ 2025 1M

We _____ tennis in the lawn when it suddenly started to rain.

Select the most appropriate option to complete the above sentence.

A. have been playing **B.** had been playing

C. could be playing **D.** would have been playing

Q25.5.3 MCQ 2025 1M

A 4×4 digital image has pixel intensities (U) as shown in the figure. The number of pixels with $U \leq 4$ is:

0	1	0	2
4	7	3	3
5	5	4	4
6	7	3	2

A. 3 **B.** 8 **C.** 11 **D.** 9

Q25.5.4 MCQ 2025 1M

In the given figure, the numbers associated with the rectangle, triangle, and ellipse are 1, 2, and 3, respectively. Which one among the given options is the most appropriate combination of P, Q, and R?

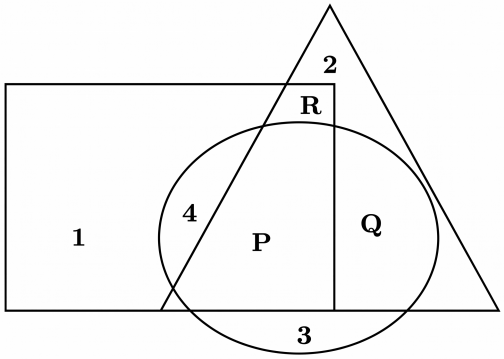


Fig. Q25.5.4

A. $P = 6; Q = 5; R = 3$ **B.** $P = 5; Q = 6; R = 3$

C. $P = 3; Q = 6; R = 6$ **D.** $P = 5; Q = 3; R = 6$

Q25.5.5 MCQ 2025 1M

A rectangle has a length L and a width W, where $L > W$. If the width, W, is increased by 10%, which one of the following statements is correct for all values of L and W?

A. Perimeter increases by 10%.

B. Length of the diagonals increases by 10%.

C. Area increases by 10%.

D. The rectangle becomes a square.

Q25.5.6 MCQ 2025 2M

Column-I has statements made by Shanthala; and, Column-II has responses given by Kanishk.

Column-I		Column-II	
P.	This house is in a mess.	1.	Alright, I won't bring it up during our conversations.
Q.	I am not happy with the marks given to me.	2.	Well, you can easily look it up.
R.	Politics is a subject I avoid talking about.	3.	No problem, let me clear it up for you.
S.	I don't know what this word means.	4.	Don't worry, I will take it up with your teacher.

Identify the option that has the correct match between Column-I and Column-II.

A. $P-2; Q-3; R-1; S-4$

B. $P-3; Q-4; R-1; S-2$

C. $P-4; Q-1; R-2; S-3$

D. $P-1; Q-2; R-4; S-3$

Q25.5.7 MCQ 2025 2M

Weight of a person can be expressed as a function of their age. The function usually varies from person to person. Suppose this function is identical for two brothers, and it monotonically increases till the age of 50 years and then it monotonically decreases. Let a_1 and a_2 (in years) denote the ages of the brothers and $a_1 < a_2$.
Which one of the following statements is correct about their age on the day when they attain the same weight?

A. $a_1 < a_2 < 50$ **B.** $a_1 < 50 < a_2$

C. $50 < a_1 < a_2$ **D.** Either $a_1 = 50$ or $a_2 = 50$

Q25.5.8 MCQ 2025 2M

A regular dodecagon (12-sided regular polygon) is inscribed in a circle of radius r cm as shown in the figure. The side of the dodecagon is d cm. All the triangles (numbered 1 to 12) in the figure are used to form squares of side r cm and each numbered triangle is used only once to form a square.
The number of squares that can be formed and the number of triangles required to form each square, respectively, are:

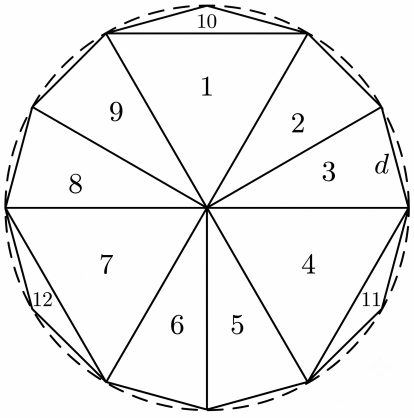


Fig. Q25.5.8

A. 3; 4 **B.** 4; 3 **C.** 3; 3 **D.** 3; 2

Q25.5.9 MCQ 2025 2M

If a real variable x satisfies $3^{x^2} = 27 \times 9^x$ then the value of $\frac{2^{x^2}}{(2^x)^2}$ is:

A. 2^{-1} **B.** 2^0 **C.** 2^3 **D.** 2^{15}

Q25.5.10 MCQ 2025 2M

The number of patients per shift (X) consulting Dr. Gita in her past 100 shifts is shown in the figure. If the amount she earns is Rs. $1000(X-0.2)$, what is the average amount (in Rs.) she has earned per shift in the past 100 shifts?
Note: The figure shown is representative.

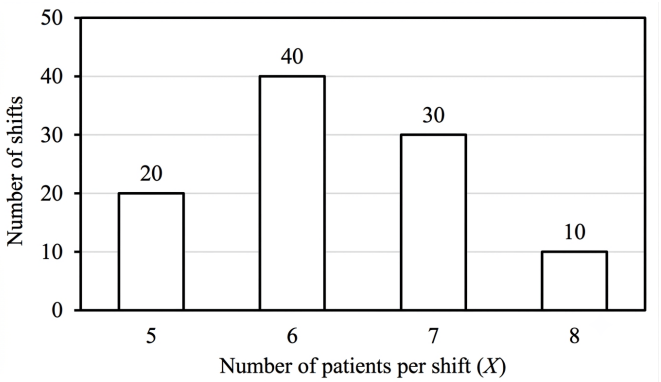


Fig. Q25.5.10

A. 6,100 **B.** 6,300 **C.** 5,800 **D.** 6,500

Session : 6 | Afternoon | 15/02/25

Q25.6.1 MCQ 2025 1M

Here are two analogous groups, Group-I and Group-II, that list words in their decreasing order of intensity. Identify the missing word in Group-II.
Group-I: Abuse → Insult → Ridicule
Group-II: _____ → Praise → Appreciate

A. Extol **B.** Prize **C.** Appropriate **D.** Espouse

Q25.6.2 MCQ 2025 1M

Had I learnt acting as a child, I _____ a famous film star. Select the most appropriate option to complete the above sentence.

A. will be **B.** can be

C. am going to be **D.** could have been

Q25.6.3 MCQ 2025 1M

The 12 musical notes are given as C, $C^{\#}$, D, $D^{\#}$, E, F, $F^{\#}$, G, $G^{\#}$, A, $A^{\#}$, and B.

Frequency of each note is $\sqrt[12]{2}$ times the frequency of the previous note. If the frequency of the note C is 130.8 Hz, then the ratio of frequencies of notes $F^{\#}$ and C is:

A. $\sqrt[6]{2}$ **B.** $\sqrt{2}$ **C.** $\sqrt[4]{2}$ **D.** 2

Q25.6.4 MCQ 2025 1M

The following figures show three curves generated using an iterative algorithm. The total length of the curve generated after 'Iteration n' is:

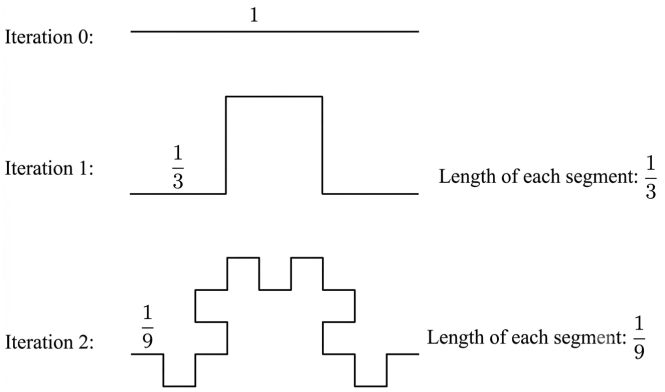
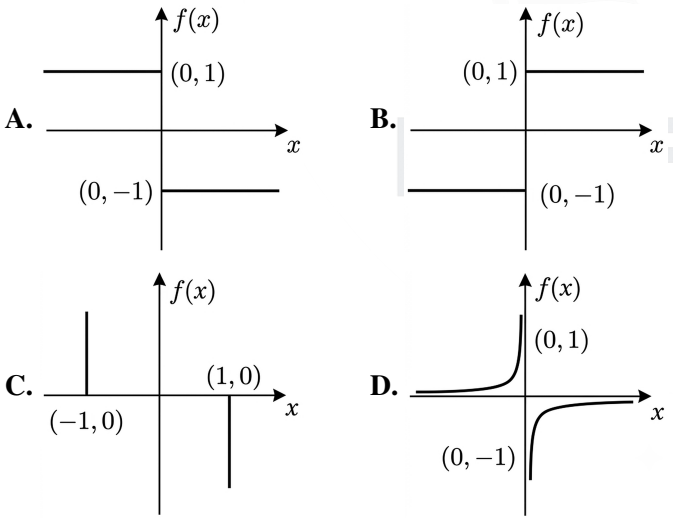


Fig. Q25.6.4

A. $\left(\frac{5}{3}\right)^{\frac{n}{2}}$ **B.** $\left(\frac{5}{3}\right)^{n}$ **C.** $\left(\frac{5}{3}\right)$ **D.** $\left(\frac{5}{3}\right)^{n(2n-1)}$

Q25.6.5 MCQ 2025 1M

Which one of the following plots represents $f(x) = -\frac{|x|}{x}$, where x is a non-zero real number?



Q25.6.6 MCQ 2025 2M

Identify the option that has the most appropriate sequence such that a coherent paragraph is formed:

P. Over time, such adaptations lead to significant evolutionary changes with the potential to shape the development of new species.

Q. In natural world, organisms constantly adapt to their environments in response to challenges and opportunities.

R. This process of adaptation is driven by the principle of natural selection, where favorable traits increase an organism's chances of survival and reproduction.

S. As environments change, organisms that can adapt their behavior, structure and physiology to such changes are more likely to survive.

A. $P \rightarrow Q \rightarrow R \rightarrow S$ **B.** $Q \rightarrow S \rightarrow R \rightarrow P$

C. $R \rightarrow S \rightarrow Q \rightarrow P$ **D.** $S \rightarrow P \rightarrow R \rightarrow Q$

Q25.6.7 MCQ 2025 2M

A stick of length one meter is broken at two locations at distances of b_1 and b_2 from the origin (0), as shown in the figure. Note that $0 < b_1 < b_2 < 1$. Which one of the following is NOT a necessary condition for forming a triangle using the three pieces?

Note: All lengths are in meter. The figure shown is representative.

Fig. Q25.6.7

A. $b_1 < 0.5$ **B.** $b_2 > 0.5$

C. $b_2 < b_1 + 0.5$ **D.** $b_1 + b_2 < 1$

Q25.6.8 MCQ 2025 2M

Eight students (P, Q, R, S, T, U, V, and W) are playing musical chairs. The figure indicates their order of position at the start of the game. They play the game by moving forward in a circle in the clockwise direction.

After the 1st round, 4th student behind P leaves the game.
After 2nd round, 5th student behind Q leaves the game.
After 3rd round, 3rd student behind V leaves the game.
After 4th round, 4th student behind U leaves the game.
Who all are left in the game after the 4th round?

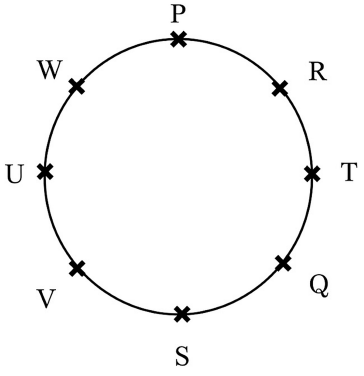


Fig. Q25.6.8

A. P; T; Q; S **B.** V; P; T; Q

C. W; R; Q; U **D.** Q; T; V; W

Q25.6.9 MCQ 2025 2M

The table lists the top 5 nations according to the number of gold medals won in a tournament; also included are the number of silver and the bronze medals won by them. Based only on the data provided in the table, which one of the following statements is INCORRECT?

Nation	Gold	Silver	Bronze
USA	40	44	41
Canada	39	27	24
Japan	20	12	13
Australia	17	19	16
France	16	26	22

A. France will occupy the third place if the list were made on the basis of the total number of medals won.

B. The order of the top two nations will not change even if the list is made on the basis of the total number of medals won.

C. USA and Canada together have less than 50% of the medals awarded to the nations in the above table.

D. Canada has won twice as many total medals as Japan.

Q25.6.10 MCQ 2025 2M

An organization allows its employees to work independently on consultancy projects but charges an overhead on the consulting fee. The overhead is 20% of the consulting fee, if the fee is up to ₹ 5,00,000. For higher fees, the overhead is ₹ 1,00,000 plus 10% of the amount by which the fee exceeds ₹ 5,00,000. The government charges a Goods and Services Tax of 18% on the total amount (the consulting fee plus the overhead). An employee of the organization charges this entire amount, i.e., the consulting fee, overhead, and tax, to the client. If the client cannot pay more than ₹ 10,00,000, what is the maximum consulting fee that the employee can charge?

A. ₹ 7,01,438 **B.** ₹ 7,24,961

C. ₹ 7,51,232 **D.** ₹ 7,75,784

Session : 7 | Forenoon | 16/02/25

Q25.7.1 MCQ 2025 1M

Is there any good show _____ television tonight?
Select the most appropriate option to complete the above sentence.

A. in **B.** at **C.** within **D.** on

Q25.7.2 MCQ 2025 1M

As the police officer was found guilty of embezzlement, he was _____ dismissed from the service in accordance with the Service Rules.
Select the most appropriate option to complete the above sentence.

A. sumptuously **B.** brazenly

C. unintentionally **D.** summarily

Q25.7.3 MCQ 2025 1M

The sum of the following infinite series is:
$\frac{1}{1!}+\frac{1}{2!}+\frac{1}{3!}+\frac{1}{4!}+\frac{1}{5!}+\cdots$

A. π **B.** $1+e$ **C.** $e-1$ **D.** e

Q25.7.4 MCQ 2025 1M

A thin wire is used to construct all the edges of a cube of 1 m side by bending, cutting and soldering the wire. If the wire is 12 m long, what is the minimum number of cuts required to construct the wire frame to form the cube?

A. 3 **B.** 4 **C.** 6 **D.** 12

Q25.7.5 MCQ 2025 1M

The figures I, II and III are parts of a sequence. Which one of the following options comes next in the sequence at IV?

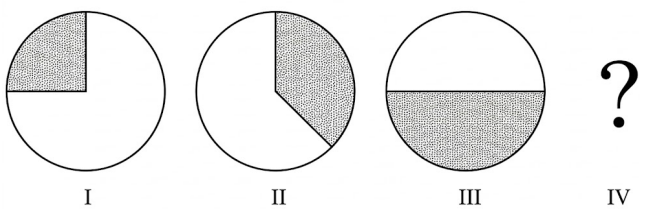
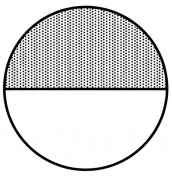
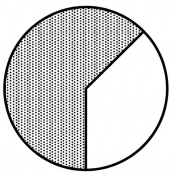
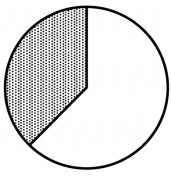
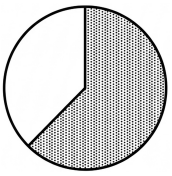


Fig. Q25.7.5

A. 

B. 

C. 

D. 

Q25.7.6 MCQ | 2025 | 2M

"Why do they pull down and do away with crooked streets, I wonder, which are my delight, and hurt no man living? Every day the wealthier nations are pulling down one or another in their capitals and their great towns: they do not know why they do it; neither do I. It ought to be enough, surely, to drive the great broad ways which commerce needs and which are the life-channels of a modern city, without destroying all history and all the humanity in between: the islands of the past."

(From Hilaire Belloc's "The Crooked Streets")

Based only on the information provided in the above passage, which one of the following statements is true?

A. The author of the passage takes delight in wondering.

B. The wealthier nations are pulling down the crooked streets in their capitals.

C. In the past, crooked streets were only built on islands.

D. Great broad ways are needed to protect commerce and history.

Q25.7.7 MCQ | 2025 | 2M

Rohit goes to a restaurant for lunch at about 1 PM. When he enters the restaurant, he notices that the hour and minute hands on the wall clock are exactly coinciding. After about an hour, when he leaves the restaurant, he notices that the clock hands are again exactly coinciding. How much time (in minutes) did Rohit spend at the restaurant?

A. $64\frac{6}{11}$ **B.** $66\frac{5}{13}$ **C.** $65\frac{5}{11}$ **D.** $66\frac{6}{13}$

Q25.7.8 MCQ | 2025 | 2M

A color model is shown in the figure with color codes: Yellow (Y), Magenta (M), Cyan (Cy), Red (R), Blue (Bl), Green (G), and Black (K).

Which one of the following options displays the color codes that are consistent with the color model?

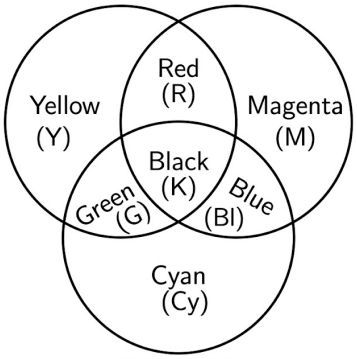
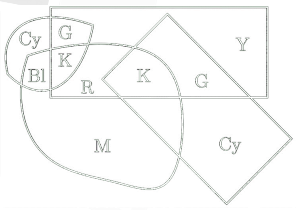
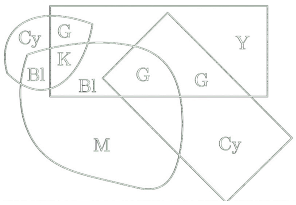
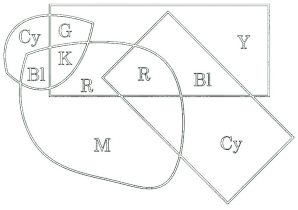
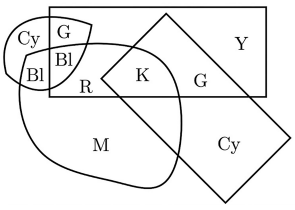


Fig. Q25.7.8

A. 

B. 

C. 

D. 

Q25.7.9 MCQ | 2025 | 2M

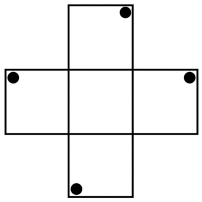
A circle with center at $(x, y) = (0.5, 0)$ and radius = 0.5 intersects with another circle with center at $(x, y) = (1, 1)$ and radius = 1 at two points. One of the points of intersection (x, y) is:

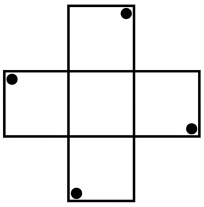
A. (0,0) **B.** (0.2, 0.4) **C.** (0.5, 0.5) **D.** (1,2)

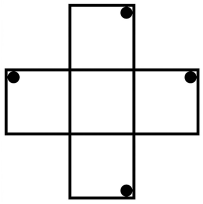
Q25.7.10 MCQ | 2025 | 2M

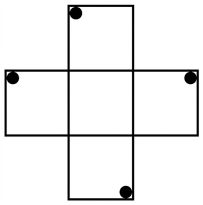
An object is said to have an n-fold rotational symmetry if the object, rotated by an angle of $\frac{2\pi}{n}$, is identical to the original.

Which one of the following objects exhibits 4-fold rotational symmetry about an axis perpendicular to the plane of the screen?

A. 

B. 

C. 

D. 

Session : 8 | Afternoon | 16/02/25

Q25.8.1 MCQ 2025 1M

Even though I had planned to go skiing with my friends, I had to _____ at the last moment because of an injury. Select the most appropriate option to complete the above sentence.

A. back up **B.** back of **C.** back on **D.** back out

Q25.8.2 MCQ 2025 1M

The President, along with the Council of Ministers, _____ to visit India next week. Select the most appropriate option to complete the above sentence.

A. wish **B.** wishes **C.** will wish **D.** is wishing

Q25.8.3 MCQ 2025 1M

An electricity utility company charges ₹ 7 per kWh (kilo watt-hour). If a 40-watt desk light is left on for 10 hours each night for 180 days, what would be the cost of energy consumption? If the desk light is on for 2 more hours each night for the 180 days, what would be the percentage-increase in the cost of energy consumption?

A. ₹ 604.8; 10% **B.** ₹ 504; 20%

C. ₹ 604.8; 12% **D.** ₹ 720; 15%

Q25.8.4 MCQ 2025 1M

In the context of the given figure, which one of the following options correctly represents the entries in the blocks labelled (i), (ii), (iii), and (iv), respectively?

N	U	F	(i)
21	14	9	6
H	L	(ii)	O
12	(iv)	15	(iii)

Fig. Q25.8.4

A. Q, M, 12, and 8 **B.** K, L, 10 and 14

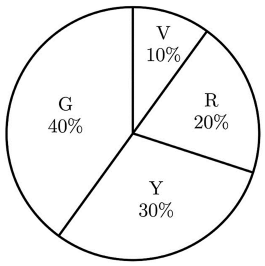
C. I, J, 10, and 8 **D.** L, K, 12 and 8

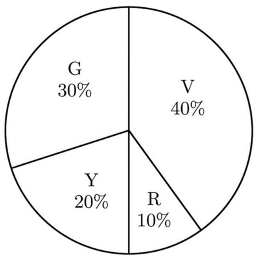
Q25.8.5 MCQ 2025 1M

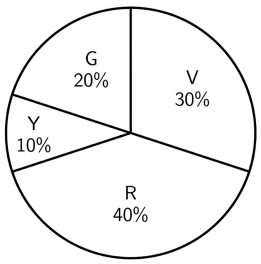
A bag contains Violet (V), Yellow (Y), Red (R), and Green (G) balls. On counting them, the following results are obtained:

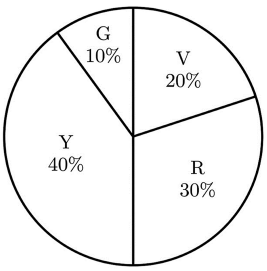
(i) The sum of Yellow balls and twice the number of Violet balls is 50.
(ii) The sum of Violet and Green balls is 50.
(iii) The sum of Yellow and Red balls is 50.
(iv) The sum of Violet and twice the number of Red balls is 50.

Which one of the following Pie charts correctly represents the balls in the bag?

A. 

B. 

C. 

D. 

Q25.8.6 MCQ 2025 2M

"His life was divided between the books, his friends, and long walks. A solitary man, he worked at all hours without much method, and probably courted his fatal illness in this way. To his own name there is not much to show; but such was his liberality that he was continually helping others, and fruits of his erudition are widely scattered, and have gone to increase many a comparative stranger's reputation."

(From E.V. Lucas's "A Funeral")

Based only on the information provided in the above passage, which one of the following statements is true?

A. The solitary man described in the passage is dead.

B. Strangers helped create a grand reputation for the solitary man described in the passage.

C. The solitary man described in the passage found joy in scattering fruits.

D. The solitary man worked in a court where he fell ill.

Q25.8.7 MCQ 2025 2M

For the clock shown in the figure, if

O^* = O Q S Z P R T, and

X^* = X Z P W O Q

then which one among the given options is most appropriate for P^*?

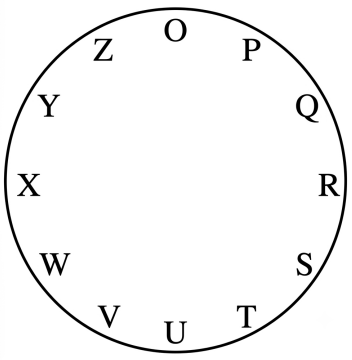


Fig. Q25.8.7

A. P U W R T V W

B. P R T O Q S U

C. P T V Q S U W

D. P S U P R T V

Q25.8.8 MCQ 2025 2M

Consider a five-digit number PQRST that has distinct digits P, Q, R, S, and T, and satisfies the following conditions:

$$P < Q$$

$$S > P > T$$

$$R < T$$

If integers 1 through 5 are used to construct such a number, the value of P is:

A. 1 **B.** 2 **C.** 3 **D.** 4

Q25.8.9 MCQ 2025 2M

A business person buys potatoes of two different varieties P and Q, mixes them in a certain ratio and sells them at ₹192 per kg. The cost of the variety P is ₹800 for 5 kg. The cost of the variety Q is ₹800 for 4 kg. If the person gets 8% profit, what is the P:Q ratio (by weight)?

A. 5:4 **B.** 3:4 **C.** 3:2 **D.** 1:1

Q25.8.10 MCQ 2025 2M

Three villages P, Q, and R are located in such a way that the distance PQ = 13 km, QR = 14 km, and RP = 15 km, as shown in the figure. A straight road joins Q and R. It is proposed to connect P to this road QR by constructing another road. What is the minimum possible length (in km) of this connecting road?

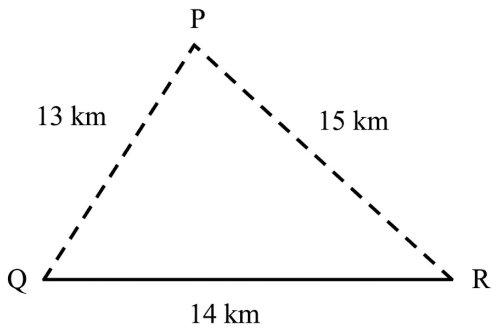


Fig. Q25.8.10

A. 10.5 **B.** 11.0 **C.** 12.0 **D.** 12.5

Answer Key

Note: MTA Marks To All awarded — see detailed solutions.

Session : 1					
Q25.1.1 A	Q25.1.2 B	Q25.1.3 C	Q25.1.4 A	Q25.1.5 C	Q25.1.6 C
Q25.1.7 A	Q25.1.8 A	Q25.1.9 A	Q25.1.10 B		
Session : 2					
Q25.2.1 C	Q25.2.2 C	Q25.2.3 A	Q25.2.4 A	Q25.2.5 B	Q25.2.6 B
Q25.2.7 B	Q25.2.8 A	Q25.2.9 D	Q25.2.10 C		
Session : 3					
Q25.3.1 A	Q25.3.2 A	Q25.3.3 C	Q25.3.4 D	Q25.3.5 A	Q25.3.6 D
Q25.3.7 D	Q25.3.8 B	Q25.3.9 B	Q25.3.10 B		
Session : 4					
Q25.4.1 D	Q25.4.2 A	Q25.4.3 A	Q25.4.4 A	Q25.4.5 C	Q25.4.6 C
Q25.4.7 C	Q25.4.8 B	Q25.4.9 C	Q25.4.10 A		
Session : 5					
Q25.5.1 A	Q25.5.2 B	Q25.5.3 C	Q25.5.4 A	Q25.5.5 C	Q25.5.6 B
Q25.5.7 B	Q25.5.8 A	Q25.5.9 C	Q25.5.10 A		
Session : 6					
Q25.6.1 A	Q25.6.2 D	Q25.6.3 B	Q25.6.4 B	Q25.6.5 A	Q25.6.6 B
Q25.6.7 D	Q25.6.8 MTA	Q25.6.9 C	Q25.6.10 B		
Session : 7					
Q25.7.1 D	Q25.7.2 D	Q25.7.3 C	Q25.7.4 A	Q25.7.5 B	Q25.7.6 B
Q25.7.7 C	Q25.7.8 A	Q25.7.9 B	Q25.7.10 B		
Session : 8					
Q25.8.1 D	Q25.8.2 B	Q25.8.3 B	Q25.8.4 C	Q25.8.5 A	Q25.8.6 A
Q25.8.7 B	Q25.8.8 C	Q25.8.9 A	Q25.8.10 C		

YEAR

2024

8 Sessions · 80 Questions

Sessions Summary

Session	Participating Branches	Total Qs
Session 1	AR, CY, DA, ES, PI	10
Session 2	GE, ME, PH	10
Session 3	BM, CE1, CH, EY, GG, XH	10
Session 4	CE2, IN, MA, MN, PE	10
Session 5	AE, AG, BT, CS1, MT, NM, TF	10
Session 6	CS2, ST, XE, XL	10
Session 7	EC	10
Session 8	EE	10

PrepFusion

Year Snapshot

Total Questions: 80 **MCQ:** 80 **MSQ:** 0 **NAT:** 0

Session : 1 | Forenoon | 03/02/24

Q24.1.1 MCQ 2024 1M

If '→' denotes increasing order of intensity, then the meaning of the words
[sick → infirm → moribund] is analogous to [silly → _____ → daft].
Which one of the given options is appropriate to fill the blank?

A. frown **B.** fawn **C.** vein **D.** vain

Q24.1.2 MCQ 2024 1M

The 15 parts of the given figure are to be painted such that no two adjacent parts with shared boundaries (excluding corners) have the same color. The minimum number of colors required is

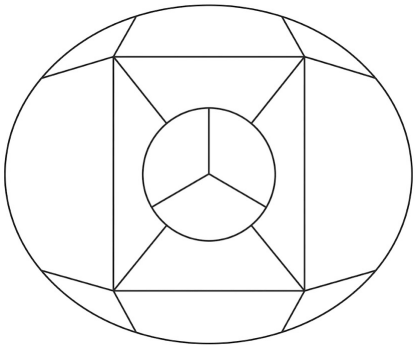


Fig. Q24.1.2

A. 4 **B.** 3 **C.** 5 **D.** 6

Q24.1.3 MCQ 2024 1M

How many 4-digit positive integers divisible by 3 can be formed using only the digits $\{1, 3, 4, 6, 7\}$, such that no digit appears more than once in a number?

A. 24 **B.** 48 **C.** 72 **D.** 12

Q24.1.4 MCQ 2024 1M

The sum of the following infinite series is

$$2 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{8} + \frac{1}{9} + \frac{1}{16} + \frac{1}{27} + \cdots$$

A. 11/3 **B.** 7/2 **C.** 13/4 **D.** 9/2

Q24.1.5 MCQ 2024 1M

In an election, the share of valid votes received by the four candidates A, B, C, and D is represented by the pie chart shown. The total number of votes cast in the election were 1,15,000, out of which 5,000 were invalid.

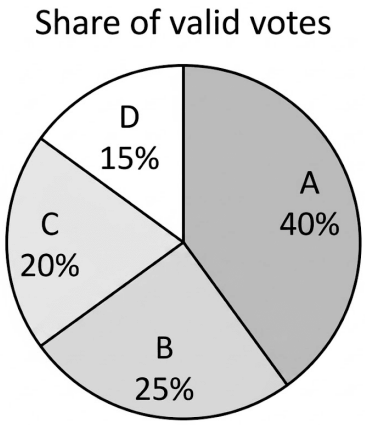


Fig. Q24.1.5

Based on the data provided, the total number of valid votes received by the candidates B and C is .

A. 45,000 **B.** 49,500 **C.** 51,750 **D.** 54,000

Q24.1.6 MCQ 2024 2M

Thousands of years ago, some people began dairy farming. This coincided with a number of mutations in a particular gene that resulted in these people developing the ability to digest dairy milk.
Based on the given passage, which of the following can be inferred?

A. All human beings can digest dairy milk.

B. No human being can digest dairy milk.

C. Digestion of dairy milk is essential for human beings.

D. In human beings, digestion of dairy milk resulted from a mutated gene.

Q24.1.7 MCQ 2024 2M

The probability of a boy or a girl being born is 1/2. For a family having only three children, what is the probability of having two girls and one boy?

A. 3/8 **B.** 1/8 **C.** 1/4 **D.** 1/2

Q24.1.8 MCQ 2024 2M

Person 1 and Person 2 invest in three mutual funds A, B, and C. The amounts they invest in each of these mutual funds are given in the table.

	Mutual fund		
	A	B	C
Person 1	₹10,000	₹20,000	₹20,000
Person 2	₹20,000	₹15,000	₹15,000

At the end of one year, the total amount that Person 1 gets is Rs 500 more than Person 2. The annual rate of return for the mutual funds B and C is 15% each. What is the annual rate of return for the mutual fund A?

A. 7.5% **B.** 10% **C.** 15% **D.** 20%

Q24.1.9 MCQ 2024 2M

Three different views of a dice are shown in the figure below.

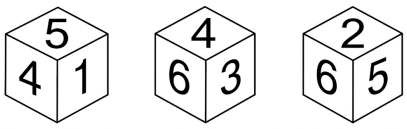


Fig. Q24.1.9

The piece of paper that can be folded to make this dice is .

A.

5	1
4	
6	
2	3

B.

5	1
4	
2	
6	3

C.

5	1
3	
2	
4	6

D.

5	1
4	
6	
3	2

Q24.1.10 MCQ 2024 2M

Visualize two identical right circular cones such that one is inverted over the other and they share a common circular base. If a cutting plane passes through the vertices of the assembled cones, what shape does the outer boundary of the resulting cross-section make?

A. A rhombus **B.** A triangle

C. An ellipse **D.** A hexagon

Session : 2 | Afternoon | 03/02/24

Q24.2.1 MCQ 2024 1M

If '→' denotes increasing order of intensity, then the meaning of the words

[smile → giggle → laugh] is analogous to [disapprove → _____ → chide].

Which one of the given options is appropriate to fill the blank?

A. reprove **B.** praise **C.** reprise **D.** grieve

Q24.2.2 MCQ 2024 1M

Find the odd one out in the set:

{19, 37, 21, 17, 23, 29, 31, 11}

A. 21 **B.** 29 **C.** 37 **D.** 23

Q24.2.3 MCQ 2024 1M

In the following series, identify the number that needs to be changed to form the Fibonacci series.

1, 1, 2, 3, 6, 8, 13, 21,...

A. 8 **B.** 21 **C.** 6 **D.** 13

Q24.2.4 MCQ 2024 1M

The real variables x, y, z, and the real constants p, q, r satisfy

$$\frac{x}{pq - r^2} = \frac{y}{qr - p^2} = \frac{z}{rp - q^2}$$

Given that the denominators are non-zero, the value of $px + qy + rz$ is

A. 0 **B.** 1 **C.** pqr **D.** $p^2 + q^2 + r^2$

Q24.2.5 MCQ 2024 1M

Take two long dice (rectangular parallelepiped), each having four rectangular faces labelled as 2, 3, 5, and 7. If thrown, the long dice cannot land on the square faces and has 1/4 probability of landing on any of the four rectangular faces. The label on the top face of the dice is the score of the throw.

If thrown together, what is the probability of getting the sum of the two long dice scores greater than 11?

A. 3/8 **B.** 1/8 **C.** 1/16 **D.** 3/16

Q24.2.6 MCQ 2024 2M

In the given text, the blanks are numbered (i)-(iv). Select the best match for all the blanks.

Prof. P (i) merely a man who narrated funny stories (ii) in his blackest moments he was capable of self-deprecating humor.

Prof. Q (iii) a man who hardly narrated funny stories (iv) in his blackest moments was he able to find humor.

A. (i) was, (ii) Only, (iii) wasn't, (iv) Even

B. (i) wasn't, (ii) Even, (iii) was, (iv) Only

C. (i) was, (ii) Even, (iii) wasn't, (iv) Only

D. (i) wasn't, (ii) Only, (iii) was, (iv) Even

Q24.2.7 MCQ 2024 2M

How many combinations of non-null sets A, B, C are possible from the subsets of {2, 3, 5} satisfying the conditions: (i) A is a subset of B, and (ii) B is a subset of C?

A. 28 **B.** 27 **C.** 18 **D.** 19

Q24.2.8 MCQ 2024 2M

The bar chart gives the batting averages of VK and RS for 11 calendar years from 2012 to 2022. Considering that 2015 and 2019 are world cup years, which one of the following options is true?

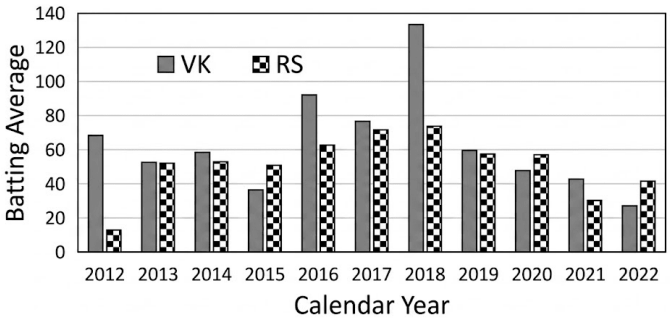


Fig. Q24.2.8

A. RS has a higher yearly batting average than that of VK in every world cup year.

B. VK has a higher yearly batting average than that of RS in every world cup year.

C. VK's yearly batting average is consistently higher than that of RS between the two world cup years.

D. RS's yearly batting average is consistently higher than that of VK in the last three calendar years.

Q24.2.9 MCQ 2024 2M

A planar rectangular paper has two V-shaped pieces attached as shown below.

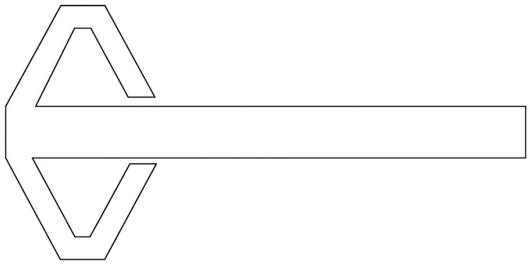


Fig. Q24.2.9

This piece of paper is folded to make the following closed three-dimensional object.

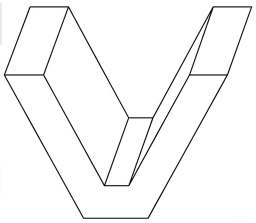


Fig. Q24.2.9

The number of folds required to form the above object is

A. 9 **B.** 7 **C.** 11 **D.** 8

Q24.2.10 MCQ 2024 2M

Four equilateral triangles are used to form a regular closed three-dimensional object by joining along the edges. The angle between any two faces is

A. 30° **B.** 60° **C.** 45° **D.** 90°

Session : 3 | Forenoon | 04/02/24

Q24.3.1 MCQ 2024 1M

If '→' denotes increasing order of intensity, then the meaning of the words

[simmer → seethe → smolder] is analogous to [break → _____ → raze].

Which one of the given options is appropriate to fill the blank?

A. obfuscate **B.** obliterate **C.** fracture **D.** fissure

Q24.3.2 MCQ 2024 1M

In a locality, the houses are numbered in the following way: The house-numbers on one side of a road are consecutive odd integers starting from 301, while the house-numbers on the other side of the road are consecutive even numbers starting from 302. The total number of houses is the same on both sides of the road.

If the difference of the sum of the house-numbers between the two sides of the road is 27, then the number of houses on each side of the road is

A. 27 **B.** 52 **C.** 54 **D.** 26

Q24.3.3 MCQ 2024 1M

For positive integers p and q, with $\frac{p}{q} \neq 1$, $\left(\frac{p}{q}\right)^{\frac{p}{q}} = p^{\left(\frac{p}{q}-1\right)}$. Then,

A. $q^p = p^q$ **B.** $q^p = p^{2q}$

C. $\sqrt{q} = \sqrt{p}$ **D.** $\sqrt[q]{q} = \sqrt[q]{p}$

Q24.3.4 MCQ 2024 1M

Which one of the given options is a possible value of x in the following sequence?

3, 7, 15, x, 63, 127, 255

A. 35 **B.** 40 **C.** 45 **D.** 31

Q24.3.5 MCQ 2024 1M

On a given day, how many times will the second-hand and the minute-hand of a clock cross each other during the clock time 12:05:00 hours to 12:55:00 hours?

A. 51 **B.** 49 **C.** 50 **D.** 55

Q24.3.6 MCQ 2024 2M

In the given text, the blanks are numbered (i)-(iv). Select the best match for all the blanks.

From the ancient Athenian arena to the modern Olympic stadiums, athletics (i) the potential for a spectacle. The crowd (ii) with bated breath as the Olympian artist twists his body, stretching the javelin behind him. Twelve strides in, he begins to cross-step. Six cross-steps (iii) in an abrupt stop on his left foot. As his body (iv) like a door turning on a hinge, the javelin is launched skyward at a precise angle.

A. (i) hold, (ii) waits, (iii) culminates, (iv) pivot

B. (i) holds, (ii) wait, (iii) culminates, (iv) pivot

C. (i) hold, (ii) wait, (iii) culminate, (iv) pivots

D. (i) holds, (ii) waits, (iii) culminate, (iv) pivots

Q24.3.7 MCQ 2024 2M

Three distinct sets of indistinguishable twins are to be seated at a circular table that has 8 identical chairs. Unique seating arrangements are defined by the relative positions of the people.

How many unique seating arrangements are possible such that each person is sitting next to their twin?

A. 12 **B.** 14 **C.** 10 **D.** 28

Q24.3.8 MCQ 2024 2M

The chart given below compares the Installed Capacity (MW) of four power generation technologies, T1, T2, T3, and T4, and their Electricity Generation (MWh) in a time of 1000 hours (h).

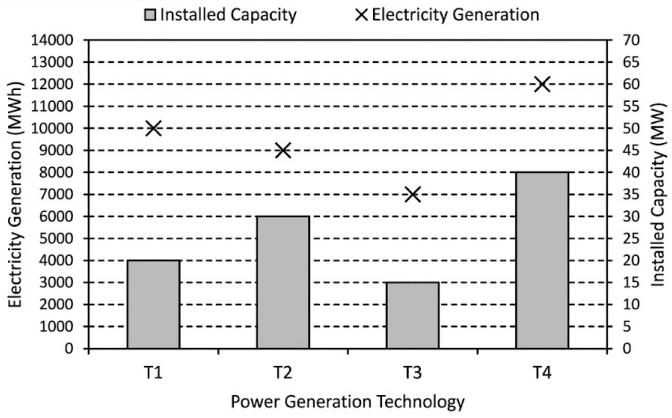


Fig. Q24.3.8

The Capacity Factor of a power generation technology is:

$$\text{Capacity Factor} = \frac{\text{Electricity Generation (MWh)}}{\text{Installed Capacity (MW)} \times 1000\ \text{(h)}}$$

Which one of the given technologies has the highest Capacity Factor?

A. T1 **B.** T2 **C.** T3 **D.** T4

Q24.3.9 MCQ 2024 2M

In the 4×4 array shown below, each cell of the first three columns has either a cross (X) or a number, as per the given rule.

1	1	2	
2	X	3	
2	X	4	
1	2	X	

Fig. Q24.3.9

Rule: The number in a cell represents the count of crosses around its immediate neighboring cells (left, right, top, bottom, diagonals).
As per this rule, the **maximum** number of crosses possible in the empty column is

A. 0 **B.** 1 **C.** 2 **D.** 3

Q24.3.10 MCQ 2024 2M

During a half-moon phase, the Earth-Moon-Sun form a right triangle. If the Moon-Earth-Sun angle at this half-moon phase is measured to be 89.85°, the ratio of the Earth-Sun and Earth-Moon distances is closest to

A. 328 **B.** 382 **C.** 238 **D.** 283

Session : 4 | Afternoon | 04/02/24

Q24.4.1 MCQ 2024 1M

If '→' denotes increasing order of intensity, then the meaning of the words
[drizzle → rain → downpour] is analogous to [_____ → quarrel → feud].
Which one of the given options is appropriate to fill the blank?

A. bicker **B.** bog **C.** dither **D.** dodge

Q24.4.2 MCQ 2024 1M

Statements:

1. All heroes are winners.
2. All winners are lucky people.

Inferences:

I. All lucky people are heroes.
II. Some lucky people are heroes.
III. Some winners are heroes.

Which of the above inferences can be logically deduced from statements 1 and 2?

A. Only I and II **B.** Only II and III
C. Only I and III **D.** Only III

Q24.4.3 MCQ 2024 1M

A student was supposed to **multiply** a positive real number p with another positive real number q. Instead, the student **divided** p by q. If the percentage error in the student's answer is 80%, the value of q is

A. 5 **B.** $\sqrt{2}$ **C.** 2 **D.** $\sqrt{5}$

Q24.4.4 MCQ 2024 1M

If the sum of the first 20 consecutive positive odd numbers is divided by 20^2, the result is

A. 1 **B.** 20 **C.** 2 **D.** 1/2

Q24.4.5 MCQ 2024 1M

The ratio of the number of girls to boys in class VIII is the same as the ratio of the number of boys to girls in class IX. The total number of students (boys and girls) in classes VIII and IX is 450 and 360, respectively. If the number of girls in classes VIII and IX is the same, then the number of girls in each class is

A. 150 **B.** 200 **C.** 250 **D.** 175

Q24.4.6 MCQ 2024 2M

In the given text, the blanks are numbered (i)-(iv). Select the best match for all the blanks.
Yoko Roi stands (i) as an author for standing (ii) as an honorary fellow, after she stood (iii) her writings that stand (iv) the freedom of speech.

A. (i) out, (ii) down, (iii) in, (iv) for

B. (i) down, (ii) out, (iii) by, (iv) in

C. (i) down, (ii) out, (iii) for, (iv) in

D. (i) out, (ii) down, (iii) by, (iv) for

Q24.4.7 MCQ 2024 2M

Seven identical cylindrical chalk-sticks are fitted tightly in a cylindrical container. The figure below shows the arrangement of the chalk-sticks inside the cylinder.

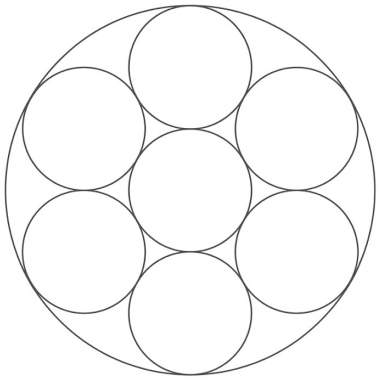


Fig. Q24.4.7

The length of the container is equal to the length of the chalk-sticks. The ratio of the occupied space to the empty space of the container is

A. 5/2 **B.** 7/2 **C.** 9/2 **D.** 3

Q24.4.8 MCQ 2024 2M

The plot below shows the relationship between the mortality risk of cardiovascular disease and the number of steps a person walks per day. Based on the data, which one of the following options is true?

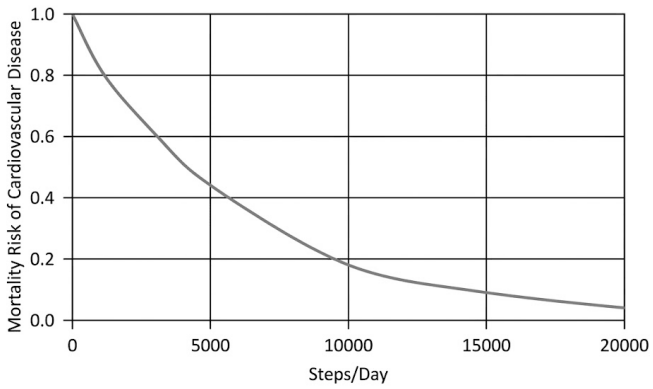


Fig. Q24.4.8

A. The risk reduction on increasing the steps/day from 0 to 10000 is less than the risk reduction on increasing the steps/day from 10000 to 20000.

B. The risk reduction on increasing the steps/day from 0 to 5000 is less than the risk reduction on increasing the steps/day from 15000 to 20000.

C. For any 5000 increment in steps/day the largest risk reduction occurs on going from 0 to 5000.

D. For any 5000 increment in steps/day the largest risk reduction occurs on going from 15000 to 20000.

Q24.4.9 MCQ 2024 2M

Five cubes of identical size and another smaller cube are assembled as shown in Figure A. If viewed from direction X, the planar image of the assembly appears as Figure B.

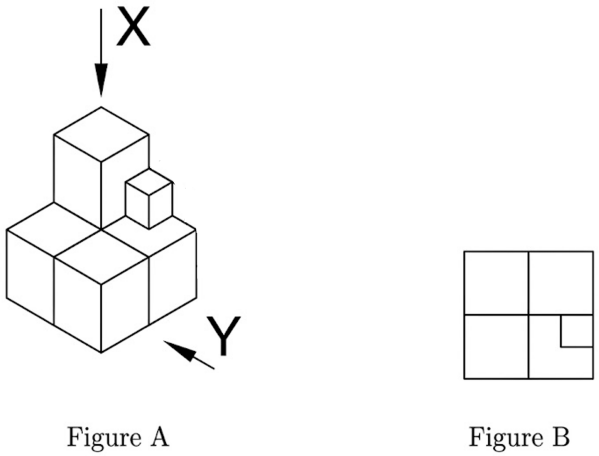
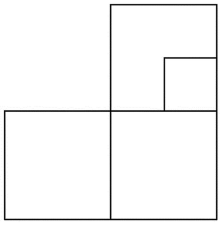


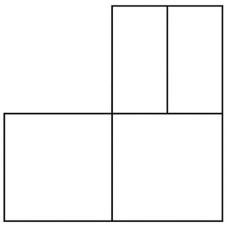
Fig. Q24.4.9

If viewed from direction Y, the planar image of the assembly appears as

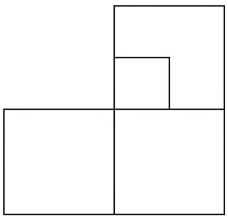
A.



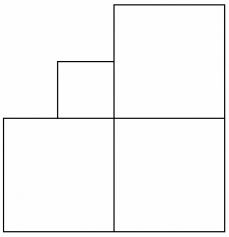
B.



C.



D.



Q24.4.10 MCQ 2024 2M

Visualize a cube that is held with one of the four body diagonals aligned to the vertical axis. Rotate the cube about this axis such that its view remains unchanged. The magnitude of the minimum angle of rotation is

A. 120° **B.** 60° **C.** 90° **D.** 180°

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Q24.5.1 MCQ 2024 1M

If '→' denotes increasing order of intensity, then the meaning of the words
[dry → arid → parched] is analogous to [diet → fast → _____].
Which one of the given options is appropriate to fill the blank?

A. starve **B.** reject **C.** feast **D.** deny

Q24.5.2 MCQ 2024 1M

If two distinct non-zero real variables x and y are such that $(x + y)$ is proportional to $(x - y)$ then the value of $\frac{x}{y}$

A. depends on xy

B. depends only on x and not on y

C. depends only on y and not on x

D. is a constant

Q24.5.3 MCQ 2024 1M

Consider the following sample of numbers:

$$9, 18, 11, 14, 15, 17, 10, 69, 11, 13$$

The median of the sample is

A. 13.5 **B.** 14 **C.** 11 **D.** 18.7

Q24.5.4 MCQ 2024 1M

The number of coins of ₹1, ₹5, and ₹10 denominations that a person has are in the ratio 5:3:13. Of the total amount, the percentage of money in ₹5 coins is _____.

A. 21% **B.** $14\frac{2}{7}\%$ **C.** 10% **D.** 30%

Q24.5.5 MCQ 2024 1M

For positive non-zero real variables p and q, if

$$\log(p^2 + q^2) = \log p + \log q + 2\log 3$$

then, the value of $\frac{p^4 + q^4}{p^2 q^2}$ is _____.

A. 79 **B.** 81 **C.** 9 **D.** 83

Q24.5.6 MCQ 2024 2M

In the given text, the blanks are numbered (i)-(iv). Select the best match for all the blanks.
Steve was advised to keep his head __(i)__ before heading __(ii)__ to bat;for, while he had a head __(iii)__ batting, he could only do so with a cool head __(iv)__ his shoulders.

A. (i) down, (ii) down, (iii) on, (iv) for

B. (i) on, (ii) down, (iii) for, (iv) on

C. (i) down, (ii) out, (iii) for, (iv) on

D. (i) on, (ii) out, (iii) on, (iv) for

Q24.5.7 MCQ 2024 2M

A rectangular paper sheet of dimensions 54 cm x 4 cm is taken. The two longer edges of the sheet are joined together to create a cylindrical tube. A cube whose surface area is equal to the area of the sheet is also taken.
Then, the ratio of the volume of the cylindrical tube to the volume of the cube is

A. $1/\pi$ **B.** $2/\pi$ **C.** $3/\pi$ **D.** $4/\pi$

Q24.5.8 MCQ 2024 2M

The pie chart presents the percentage contribution of different macronutrients to a typical 2,000 kcal diet of a person.

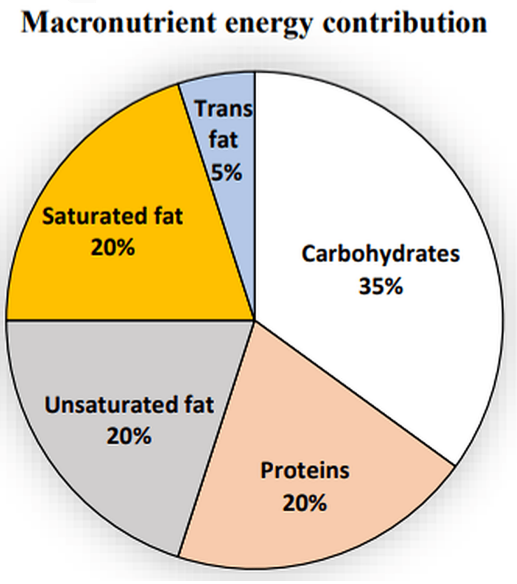

Fig. Q24.5.8

The typical energy density (kcal/g) of these macronutrients is given in the table.

Macronutrient	Energy density (kcal/g)
Carbohydrates	4
Proteins	4
Unsaturated fat	9
Saturated fat	9
Trans fat	9

The total fat (all three types), in grams, this person consumes is

A. 44.4 **B.** 77.8 **C.** 100 **D.** 3,600

Q24.5.9 MCQ 2024 2M

A rectangular paper of 20 cm × 8 cm is folded 3 times. Each fold is made along the line of symmetry, which is perpendicular to its long edge. The perimeter of the final folded sheet (in cm) is

A. 18 **B.** 24 **C.** 20 **D.** 21

Q24.5.10 MCQ 2024 2M

The least number of squares to be added in the figure to make AB a line of symmetry is

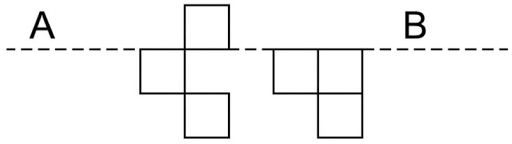

Fig. Q24.5.10

A. 6 **B.** 4 **C.** 5 **D.** 7

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Q24.6.1 MCQ 2024 1M

If '→' denotes increasing order of intensity, then the meaning of the words [walk → jog → sprint] is analogous to [bothered → _____ → daunted].
Which one of the given options is appropriate to fill the blank?

A. phased **B.** phrased **C.** fazed **D.** fused

Q24.6.2 MCQ 2024 1M

Two wizards try to create a spell using all the four elements, *water, air, fire*, and *earth*. For this, they decide to mix all these elements in all possible orders. They also decide to work independently. After trying all possible combination of elements, they conclude that the spell does not work.
How many attempts does each wizard make before coming to this conclusion, independently?

A. 24 **B.** 48 **C.** 16 **D.** 12

Q24.6.3 MCQ 2024 1M

In an engineering college of 10,000 students, 1,500 like neither their core branches nor other branches. The number of students who like their core branches is $1/4^{th}$ of the number of students who like other branches. The number of students who like both their core and other branches is 500.
The number of students who like their core branches is

A. 1,800 **B.** 3,500 **C.** 1,600 **D.** 1,500

Q24.6.4 MCQ 2024 1M

For positive non-zero real variables x and y, if

$$\ln\left(\frac{x+y}{2}\right) = \frac{1}{2}[\ln(x) + \ln(y)]$$

then, the value of $\frac{x}{y} + \frac{y}{x}$ is .

A. 1 **B.** 1/2 **C.** 2 **D.** 4

Q24.6.5 MCQ 2024 1M

In the sequence 6, 9, 14, x, 30, 41, a possible value of x is

A. 25 **B.** 21 **C.** 18 **D.** 20

Q24.6.6 MCQ 2024 2M

Sequence the following sentences in a coherent passage.

P: This fortuitous geological event generated a colossal amount of energy and heat that resulted in the rocks rising to an average height of 4 km across the contact zone.
Q: Antarctic Plate collided with the massive Eurasian Plate.
R: Thus, the geophysicists tend to think of the Himalayas as an active geological event rather than as a static geological feature.
S: Many millennia ago, a breakaway chunk of bedrock from the Antarctic Plate collided with the massive Eurasian Plate.

A. QPSR **B.** QSPR **C.** SPRQ **D.** SRPQ

Q24.6.7 MCQ 2024 2M

A person sold two different items at the same price. He made 10% profit in one item, and 10% loss in the other item. In selling these two items, the person made a total of

A. 1% profit **B.** 2% profit **C.** 1% loss **D.** 2% loss

Q24.6.8 MCQ 2024 2M

The pie charts depict the shares of various power generation technologies in the total electricity generation of a country for the years 2007 and 2023.

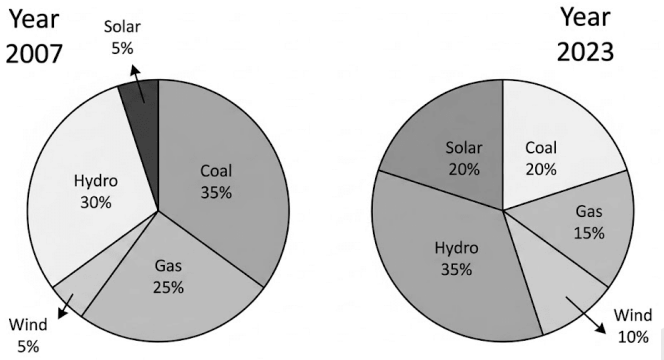

Fig. Q24.6.8

The renewable sources of electricity generation consist of Hydro, Solar and Wind. Assuming that the total electricity generated remains the same from 2007 to 2023, what is the percentage increase in the share of the renewable sources of electricity generation over this period?

A. 25% **B.** 50% **C.** 77.5% **D.** 62.5%

Q24.6.9 MCQ 2024 2M

A cube is to be cut into 8 pieces of equal size and shape. Here, each cut should be straight and it should not stop till it reaches the other end of the cube.

The minimum number of such cuts required is

A. 3 **B.** 4 **C.** 7 **D.** 8

Q24.6.10 MCQ 2024 2M

In the 4x4 array shown below, each cell of the first three rows has either a cross (X) or a number.

1	X	4	3
X	5	5	4
3	X	6	X

Fig. Q24.6.10

The number in a cell represents the count of the immediate neighboring cells (left, right, top, bottom, diagonals) NOT having a cross (X).

Given that the last row has no crosses (X), the sum of the four numbers to be filled in the last row is

A. 11 **B.** 10 **C.** 12 **D.** 9

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Q24.7.1 MCQ 2024 1M

If '→' denotes increasing order of intensity, then the meaning of the words

[charm → enamor → bewitch] is analogous to [bored → _____ → weary].

Which one of the given options is appropriate to fill the blank?

A. jaded **B.** baffled **C.** dead **D.** worsted

Q24.7.2 MCQ 2024 1M

P, Q, R, S, and T have launched a new startup. Two of them are siblings. The office of the startup has just three rooms. All of them agree that the siblings should not share the same room.

If S and Q are single children, and the room allocations shown below are acceptable to all:

P R	T S	Q

P Q	R T	S

Fig. Q24.7.2

then, which one of the given options is the siblings?

A. P and T **B.** P and S **C.** T and Q **D.** T and R

Q24.7.3 MCQ 2024 1M

Five years ago, the ratio of Aman's age to his father's age was 1:4, and five years from now, the ratio will be 2:5. What was his father's age when Aman was born?

A. 28 years **B.** 30 years **C.** 35 years **D.** 32 years

Q24.7.4 MCQ 2024 1M

For a real number $x > 1$,

$$\frac{1}{\log_2 x} + \frac{1}{\log_3 x} + \frac{1}{\log_4 x} = 1$$

The value of x is .

A. 4 **B.** 12 **C.** 24 **D.** 36

Q24.7.5 MCQ 2024 1M

The greatest prime factor of $(3^{199} - 3^{196})$ is

A. 13 **B.** 17 **C.** 3 **D.** 11

Q24.7.6 MCQ 2024 2M

Sequence the following sentences (P, Q, R, S) in a coherent passage:

P: Shifu's student exclaimed, "Why do you run since the bull is an illusion?"
Q: Shifu said, "Surely my running away from the bull is also an illusion."
R: Shifu once proclaimed that all life is illusion.
S: One day, when a bull gave him chase, Shifu began running for his life.

A. SPRQ **B.** SRPQ **C.** RSPQ **D.** RPQS

Q24.7.7 MCQ 2024 2M

Four identical cylindrical chalk-sticks, each of radius $r = 0.5$ cm and length $l = 10$ cm, are bound tightly together using a duct tape as shown.

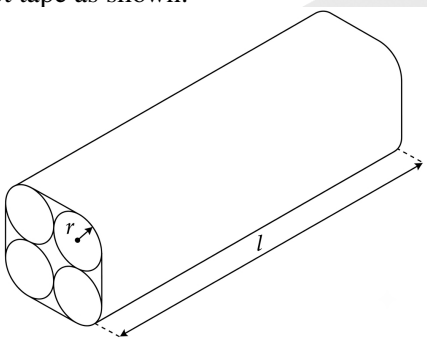


Fig. Q24.7.7

The width of the duct tape is equal to the length of the chalk-stick. The area (in cm^2) of the duct tape required to wrap the bundle of chalk-sticks once is

A. $20(4+\pi)$ **B.** $20(8+\pi)$ **C.** $10(8+\pi)$ **D.** $10(4+\pi)$

Q24.7.8 MCQ 2024 2M

The bar chart shows the data for the percentage of population falling into different categories based on Body Mass Index (BMI) in 2003 and 2023.

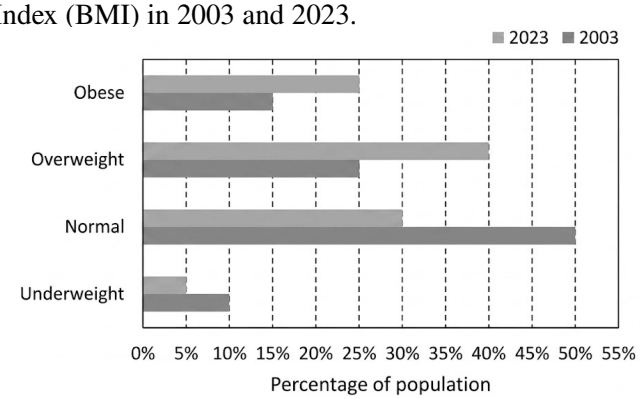


Fig. Q24.7.8

Based on the data provided, which one of the following options is INCORRECT?

A. The ratio of the percentage of population falling into overweight category to the percentage of population falling into normal category has increased in 20 years.

B. The ratio of the percentage of population falling into underweight category to the percentage of population falling into normal category has decreased in 20 years.

C. The ratio of the percentage of population falling into obese category to the percentage of population falling into normal category has decreased in 20 years.

D. The percentage of population falling into normal category has decreased in 20 years.

Q24.7.9 MCQ 2024 2M

Examples of mirror and water reflections are shown in the figures below:

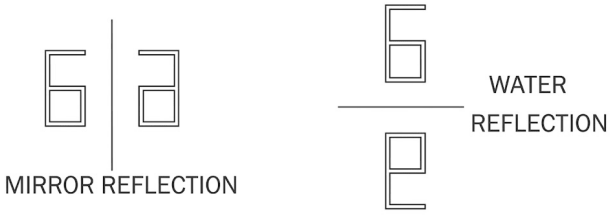


Fig. Q24.7.9

An object appears as the following image after first reflecting in a mirror and then reflecting on water.

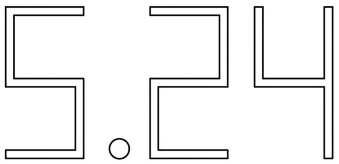
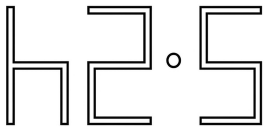
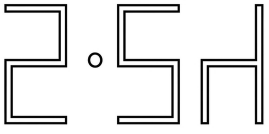
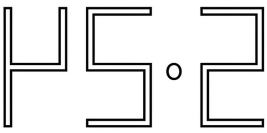


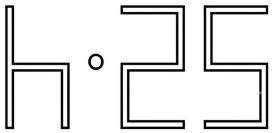
Fig. Q24.7.9

The original object is:

A. 

B. 

C. 

D. 

Q24.7.10 MCQ 2024 2M

Two identical sheets A and B, of dimensions 24 cm×16 cm, can be folded into half using two distinct operations, FO1 or FO2.

In FO1, the axis of folding remains parallel to the initial long edge, and in FO2, the axis of folding remains parallel to the initial short edge.

If sheet A is folded twice using FO1, and sheet B is folded twice using FO2, the ratio of the perimeters of the final shapes of A and B is

A. 14:11 **B.** 11:14 **C.** 18:11 **D.** 11:18

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Q24.8.1 MCQ 2024 1M

If '→' denotes increasing order of intensity, then the meaning of the words

[talk → shout → scream] is analogous to [please → _____ → pander].

Which one of the given options is appropriate to fill the blank?

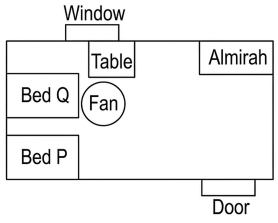
A. flatter **B.** flutter **C.** fritter **D.** frizzle

Q24.8.2 MCQ 2024 1M

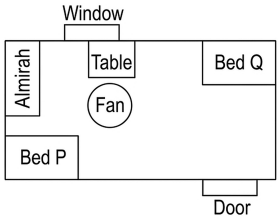
P and Q have been allotted a hostel room with two beds, a study table, and an almirah. P is an avid bird-watcher and wants to sit at the table and watch birds outside the window. Q does not mind that as long as his bed is close to the ceiling fan.

Which one of the following arrangements suits them the most?

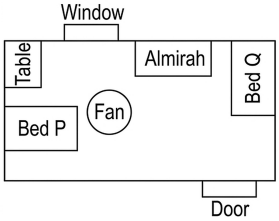
A.



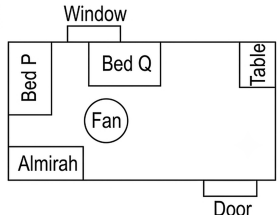
B.



C.



D.



Q24.8.3 MCQ 2024 1M

The decimal number system uses the characters 0, 1, 2, .., 8, 9, and the octal number system uses the characters 0, 1, 2, ..., 6, 7.

For example, the decimal number 12 $(= 1 \times 10^1 + 2 \times 10^0)$ is expressed as 14 $(= 1 \times 8^1 + 4 \times 8^0)$ in the octal number system.

The decimal number 108 in the octal number system is

A. 168 **B.** 108 **C.** 150 **D.** 154

Q24.8.4 MCQ 2024 1M

A shopkeeper buys shirts from a producer and sells them at 20% profit. A customer has to pay ₹3,186.00 including 18% taxes, per shirt. At what price did the shopkeeper buy each shirt from the producer?

A. ₹2,500.00 **B.** ₹1,975.40

C. ₹2,250.00 **D.** ₹2,548.80

Q24.8.5 MCQ 2024 1M

If, for non-zero real variables x, y, and real parameter $a > 1$,

$$x : y = (a + 1) : (a - 1),$$

then the ratio $(x^2 - y^2) : (x^2 + y^2)$ is

A. $2a : (a^2 + 1)$ **B.** $a : (a^2 + 1)$

C. $2a : (a^2 - 1)$ **D.** $a : (a^2 - 1)$

Q24.8.6 MCQ 2024 2M

In the given text, the blanks are numbered (i)-(iv). Select the best match for all the blanks.

Following a row (i) the shopkeeper (ii) the price of a frying pan, the cook stood (iii) a row to withdraw cash (iv) the ATM booth.

A. (i) with, (ii) over, (iii) at, (iv) with

B. (i) at, (ii) over, (iii) over, (iv) at

C. (i) with, (ii) over, (iii) at, (iv) at

D. (i) over, (ii) with, (iii) over, (iv) at

Q24.8.7 MCQ 2024 6M

In the following figure,

$CD = 5$ cm, $BE = 10$ cm, $AE = 12$ cm,
$\angle DAB = \angle DCB$, and $\angle DAE = \angle DBC = 90°$.

Points AFCD create a rhombus.

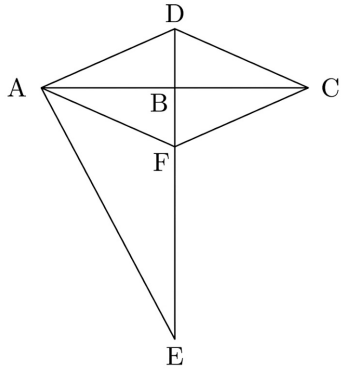


Fig. Q24.8.7

The length of BF (in cm) is

A. 3 **B.** 2 **C.** 4 **D.** 6

Q24.8.8 MCQ 2024 2M

The chart below shows the data of the number of cars bought by Millennials and Gen X people in a country from the year 2010 to 2020 as well as the yearly fuel consumption of the country (in Million liters)

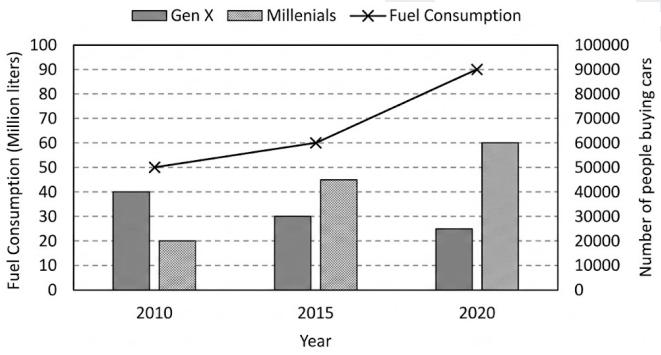


Fig. Q24.8.8

Considering the data presented in the chart, which one of the following options is true?

A. The percentage increase in fuel consumption from 2010 to 2015 is more than the percentage increase in fuel consumption from 2015 to 2020.

B. The increase in the number of Millennial car buyers from 2015 to 2020 is less than the decrease in the number of Gen X car buyers from 2010 to 2015.

C. The increase in the number of Millennial car buyers from 2010 to 2015 is more than the decrease in the number of Gen X car buyers from 2010 to 2015.

D. The decrease in the number of Gen X car buyers from 2015 to 2020 is more than the increase in the number of Millennial car buyers from 2010 to 2015.

Q24.8.9 MCQ 2024 2M

The assembly shown below has three teethed circular objects (Pinions) and two teethed flat objects (Racks), which are perfectly mating with each other. Pinions can only rotate clockwise or anti-clockwise staying at its own center. Racks can translate towards the left (←) or the right (→) direction.

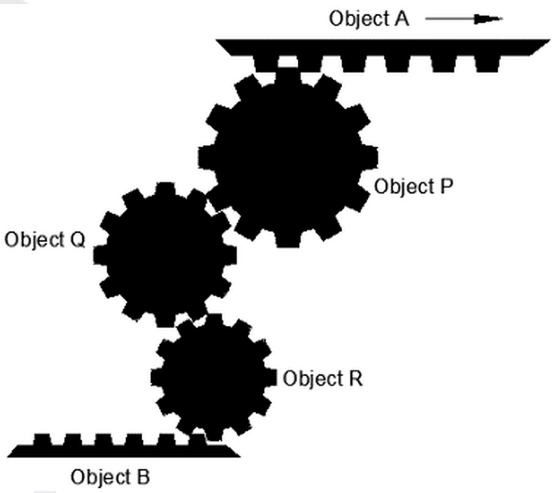


Fig. Q24.8.9

If the object A (Rack) is translating towards the right (→) direction, the correct statement among the following is

A. Object B translates towards the right direction.

B. Object B translates towards the left direction.

C. Object R rotates in the anticlockwise direction.

D. Object Q rotates in the clockwise direction.

Q24.8.10 MCQ 2024 2M

A surveyor has to measure the horizontal distance from her position to a distant reference point C. Using her position as the center, a 200 m horizontal line segment is drawn with the two endpoints A and B. Points A, B, and C are not collinear. Each of the angles $\angle CAB$ and $\angle CBA$ are measured as 87.8°. The distance (in m) of the reference point C from her position is nearest to

A. 2603 **B.** 2606 **C.** 2306 **D.** 2063

Answer Key

Note: MTA Marks To All awarded — see detailed solutions.

Session : 1					
Q24.1.1 D	Q24.1.2 A	Q24.1.3 B	Q24.1.4 B	Q24.1.5 B	Q24.1.6 D
Q24.1.7 A	Q24.1.8 B	Q24.1.9 A	Q24.1.10 A		
Session : 2					
Q24.2.1 A	Q24.2.2 A	Q24.2.3 C	Q24.2.4 A	Q24.2.5 D	Q24.2.6 B
Q24.2.7 MTA	Q24.2.8 C	Q24.2.9 A	Q24.2.10 MTA		
Session : 3					
Q24.3.1 B	Q24.3.2 A	Q24.3.3 A	Q24.3.4 D	Q24.3.5 C	Q24.3.6 D
Q24.3.7 A	Q24.3.8 A	Q24.3.9 C	Q24.3.10 B		
Session : 4					
Q24.4.1 A	Q24.4.2 B	Q24.4.3 D	Q24.4.4 A	Q24.4.5 B	Q24.4.6 D
Q24.4.7 B	Q24.4.8 C	Q24.4.9 A	Q24.4.10 A		
Session : 5					
Q24.5.1 A	Q24.5.2 D	Q24.5.3 A	Q24.5.4 C	Q24.5.5 A	Q24.5.6 C
Q24.5.7 A	Q24.5.8 C	Q24.5.9 A	Q24.5.10 A		
Session : 6					
Q24.6.1 C	Q24.6.2 A	Q24.6.3 A	Q24.6.4 C	Q24.6.5 B	Q24.6.6 C
Q24.6.7 C	Q24.6.8 D	Q24.6.9 A	Q24.6.10 A		
Session : 7					
Q24.7.1 A	Q24.7.2 A	Q24.7.3 B	Q24.7.4 C	Q24.7.5 A	Q24.7.6 C
Q24.7.7 D	Q24.7.8 C	Q24.7.9 A	Q24.7.10 A		
Session : 8					
Q24.8.1 A	Q24.8.2 A	Q24.8.3 D	Q24.8.4 C	Q24.8.5 A	Q24.8.6 C
Q24.8.7 MTA	Q24.8.8 C	Q24.8.9 B	Q24.8.10 A		

YEAR

2023

8 Sessions · 80 Questions

Sessions Summary

Session	Participating Branches	Total Qs
Session 1	CS	10
Session 2	AR, ME	10
Session 3	EE, ES, XH	10
Session 4	BM, CY, EC	10
Session 5	GG, IN, MA, PE, XE, XL	10
Session 6	AE, AG, BT, CH, EY, GE, MT, NM, PH, PI, TF	10
Session 7	CE1, ST	10
Session 8	CE2, MN	10

PrepFusion

Year Snapshot

Total Questions: 80 **MCQ:** 80 **MSQ:** 0 **NAT:** 0

Session : 1 | Forenoon | 04/02/23

Q23.1.1 MCQ 2023 1M

We reached the station late, and _____ missed the train.

A. near **B.** nearly **C.** utterly **D.** mostly

Q23.1.2 MCQ 2023 1M

Kind: _____ :: Often: Frequently
(By word meaning)

A. Mean **B.** Type **C.** Cruel **D.** Kindly

Q23.1.3 MCQ 2023 1M

A series of natural numbers $F_1, F_2, F_3, F_4, F_5, F_6, F_7, \ldots$ obeys $F_{n+1} = F_n + F_{n-1}$ for all integers $n \geq 2$.
If $F_6 = 37$ and $F_7 = 60$, then what is F_1?

A. 4 **B.** 5 **C.** 8 **D.** 9

Q23.1.4 MCQ 2023 1M

A survey for a certain year found that 90% of pregnant women received medical care at least once before giving birth. Of these women, 60% received medical care from doctors, while 40% received medical care from other healthcare providers.
Given this information, which one of the following statements can be inferred with *certainty*?

A. More than half of the pregnant women received medical care at least once from a doctor.

B. Less than half of the pregnant women received medical care at least once from a doctor.

C. More than half of the pregnant women received medical care at most once from a doctor.

D. Less than half of the pregnant women received medical care at most once from a doctor.

Q23.1.5 MCQ 2023 1M

Looking at the surface of a smooth 3-dimensional object from the outside, which one of the following options is TRUE?

A. The surface of the object must be concave everywhere.

B. The surface of the object must be convex everywhere.

C. The surface of the object may be concave in some places and convex in other places.

D. The object can have edges, but no corners.

Q23.1.6 MCQ 2023 2M

The country of Zombieland is in distress since more than 75% of its working population is suffering from serious health issues. Studies conducted by competent health experts concluded that a complete lack of physical exercise among its working population was one of the leading causes of their health issues. As one of the measures to address the problem, the Government of Zombieland has decided to provide monetary incentives to those who ride bicycles to work.
Based only on the information provided above, which one of the following statements can be logically inferred with *certainty*?

A. All the working population of Zombieland will henceforth ride bicycles to work.

B. Riding bicycles will ensure that all of the working population of Zombieland is free of health issues.

C. The health experts suggested to the Government of Zombieland to declare riding bicycles as mandatory.

D. The Government of Zombieland believes that riding bicycles is a form of physical exercise.

Q23.1.7 MCQ 2023 2M

Consider two functions of time (t),

$$f(t) = 0.01t^2$$
$$g(t) = 4t$$

where $0 < t < \infty$.
Now consider the following two statements:

(i) For some $t > 0$, $g(t) > f(t)$.
(ii) There exists a T, such that $f(t) > g(t)$ for all $t > T$.

Which one of the following options is TRUE?

A. only (i) is correct

B. only (ii) is correct

C. both (i) and (ii) are correct

D. neither (i) nor (ii) is correct

Q23.1.8 MCQ 2023 2M

Which one of the following sentence sequences creates a coherent narrative?

(i) Once on the terrace, on her way to her small room in the corner, she notices the man right away.

(ii) She begins to pant by the time she has climbed all the stairs.
(iii) Mina has bought vegetables and rice at the market, so her bags are heavy.
(iv) He was leaning against the parapet, watching the traffic below.

A. (i), (ii), (iv), (iii) **B.** (ii), (iii), (i), (iv)

C. (iv), (ii), (i), (iii) **D.** (iii), (ii), (i), (iv)

Q23.1.9 MCQ | 2023 | 2M

$f(x)$ and $g(y)$ are functions of x and y, respectively, and $f(x) = g(y)$ for all real values of x and y. Which one of the following options is *necessarily* TRUE for all x and y?

A. $f(x) = 0$ and $g(y) = 0$

B. $f(x) = g(y) =$ constant

C. $f(x) \neq$ constant and $g(y) \neq$ constant

D. $f(x) + g(y) = f(x) - g(y)$

Q23.1.10 MCQ | 2023 | 2M

Which one of the options best describes the transformation of the 2-dimensional figure **P** to **Q**, and then to **R**, as shown?

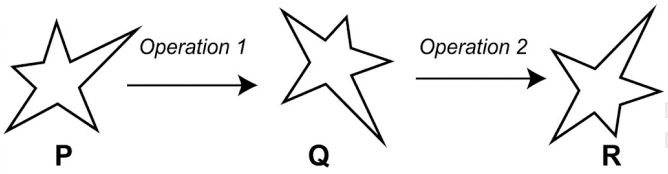


Fig. Q23.1.10

A. *Operation 1:* A clockwise rotation by 90° about an axis perpendicular to the plane of the figure;
Operation 2: A reflection along a horizontal line

B. *Operation 1:* A counter clockwise rotation by 90° about an axis perpendicular to the plane of the figure;
Operation 2: A reflection along a horizontal line

C. *Operation 1:* A clockwise rotation by 90° about an axis perpendicular to the plane of the figure;
Operation 2: A reflection along a vertical line

D. *Operation 1:* A counter clockwise rotation by 180° about an axis perpendicular to the plane of the figure;
Operation 2: A reflection along a vertical line

Session : 2 | Afternoon | 04/02/23

Q23.2.1 MCQ | 2023 | 1M

He did not manage to fix the car himself, so he _____ in the garage.

A. got it fixed **B.** getting it fixed

C. gets fixed **D.** got fixed

Q23.2.2 MCQ | 2023 | 1M

Planting: Seed :: Raising: _____
(By word meaning)

A. Child **B.** Temperature **C.** Height **D.** Lift

Q23.2.3 MCQ | 2023 | 1M

A certain country has 504 universities and 25951 colleges. These are categorised into Grades I, II, and III as shown in the given pie charts.
What is the percentage, correct to one decimal place, of higher education institutions (colleges and universities) that fall into Grade III?

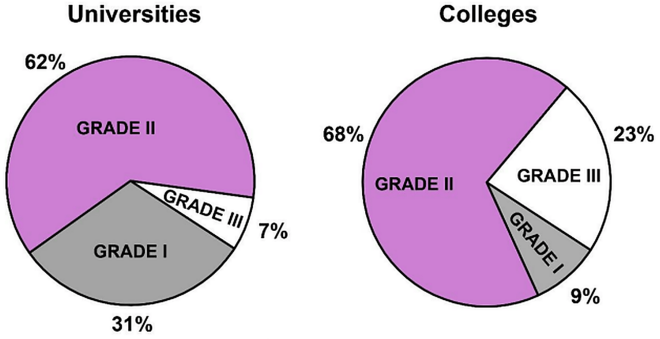


Fig. Q23.2.3

A. 22.7 **B.** 23.7 **C.** 15.0 **D.** 66.8

Q23.2.4 MCQ | 2023 | 1M

The minute-hand and second-hand of a clock cross each other ____ times between 09:15:00 AM and 09:45:00 AM on a day.

A. 30 **B.** 15 **C.** 29 **D.** 31

Q23.2.5 MCQ 2023 1M

The symbols ◯, ∗, Δ and ☐ are to be filled, one in each box, as shown below. The rules for filling in the four symbols are as follows.

1) Every row and every column must contain each of the four symbols.
2) Every 2×2 square delineated by bold lines must contain each of the four symbols.

Which symbol will occupy the box marked with '**?**' in the partially filled figure?

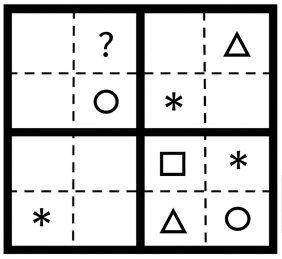


Fig. Q23.2.5

A. ◯ **B.** ∗ **C.** Δ **D.** ☐

Q23.2.6 MCQ 2023 2M

In a recently held parent-teacher meeting, the teachers had very few complaints about Ravi. After all, Ravi was a hardworking and kind student. Incidentally, almost all of Ravi's friends at school were hardworking and kind too. But the teachers drew attention to Ravi's complete lack of interest in sports. The teachers believed that, along with some of his friends who showed similar disinterest in sports, Ravi needed to engage in some sports for his overall development.

Based only on the information provided above, which one of the following statements can be logically inferred with *certainty*?

A. All of Ravi's friends are hardworking and kind.

B. No one who is not a friend of Ravi is hardworking and kind.

C. None of Ravi's friends are interested in sports.

D. Some of Ravi's friends are hardworking and kind.

Q23.2.7 MCQ 2023 2M

Consider the following inequalities:

$$p^2 - 4q < 4$$
$$3p + 2q < 6$$

where p and q are positive integers.

The value of $(p + q)$ is _____.

A. 2 **B.** 1 **C.** 3 **D.** 4

Q23.2.8 MCQ 2023 2M

Which one of the sentence sequences in the given options creates a coherent narrative?

(i) I could not bring myself to knock.
(ii) There was a murmur of unfamiliar voices coming from the big drawing room and the door was firmly shut.
(iii) The passage was dark for a bit, but then it suddenly opened into a bright kitchen.
(iv) I decided I would rather wander down the passage.

A. (iv), (i), (iii), (ii) **B.** (iii), (i), (ii), (iv)

C. (ii), (i), (iv), (iii) **D.** (i), (iii), (ii), (iv)

Q23.2.9 MCQ 2023 2M

How many pairs of sets (S, T) are possible among the subsets of {1, 2, 3, 4, 5, 6} that satisfy the condition that S is a subset of T?

A. 729 **B.** 728 **C.** 665 **D.** 664

Q23.2.10 MCQ 2023 2M

An opaque pyramid shown below, with a square base and isosceles faces, is suspended in the path of a parallel beam of light, such that its shadow is cast on a screen oriented perpendicular to the direction of the light beam. The pyramid can be reoriented in any direction within the light beam. Under these conditions, which one of the shadows **P, Q, R** and **S** is NOT possible?

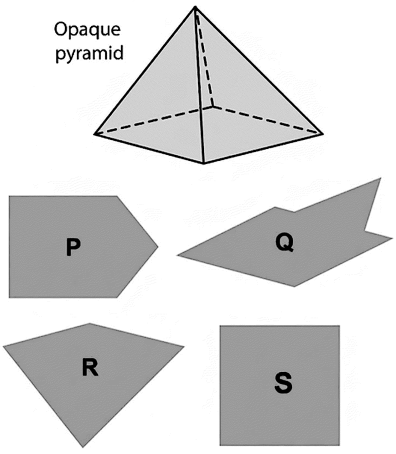


Fig. Q23.2.10

A. P **B.** Q **C.** R **D.** S

Session : 3 | Forenoon | 05/02/23

Q23.3.1 MCQ 2023 1M

Rafi told Mary, "I am thinking of watching a film this weekend."
The following reports the above statement in indirect speech:
Rafi told Mary that he _____ of watching a film that weekend.

A. thought **B.** is thinking
C. am thinking **D.** was thinking

Q23.3.2 MCQ 2023 1M

Permit: _____ :: Enforce: Relax
(By word meaning)

A. Allow **B.** Forbid **C.** License **D.** Reinforce

Q23.3.3 MCQ 2023 1M

Given a fair six-faced dice where the faces are labelled '1', '2', '3', '4', '5', and '6', what is the probability of getting a '1' on the first roll of the dice and a '4' on the second roll?

A. $\frac{1}{36}$ **B.** $\frac{1}{6}$ **C.** $\frac{5}{6}$ **D.** $\frac{1}{3}$

Q23.3.4 MCQ 2023 1M

A recent survey shows that 65% of tobacco users were advised to stop consuming tobacco. The survey also shows that 3 out of 10 tobacco users attempted to stop using tobacco.
Based only on the information in the above passage, which one of the following options can be logically inferred with *certainty*?

A. A majority of tobacco users who were advised to stop consuming tobacco made an attempt to do so.

B. A majority of tobacco users who were advised to stop consuming tobacco did not attempt to do so.

C. Approximately 30% of tobacco users successfully stopped consuming tobacco.

D. Approximately 65% of tobacco users successfully stopped consuming tobacco.

Q23.3.5 MCQ 2023 1M

How many triangles are present in the given figure?

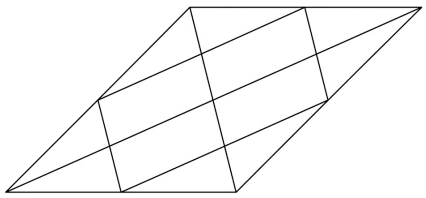


Fig. Q23.3.5

A. 12 **B.** 16 **C.** 20 **D.** 24

Q23.3.6 MCQ 2023 2M

Students of all the departments of a college who have successfully completed the registration process are eligible to vote in the upcoming college elections. However, by the time the due date for registration was over, it was found that surprisingly none of the students from the Department of Human Sciences had completed the registration process.
Based only on the information provided above, which one of the following sets of statement(s) can be logically inferred with *certainty*?

(i) All those students who would not be eligible to vote in the college elections would certainly belong to the Department of Human Sciences.
(ii) None of the students from departments other than Human Sciences failed to complete the registration process within the due time.
(iii) All the eligible voters would certainly be students who are not from the Department of Human Sciences.

A. (i) and (ii) **B.** (i) and (iii) **C.** only (i) **D.** only (iii)

Q23.3.7 MCQ 2023 2M

Which one of the following options represents the given graph?

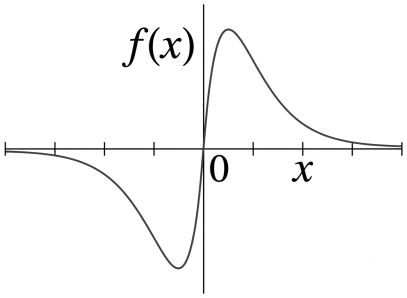


Fig. Q23.3.7

A. $f(x) = x^2 2^{-|x|}$ **B.** $f(x) = x2^{-|x|}$
C. $f(x) = |x|2^{-x}$ **D.** $f(x) = x2^{-x}$

Q23.3.8 MCQ 2023 2M

Which one of the options does NOT describe the passage below or follow from it?

We tend to think of cancer as a 'modern' illness because its metaphors are so modern. It is a disease of overproduction, of sudden growth, a growth that is unstoppable, tipped into the abyss of no control. Modern cell biology encourages us to imagine the cell as a molecular machine. Cancer is that machine unable to quench its initial command (to grow) and thus transform into an indestructible, self-propelled automaton.

[Adapted from *The Emperor of All Maladies* by Siddhartha Mukherjee]

A. It is a reflection of why cancer seems so modern to most of us.

B. It tells us that modern cell biology uses and promotes metaphors of machinery.

C. Modern cell biology encourages metaphors of machinery, and cancer is often imagined as a machine.

D. Modern cell biology never uses figurative language, such as metaphors, to describe or explain anything.

Q23.3.9 MCQ 2023 2M

The digit in the unit's place of the product $3^{999} \times 7^{1000}$ is

A. 7 **B.** 1 **C.** 3 **D.** 9

Q23.3.10 MCQ 2023 2M

A square with sides of length 6 cm is given. The boundary of the shaded region is defined by two semi-circles whose diameters are the sides of the square, as shown.

The area of the shaded region is _____ cm^2.

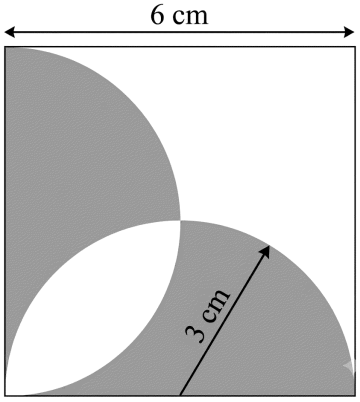


Fig. Q23.3.10

A. 6π **B.** 18 **C.** 20 **D.** 9π

Session : 4 | Afternoon | 05/02/23

Q23.4.1 MCQ 2023 1M

"I cannot support this proposal. My _____ will not permit it."

A. conscious **B.** consensus **C.** conscience **D.** consent

Q23.4.2 MCQ 2023 1M

Courts: _____ :: Parliament: Legislature

(By word meaning)

A. Judiciary **B.** Executive **C.** Governmental **D.** Legal

Q23.4.3 MCQ 2023 1M

What is the smallest number with distinct digits whose digits add up to 45?

A. 123555789 **B.** 123457869

C. 123456789 **D.** 99999

Q23.4.4 MCQ 2023 1M

In a class of 100 students,

(i) there are 30 students who neither like romantic movies nor comedy movies,

(ii) the number of students who like romantic movies is twice the number of students who like comedy movies, and

(iii) the number of students who like both romantic movies and comedy movies is 20.

How many students in the class like romantic movies?

A. 40 **B.** 20 **C.** 60 **D.** 30

Q23.4.5 MCQ 2023 1M

How many rectangles are present in the given figure?

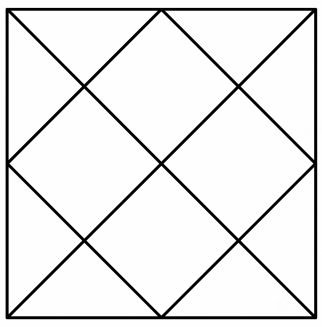


Fig. Q23.4.5

A. 8 **B.** 9 **C.** 10 **D.** 12

Q23.4.6 MCQ 2023 2M

Forestland is a planet inhabited by different kinds of creatures. Among other creatures, it is populated by animals all of whom are ferocious. There are also creatures that have claws, and some that do not. All creatures that have claws are ferocious.

Based only on the information provided above, which one of the following options can be logically inferred with *certainty*?

A. All creatures with claws are animals.

B. Some creatures with claws are non-ferocious.

C. Some non-ferocious creatures have claws.

D. Some ferocious creatures are creatures with claws.

Q23.4.7 MCQ 2023 2M

Which one of the following options represents the given graph?

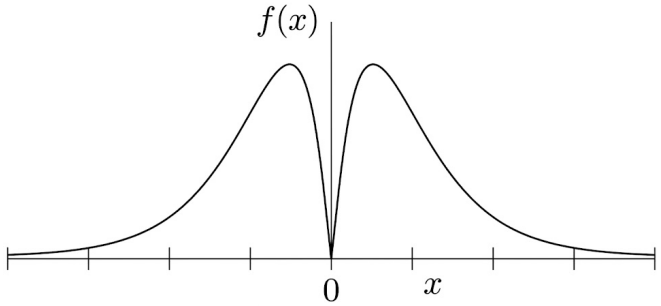


Fig. Q23.4.7

A. $f(x) = x^2 2^{-|x|}$ **B.** $f(x) = x2^{-|x|}$

C. $f(x) = |x|2^{-x}$ **D.** $f(x) = x2^{-x}$

Q23.4.8 MCQ 2023 2M

Which one of the following options can be inferred from the given passage alone?

When I was a kid, I was partial to stories about other worlds and interplanetary travel. I used to imagine that I could just gaze off into space and be whisked to another planet.

[Excerpt from The Truth about Stories by T. King]

A. It is a child's description of what he or she likes.

B. It is an adult's memory of what he or she liked as a child.

C. The child in the passage read stories about interplanetary travel only in parts.

D. It teaches us that stories are good for children.

Q23.4.9 MCQ 2023 2M

Out of 1000 individuals in a town, 100 unidentified individuals are covid positive. Due to lack of adequate covid-testing kits, the health authorities of the town devised a strategy to identify these covid-positive individuals. The strategy is to:

(i) Collect saliva samples from all 1000 individuals and randomly group them into sets of 5.

(ii) Mix the samples within each set and test the mixed sample for covid.

(iii) If the test done in (ii) gives a negative result, then declare all the 5 individuals to be covid negative.

(iv) If the test done in (ii) gives a positive result, then all the 5 individuals are separately tested for covid.

Given this strategy, no more than _____ testing kits will be required to identify all the 100 covid positive individuals irrespective of how they are grouped.

A. 700 **B.** 600 **C.** 800 **D.** 1000

Q23.4.10 MCQ 2023 2M

A 100 cm × 32 cm rectangular sheet is folded 5 times. Each time the sheet is folded, the long edge aligns with its opposite side. Eventually, the folded sheet is a rectangle of dimensions 100 cm × 1 cm.

The total number of creases visible when the sheet is unfolded is _____.

A. 32 **B.** 5 **C.** 31 **D.** 63

Session : 5 | Forenoon | 11/02/23

Q23.5.1 MCQ 2023 1M

The village was nestled in a green spot, _____ the ocean and the hills.

A. through **B.** in **C.** at **D.** between

Q23.5.2 MCQ 2023 1M

Disagree: Protest :: Agree: _____

(By word meaning)

A. Refuse **B.** Pretext **C.** Recommend **D.** Refute

Q23.5.3 MCQ 2023 1M

A 'frabjous' number is defined as a 3 digit number with all digits odd, and no two adjacent digits being the same. For example, 137 is a frabjous number, while 133 is not. How many such frabjous numbers exist?

A. 125 **B.** 720 **C.** 60 **D.** 80

Q23.5.4 MCQ 2023 1M

Which one among the following statements must be TRUE about the mean and the median of the scores of all candidates appearing for GATE 2023?

A. The median is at least as large as the mean.

B. The mean is at least as large as the median.

C. At most half the candidates have a score that is larger than the median.

D. At most half the candidates have a score that is larger than the mean.

Q23.5.5 MCQ 2023 1M

In the given diagram, ovals are marked at different heights (h) of a hill. Which one of the following options **P**, **Q**, **R**, and **S** depicts the top view of the hill?

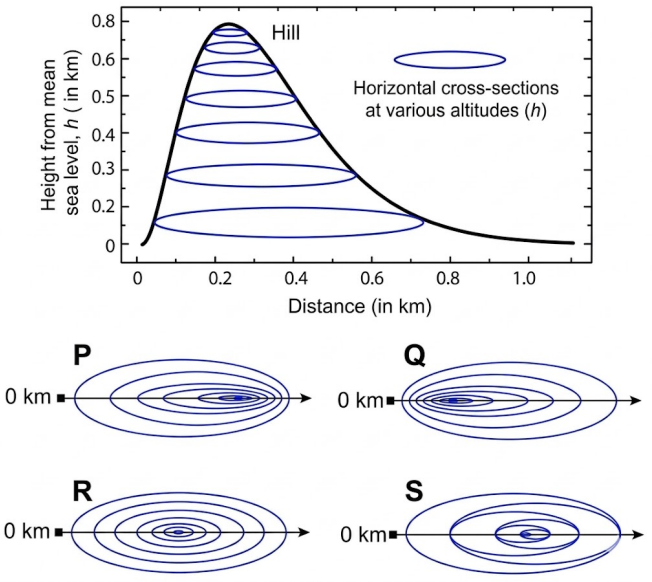


Fig. Q23.5.5

A. P **B.** Q **C.** R **D.** S

Q23.5.6 MCQ 2023 2M

Residency is a famous housing complex with many well-established individuals among its residents. A recent survey conducted among the residents of the complex revealed that all of those residents who are well established in their respective fields happen to be academicians. The survey also revealed that most of these academicians are authors of some best-selling books.

Based only on the information provided above, which one of the following statements can be logically inferred with *certainty*?

A. Some residents of the complex who are well established in their fields are also authors of some best-selling books.

B. All academicians residing in the complex are well established in their fields.

C. Some authors of best-selling books are residents of the complex who are well established in their fields.

D. Some academicians residing in the complex are well established in their fields.

Q23.5.7 MCQ 2023 2M

Ankita has to climb 5 stairs starting at the ground, while respecting the following rules:

1. At any stage, Ankita can move either one or two stairs up.
2. At any stage, Ankita cannot move to a lower step.

Let $F(N)$ denote the number of possible ways in which Ankita can reach the N^{th} stair.

For example, $F(1) = 1$, $F(2) = 2$, $F(3) = 3$.

The value of $F(5)$ is _____.

A. 8 **B.** 7 **C.** 6 **D.** 5

Q23.5.8 MCQ 2023 2M

The information contained in DNA is used to synthesize proteins that are necessary for the functioning of life. DNA is composed of four nucleotides: Adenine (A), Thymine (T), Cytosine (C), and Guanine (G). The information contained in DNA can then be thought of as a sequence of these four nucleotides: A, T, C, and G. DNA has coding and non-coding regions. Coding regions—where the sequence of these nucleotides are read in groups of three to produce individual amino acids—constitute only about 2% of human DNA. For example, the triplet of nucleotides CCG codes for the amino acid glycine, while the triplet GGA codes for the amino acid proline. Multiple amino acids are then assembled to form a protein.

Based only on the information provided above, which of the following statements can be logically inferred with certainty?

(i) The majority of human DNA has no role in the synthesis of proteins.

(ii) The function of about 98% of human DNA is not understood.

A. only (i) **B.** only (ii)

C. both (i) and (ii) **D.** neither (i) nor (ii)

Q23.5.9 MCQ | 2023 | 2M

Which one of the given figures P, Q, R and S represents the graph of the following function?

$$f(x) = ||x + 2| - |x - 1||$$

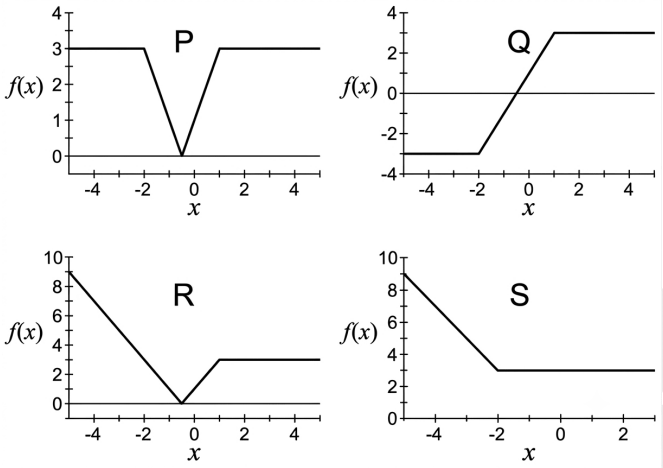


Fig. Q23.5.9

A. P **B.** Q **C.** R **D.** S

Q23.5.10 MCQ | 2023 | 2M

An opaque cylinder is suspended in the path of a parallel beam of light, such that its shadow is cast on a screen oriented perpendicular to the direction of the light beam. The cylinder can be reoriented in any direction within the light beam. Under these conditions, which one of the shadows **P, Q, R**, and **S** is NOT possible?

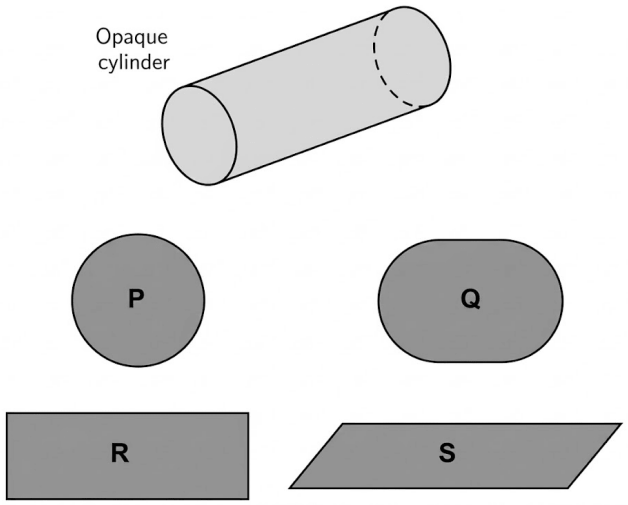


Fig. Q23.5.10

A. P **B.** Q **C.** R **D.** S

Session : 6 | Afternoon | 11/02/23

Q23.6.1 MCQ | 2023 | 1M

"You are delaying the completion of the task. Send _____ contributions at the earliest."

A. you are **B.** your **C.** you're **D.** yore

Q23.6.2 MCQ | 2023 | 1M

References:_____ :: Guidelines: Implement
(By word meaning)

A. Sight **B.** Site **C.** Cite **D.** Plagiarise

Q23.6.3 MCQ | 2023 | 1M

In the given figure, PQRS is a parallelogram with PS=7 cm, PT=4 cm and PV=5 cm. What is the length of RS in cm? (The diagram is representative.)

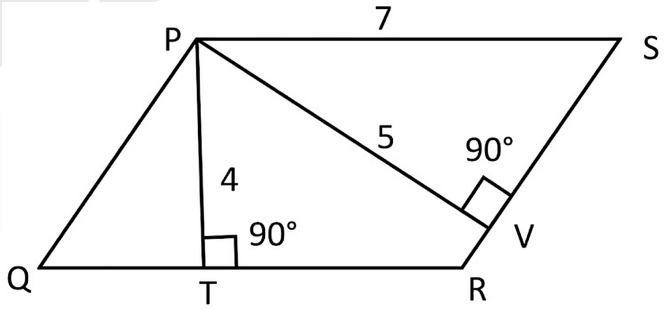


Fig. Q23.6.3

A. $\frac{20}{7}$ **B.** $\frac{28}{5}$ **C.** $\frac{9}{2}$ **D.** $\frac{35}{4}$

Q23.6.4 MCQ | 2023 | 1M

In 2022, June Huh was awarded the Fields medal, which is the highest prize in Mathematics.
When he was younger, he was also a poet. He did not win any medals in the International Mathematics Olympiads. He dropped out of college.
Based only on the above information, which one of the following statements can be logically inferred with *certainty*?

A. Every Fields medalist has won a medal in an International Mathematics Olympiad.

B. Everyone who has dropped out of college has won the Fields medal.

C. All Fields medalists are part-time poets.

D. Some Fields medalists have dropped out of college.

Q23.6.5 MCQ 2023 1M

A line of symmetry is defined as a line that divides a figure into two parts in a way such that each part is a mirror image of the other part about that line.

The given figure consists of 16 unit squares arranged as shown. In addition to the three black squares, what is the minimum number of squares that must be coloured black, such that both PQ and MN form lines of symmetry? (The figure is representative)

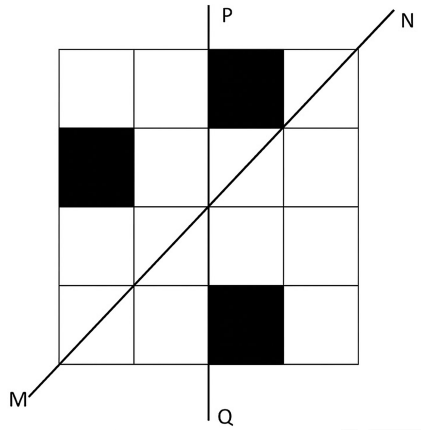


Fig. Q23.6.5

A. 3 **B.** 4 **C.** 5 **D.** 6

Q23.6.6 MCQ 2023 2M

Human beings are one among many creatures that inhabit an imagined world. In this imagined world, some creatures are cruel. If in this imagined world, it is given that the statement "Some human beings are not cruel creatures" is FALSE, then which of the following set of statement(s) can be logically inferred with *certainty*?

(i) All human beings are cruel creatures.
(ii) Some human beings are cruel creatures.
(iii) Some creatures that are cruel are human beings.
(iv) No human beings are cruel creatures.

A. only (i) **B.** only (iii) and (iv)

C. only (i) and (ii) **D.** (i), (ii) and (iii)

Q23.6.7 MCQ 2023 2M

To construct a wall, sand and cement are mixed in the ratio of 3:1. The cost of sand and that of cement are in the ratio of 1:2.

If the total cost of sand and cement to construct the wall is 1000 rupees, then what is the cost (in rupees) of cement used?

A. 400 **B.** 600 **C.** 800 **D.** 200

Q23.6.8 MCQ 2023 2M

The World Bank has declared that it does not plan to offer new financing to Sri Lanka, which is battling its worst economic crisis in decades, until the country has an adequate macroeconomic policy framework in place. In a statement, the World Bank said Sri Lanka needed to adopt structural reforms that focus on economic stabilisation and tackle the root causes of its crisis. The latter has starved it of foreign exchange and led to shortages of food, fuel, and medicines. The bank is repurposing resources under existing loans to help alleviate shortages of essential items such as medicine, cooking gas, fertiliser, meals for children, and cash for vulnerable households.

Based only on the above passage, which one of the following statements can be inferred with *certainty*?

A. According to the World Bank, the root cause of Sri Lanka's economic crisis is that it does not have enough foreign exchange.

B. The World Bank has stated that it will advise the Sri Lankan government about how to tackle the root causes of its economic crisis.

C. According to the World Bank, Sri Lanka does not yet have an adequate macroeconomic policy framework.

D. The World Bank has stated that it will provide Sri Lanka with additional funds for essentials such as food, fuel, and medicines.

Q23.6.9 MCQ 2023 2M

The coefficient of x^4 in the polynomial $(x-1)^3(x-2)^3$ is equal to _____.

A. 33 **B.** -3 **C.** 30 **D.** 21

Q23.6.10 MCQ 2023 2M

Which one of the following shapes can be used to tile (completely cover by repeating) a flat plane, extending to infinity in all directions, without leaving any empty spaces in between them? The copies of the shape used to tile are identical and are not allowed to overlap.

A. circle **B.** regular octagon

C. regular pentagon **D.** rhombus

Session : 7 | Forenoon | 12/02/23

Q23.7.1 MCQ 2023 1M

"I have not yet decided what I will do this evening; I _____ visit a friend."

A. mite **B.** would **C.** might **D.** didn't

Q23.7.2 MCQ 2023 1M

Eject: Insert :: Advance: _____
(By word meaning)

A. Advent **B.** Progress **C.** Retreat **D.** Loan

Q23.7.3 MCQ 2023 1M

In the given figure, PQRSTV is a regular hexagon with each side of length 5 cm. A circle is drawn with its centre at V such that it passes through P. What is the area (in cm^2) of the shaded region?

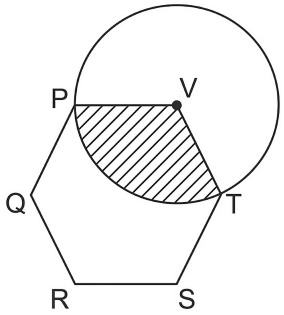


Fig. Q23.7.3

A. $\frac{25\pi}{3}$ **B.** $\frac{20\pi}{3}$ **C.** 6π **D.** 7π

Q23.7.4 MCQ 2023 1M

A duck named Donald Duck says "All ducks always lie." Based only on the information above, which one of the following statements can be logically inferred with *certainty*?

A. Donald Duck always lies.

B. Donald Duck always tells the truth.

C. Donald Duck's statement is true.

D. Donald Duck's statement is false.

Q23.7.5 MCQ 2023 1M

A line of symmetry is defined as a line that divides a figure into two parts in a way such that each part is a mirror image of the other part about that line.
The figure below consists of 20 unit squares arranged as shown. In addition to the given black squares, upto 5 more may be coloured black. Which one among the following options depicts the minimum number of boxes that must be coloured black to achieve two lines of symmetry?

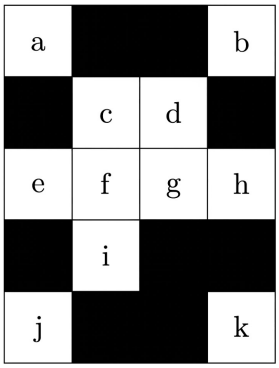


Fig. Q23.7.5

A. d **B.** c, d, i **C.** c, i **D.** c, d, i, f, g

Q23.7.6 MCQ 2023 2M

Based only on the truth of the statement 'Some humans are intelligent', which one of the following options can be logically inferred with *certainty*?

A. No human is intelligent.

B. All humans are intelligent.

C. Some non-humans are intelligent.

D. Some intelligent beings are humans.

Q23.7.7 MCQ 2023 2M

Which one of the options can be inferred about the mean, median, and mode for the given probability distribution (i.e. probability mass function), $P(x)$ of a variable x?

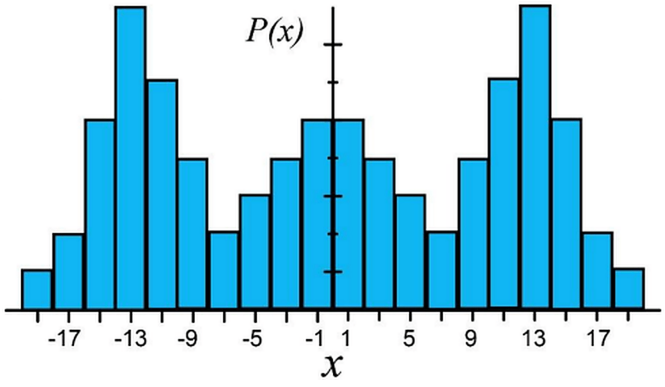


Fig. Q23.7.7

A. mean = median ≠ mode **B.** mean = median = mode

C. mean ≠ median = mode **D.** mean ≠ mode = median

Q23.7.8 MCQ | 2023 | 2M

The James Webb telescope, recently launched in space, is giving humankind unprecedented access to the depths of time by imaging very old stars formed almost 13 billion years ago. Astrophysicists and cosmologists believe that this odyssey in space may even shed light on the existence of dark matter. Dark matter is supposed to interact only via the gravitational interaction and not through the electromagnetic-, the weak- or the strong-interaction. This may justify the epithet "dark" in dark matter.
Based on the above paragraph, which one of the following statements is FALSE?

A. No other telescope has captured images of stars older than those captured by the James Webb telescope.

B. People other than astrophysicists and cosmologists may also believe in the existence of dark matter.

C. The James Webb telescope could be of use in the research on dark matter.

D. If dark matter was known to interact via the strong-interaction, then the epithet "dark" would be justified.

Q23.7.9 MCQ | 2023 | 2M

Let $a = 30!$, $b = 50!$ and $c = 100!$. Consider the following numbers:

$$log_a c, \quad log_c a, \quad log_b a, \quad log_a b.$$

Which one of the following inequalities is CORRECT?

A. $log_c a < log_b a < log_a b < log_a c$

B. $log_c a < log_a b < log_b a < log_b c$

C. $log_c a < log_b a < log_a c < log_a b$

D. $log_b a < log_c a < log_a b < log_a c$

Q23.7.10 MCQ | 2023 | 2M

A square of side length 4 cm is given. The boundary of the shaded region is defined by one semi-circle on the top and two circular arcs at the bottom, each of radius 2 cm, as shown.

The area of the shaded region is _____ cm^2.

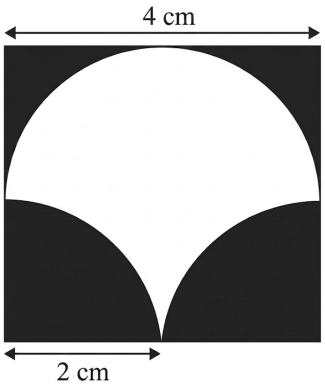


Fig. Q23.7.10

A. 8 **B.** 4 **C.** 12 **D.** 10

Session : 8 | Afternoon | 12/02/23

Q23.8.1 MCQ | 2023 | 1M

The line ran _____ the page, right through the centre, and divided the page into two.

A. across **B.** of **C.** between **D.** about

Q23.8.2 MCQ | 2023 | 1M

Kind: _____ :: Often: Seldom
(By word meaning)

A. Cruel **B.** Variety **C.** Type **D.** Kindred

Q23.8.3 MCQ | 2023 | 1M

In how many ways can cells in a 3×3 grid be shaded, such that each row and each column have exactly one shaded cell? An example of one valid shading is shown.

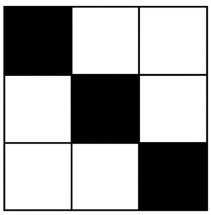


Fig. Q23.8.3

A. 2 **B.** 9 **C.** 3 **D.** 6

Q23.8.4 MCQ | 2023 | 1M

There are 4 red, 5 green, and 6 blue balls inside a box. If N number of balls are picked simultaneously, what is the smallest value of N that guarantees there will be at least two balls of the same colour?
One cannot see the colour of the balls until they are picked.

A. 4 **B.** 15 **C.** 5 **D.** 2

Q23.8.5 MCQ 2023 1M

Consider a circle with its centre at the origin (O), as shown. Two operations are allowed on the circle.

Operation 1: Scale independently along the x and y axes.
Operation 2: Rotation in any direction about the origin.

Which figure among the options can be achieved through a combination of these two operations on the given circle?

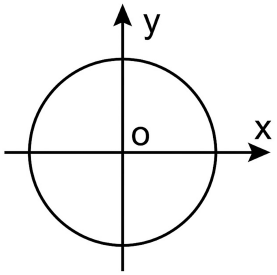
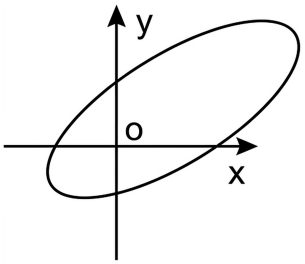
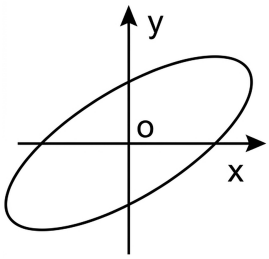
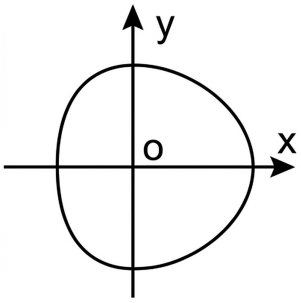
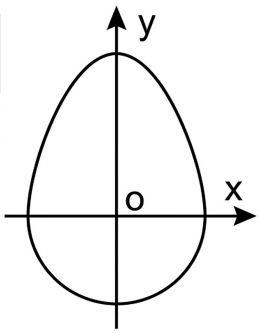


Fig. Q23.8.5

A.


B.


C.


D.


Q23.8.6 MCQ 2023 2M

Elvesland is a country that has peculiar beliefs and practices. They express almost all their emotions by gifting flowers. For instance, if anyone gifts a white flower to someone, then it is always taken to be a declaration of one's love for that person.In a similar manner, the gifting of a yellow flower to someone often means that one is angry with that person.

Based only on the information provided above, which one of the following sets of statement(s) can be logically inferred with certainty?

(i) In Elvesland, one always declares one's love by gifting a white flower.
(ii) In Elvesland, all emotions are declared by gifting flowers.
(iii) In Elvesland, sometimes one expresses one's anger by gifting a flower that is not yellow.
(iv) In Elvesland, sometimes one expresses one's love by gifting a white flower.

A. only (ii)
B. (i), (ii) and (iii)
C. (i), (iii) and (iv)
D. only (iv)

Q23.8.7 MCQ 2023 2M

Three husband-wife pairs are to be seated at a circular table that has six identical chairs. Seating arrangements are defined only by the relative position of the people. How many seating arrangements are possible such that every husband sits next to his wife?

A. 16 **B.** 4 **C.** 120 **D.** 720

Q23.8.8 MCQ 2023 2M

Based only on the following passage, which one of the options can be inferred with *certainty*? When the congregation sang together, Apenyo would also join, though her little screams were not quite audible because of the group singing. But whenever there was a special number, trouble would begin; Apenyo would try singing along, much to the embarrassment of her mother. After two or three such mortifying Sunday evenings, the mother stopped going to church altogether until Apenyo became older and learnt to behave.

At home too, Apenyo never kept quiet; she hummed or made up silly songs to sing by herself, which annoyed her mother at times but most often made her become pensive. She was by now convinced that her daughter had inherited her love of singing from her father who had died unexpectedly away from home.

[Excerpt from *These Hills Called Home* by Temsula Ao]

A. The mother was embarrassed about her daughter's singing at home.

B. The mother's feelings about her daughter's singing at home were only of annoyance.

C. The mother was not sure if Apenyo had inherited her love of singing from her father.

D. When Apenyo hummed at home, her mother tended to become thoughtful.

Q23.8.9 MCQ 2023 2M

If x satisfies the equation $4^{8^x} = 256$, then x is equal to

A. 1/2 **B.** $log_{16}8$ **C.** 2/3 **D.** $log_4 8$

Q23.8.10 MCQ 2023 2M

Consider a spherical globe rotating about an axis passing through its poles. There are three points *P, Q,* and *R* situated respectively on the equator, the north pole, and midway between the equator and the north pole in the northern hemisphere. Let *P, Q,* and *R* move with speeds v_P, v_Q, and v_R, respectively.
Which one of the following options is CORRECT?

A. $v_P < v_R < v_Q$ **B.** $v_P < v_Q < v_R$

C. $v_P > v_R > v_Q$ **D.** $v_P = v_R \neq v_Q$

PrepFusion

Answer Key

Note: MTA Marks To All awarded — see detailed solutions.

Session : 1					
Q23.1.1 B	Q23.1.2 MTA	Q23.1.3 A	Q23.1.4 A	Q23.1.5 C	Q23.1.6 D
Q23.1.7 C	Q23.1.8 D	Q23.1.9 B	Q23.1.10 A		
Session : 2					
Q23.2.1 A	Q23.2.2 A	Q23.2.3 A	Q23.2.4 A	Q23.2.5 B	Q23.2.6 D
Q23.2.7 A	Q23.2.8 C	Q23.2.9 A	Q23.2.10 B		
Session : 3					
Q23.3.1 D	Q23.3.2 B	Q23.3.3 A	Q23.3.4 B	Q23.3.5 C	Q23.3.6 D
Q23.3.7 B	Q23.3.8 D	Q23.3.9 A	Q23.3.10 B		
Session : 4					
Q23.4.1 C	Q23.4.2 A	Q23.4.3 C	Q23.4.4 C	Q23.4.5 C	Q23.4.6 D
Q23.4.7 A	Q23.4.8 B	Q23.4.9 A	Q23.4.10 C		
Session : 5					
Q23.5.1 D	Q23.5.2 C	Q23.5.3 D	Q23.5.4 C	Q23.5.5 B	Q23.5.6 MTA
Q23.5.7 A	Q23.5.8 D	Q23.5.9 A	Q23.5.10 D		
Session : 6					
Q23.6.1 B	Q23.6.2 C	Q23.6.3 B	Q23.6.4 D	Q23.6.5 C	Q23.6.6 D
Q23.6.7 A	Q23.6.8 C	Q23.6.9 A	Q23.6.10 D		
Session : 7					
Q23.7.1 C	Q23.7.2 C	Q23.7.3 A	Q23.7.4 D	Q23.7.5 B	Q23.7.6 D
Q23.7.7 A	Q23.7.8 D	Q23.7.9 A	Q23.7.10 A		
Session : 8					
Q23.8.1 A	Q23.8.2 A	Q23.8.3 D	Q23.8.4 A	Q23.8.5 B	Q23.8.6 D
Q23.8.7 A	Q23.8.8 D	Q23.8.9 C	Q23.8.10 C		

YEAR

2022

8 Sessions · 80 Questions

Sessions Summary

Session	Participating Branches	Total Qs
Session 1	CS, BM	10
Session 2	EE, MA	10
Session 3	EC, ES, ST, NM, MT, MN	10
Session 4	CY, CH, PI, XH, IN, AG, GG, TF	10
Session 5	CE1, BT, PH, EY	10
Session 6	CE2, XE, XL	10
Session 7	ME1, PE, AR	10
Session 8	ME2, GE, AE	10

PrepFusion

Year Snapshot

Total Questions: 80 **MCQ:** 80 **MSQ:** 0 **NAT:** 0

Session : 1 | Forenoon | 05/02/22

Q22.1.1 MCQ 2022 1M

The _____ is too high for it to be considered _____.

A. fair / fare **B.** faer / fair

C. fare / fare **D.** fare / fair

Q22.1.2 MCQ 2022 1M

A function $y(x)$ is defined in the interval $[0, 1]$ on the x-axis as

$$y(x) = \begin{cases} 2 & \text{if } 0 \le x < \frac{1}{3} \\ 3 & \text{if } \frac{1}{3} \le x < \frac{3}{4} \\ 1 & \text{if } \frac{3}{4} \le x \le 1 \end{cases}$$

Which one of the following is the area under the curve for the interval $[0, 1]$ on the x-axis?

A. $\frac{5}{6}$ **B.** $\frac{6}{5}$ **C.** $\frac{13}{6}$ **D.** $\frac{6}{13}$

Q22.1.3 MCQ 2022 1M

Let r be a root of the equation $x^2 + 2x + 6 = 0$.
Then the value of the expression $(r+2)(r+3)(r+4)(r+5)$ is

A. 51 **B.** -51 **C.** 126 **D.** -126

Q22.1.4 MCQ 2022 1M

Given below are four statements.

Statement 1: All students are inquisitive.
Statement 2: Some students are inquisitive.
Statement 3: No student is inquisitive.
Statement 4: Some students are not inquisitive.

From the given four statements, find the two statements that **CANNOT BE TRUE** simultaneously, assuming that there is at least one student in the class.

A. Statement 1 and Statement 3

B. Statement 1 and Statement 2

C. Statement 2 and Statement 4

D. Statement 3 and Statement 4

Q22.1.5 MCQ 2022 1M

A palindrome is a word that reads the same forwards and backwards. In a game of words, a player has the following two plates painted with letters:

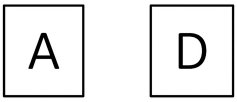
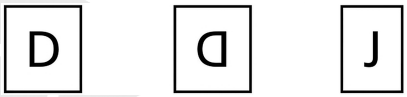


Fig. Q22.1.5

From the additional plates given in the options, which one of the combinations of additional plates would allow the player to construct a five-letter palindrome? The player should use all the five plates exactly once. The plates can be rotated in their plane.

A. 

B.

C.

D.

Q22.1.6 MCQ 2022 2M

Some people believe that "what gets measured, improves". Some others believe that "what gets measured, gets gamed". One possible reason for the difference in the beliefs is the work culture in organizations. In organizations with good work culture, metrics help improve outcomes. However, the same metrics are counterproductive in organizations with poor work culture.
Which one of the following is the CORRECT logical inference based on the information in the above passage?

A. Metrics are useful in organizations with poor work culture

B. Metrics are useful in organizations with good work culture

C. Metrics are always counterproductive in organizations with good work culture

D. Metrics are never useful in organizations with good work culture

Q22.1.7 MCQ 2022 2M

In a recently conducted national entrance test, boys constituted 65% of those who appeared for the test. Girls constituted the remaining candidates and they accounted for 60% of the qualified candidates.
Which one of the following is the correct logical inference based on the information provided in the above passage?

A. Equal number of boys and girls qualified

B. Equal number of boys and girls appeared for the test

C. The number of boys who appeared for the test is less than the number of girls who appeared

D. The number of boys who qualified the test is less than the number of girls who qualified

Q22.1.8 MCQ 2022 2M

A box contains five balls of same size and shape. Three of them are green coloured balls and two of them are orange coloured balls. Balls are drawn from the box one at a time. If a green ball is drawn, it is not replaced. If an orange ball is drawn, it is replaced with another orange ball.
First ball is drawn. What is the probability of getting an orange ball in the next draw?

A. $\frac{1}{2}$ **B.** $\frac{8}{25}$ **C.** $\frac{19}{50}$ **D.** $\frac{23}{50}$

Q22.1.9 MCQ 2022 2M

The corners and mid-points of the sides of a triangle are named using the distinct letters P, Q, R, S, T and U, but not necessarily in the same order. Consider the following statements:

- The line joining P and R is parallel to the line joining Q and S.
- P is placed on the side opposite to the corner T.
- S and U cannot be placed on the same side.

Which one of the following statements is correct based on the above information?

A. P cannot be placed at a corner

B. S cannot be placed at a corner

C. U cannot be placed at a mid-point

D. R cannot be placed at a corner

Q22.1.10 MCQ 2022 2M

A plot of land must be divided between four families. They want their individual plots to be similar in shape, not necessarily equal in area. The land has equally spaced poles, marked as dots in the below figure. Two ropes, R1 and R2, are already present and cannot be moved.
What is the least number of **additional** straight ropes needed to create the desired plots? A single rope can pass through three poles that are aligned in a straight line.

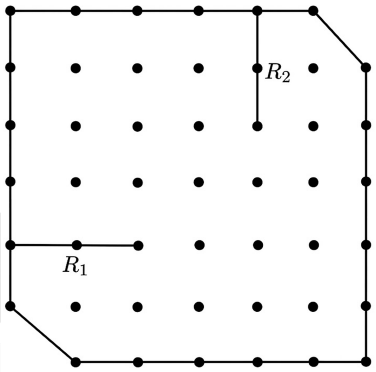


Fig. Q22.1.10

A. 2 **B.** 4 **C.** 5 **D.** 3

Session : 2 | Afternoon | 05/02/22

Q22.2.1 MCQ 2022 1M

As you grow older, an injury to your _____ may take longer to _____.

A. heel / heel **B.** heal / heel

C. heal / heal **D.** heel / heal

Q22.2.2 MCQ 2022 1M

In a 500 m race, P and Q have speeds in the ratio of 3 : 4. Q starts the race when P has already covered 140 m.
What is the distance between P and Q (in m) when P wins the race?

A. 20 **B.** 40 **C.** 60 **D.** 140

Q22.2.3 MCQ 2022 1M

Three bells P, Q, and R are rung periodically in a school. P is rung every 20 minutes; Q is rung every 30 minutes and R is rung every 50 minutes.
If all the three bells are rung at 12:00 PM, when will the three bells ring together again the next time?

A. 5:00 PM **B.** 5:30 PM **C.** 6:00 PM **D.** 6:30 PM

Q22.2.4 MCQ 2022 1M

Given below are two statements and four conclusions drawn based on the statements.

Statement 1: Some bottles are cups.
Statement 2: All cups are knives.

Conclusion I: Some bottles are knives.
Conclusion II: Some knives are cups.
Conclusion III: All cups are bottles.
Conclusion IV: All knives are cups.

Which one of the following options can be logically inferred?

A. Only conclusion I and conclusion II are correct

B. Only conclusion II and conclusion III are correct

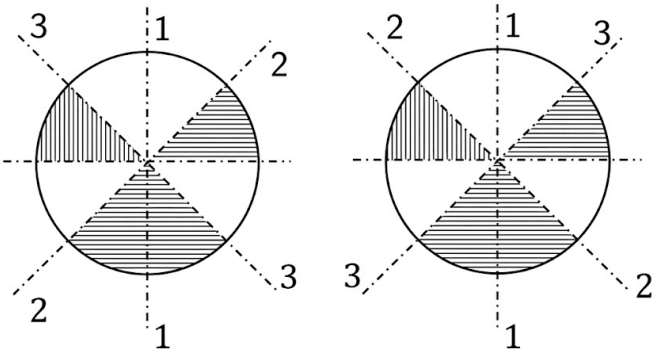
C. Only conclusion II and conclusion IV are correct

D. Only conclusion III and conclusion IV are correct

Q22.2.5 MCQ 2022 1M

The figure below shows the front and rear view of a disc, which is shaded with identical patterns. The disc is flipped once with respect to any one of the fixed axes 1-1, 2-2 or 3-3 chosen uniformly at random.

What is the probability that the disc **DOES NOT** retain the same front and rear views after the flipping operation?



Front View Rear View

Fig. Q22.2.5

A. 0 **B.** $\frac{1}{3}$ **C.** $\frac{2}{3}$ **D.** 1

Q22.2.6 MCQ 2022 2M

Altruism is the human concern for the wellbeing of others. Altruism has been shown to be motivated more by social bonding, familiarity and identification of belongingness to a group. The notion that altruism may be attributed to empathy or guilt has now been rejected.

Which one of the following is the CORRECT logical inference based on the information in the above passage?

A. Humans engage in altruism due to guilt but not empathy

B. Humans engage in altruism due to empathy but not guilt

C. Humans engage in altruism due to group identification but not empathy

D. Humans engage in altruism due to empathy but not familiarity

Q22.2.7 MCQ 2022 2M

There are two identical dice with a single letter on each of the faces. The following six letters: Q, R, S, T, U, and V, one on each of the faces. Any of the six outcomes are equally likely.

The two dice are thrown once independently at random.

What is the probability that the outcomes on the dice were composed only of any combination of the following possible outcomes: Q, U and V?

A. $\frac{1}{4}$ **B.** $\frac{3}{4}$ **C.** $\frac{1}{6}$ **D.** $\frac{5}{36}$

Q22.2.8 MCQ 2022 2M

The price of an item is 10% cheaper in an online store S compared to the price at another online store M. Store S charges ₹150 for delivery. There are no delivery charges for orders from the store M. A person bought the item from the store S and saved ₹100.

What is the price of the item at the online store S (in ₹) if there are no other charges than what is described above?

A. 2500 **B.** 2250 **C.** 1750 **D.** 1500

Q22.2.9 MCQ 2022 2M

The letters P, Q, R, S, T and U are to be placed one per vertex on a regular convex hexagon, but not necessarily in the same order.

Consider the following statements:

- The line segment joining R and S is longer than the line segment joining P and Q.

- The line segment joining R and S is perpendicular to the line segment joining P and Q.
- The line segment joining R and U is parallel to the line segment joining T and Q.

Based on the above statements, which one of the following options is CORRECT?

A. The line segment joining R and T is parallel to the line segment joining Q and S

B. The line segment joining T and Q is parallel to the line joining P and U

C. The line segment joining R and P is perpendicular to the line segment joining U and Q

D. The line segment joining Q and S is perpendicular to the line segment joining R and P

Q22.2.10 MCQ | 2022 | 2M

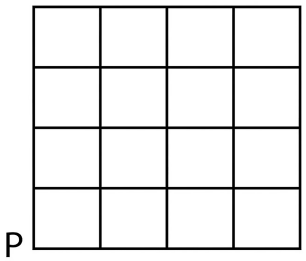
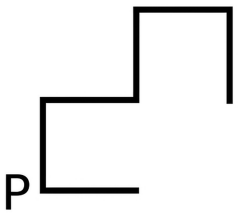


Fig. Q22.2.10

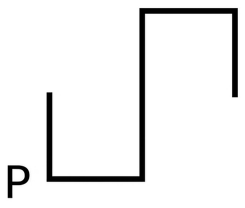
An ant is at the bottom-left corner of a grid (point P) as shown below. It aims to move to the top-right corner of the grid. The ant moves only along the lines marked in the grid such that the current distance to the top-right corner strictly decreases.

Which one of the following is a part of a possible trajectory of the ant during the movement?

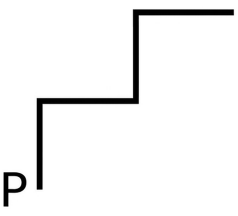
A.



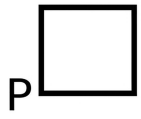
B.



C.



D.



Session : 3 | Forenoon | 06/02/22

Q22.3.1 MCQ | 2022 | 1M

Mr. X speaks _____ Japanese _____ Chinese.

A. neither / or **B.** either / nor

C. neither / nor **D.** also / but

Q22.3.2 MCQ | 2022 | 1M

A sum of money is to be distributed among P, Q, R, and S in the proportion 5:2:4:3, respectively.

If R gets ₹1000 more than S, what is the share of Q (in ₹)?

A. 500 **B.** 1000 **C.** 1500 **D.** 2000

Q22.3.3 MCQ | 2022 | 1M

A trapezium has vertices marked as P, Q, R and S (in that order anticlockwise). The side PQ is parallel to side SR. Further, it is given that, $PQ = 11$ cm, $QR = 4$ cm, $RS = 6$ cm and $SP = 3$ cm.

What is the shortest distance between PQ and SR (in cm)?

A. 1.80 **B.** 2.40 **C.** 4.20 **D.** 5.76

Q22.3.4 MCQ | 2022 | 1M

The figure shows a grid formed by a collection of unit squares. The unshaded unit square in the grid represents a hole.

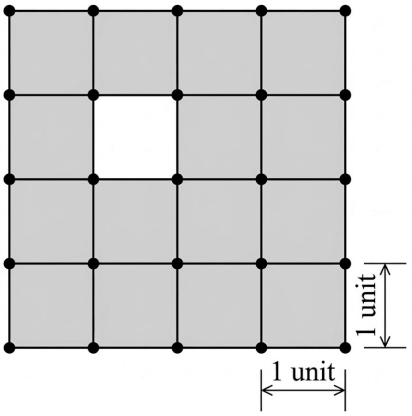


Fig. Q22.3.4

What is the maximum number of squares without a "hole in the interior" that can be formed within the 4×4 grid using the unit squares as building blocks?

A. 15 **B.** 20 **C.** 21 **D.** 26

Q22.3.5 MCQ 2022 1M

An art gallery engages a security guard to ensure that the items displayed are protected. The diagram below represents the plan of the gallery where the boundary walls are opaque. The location the security guard posted is identified such that all the inner space (shaded region in the plan) of the gallery is within the line of sight of the security guard. If the security guard does not move around the posted location and has a 360° view, which one of the following correctly represents the set of ALL possible locations among the locations P, Q, R and S, where the security guard can be posted to watch over the entire inner space of the gallery?

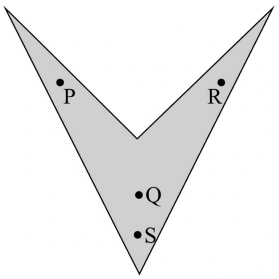


Fig. Q22.3.5

A. P and Q **B.** Q **C.** Q and S **D.** R and S

Q22.3.6 MCQ 2022 2M

Mosquitoes pose a threat to human health. Controlling mosquitoes using chemicals may have undesired consequences. In Florida, authorities have used genetically modified mosquitoes to control the overall mosquito population. It remains to be seen if this novel approach has unforeseen consequences.

Which one of the following is the correct logical inference based on the information in the above passage?

A. Using chemicals to kill mosquitoes is better than using genetically modified mosquitoes because genetic engineering is dangerous

B. Using genetically modified mosquitoes is better than using chemicals to kill mosquitoes because they do not have any side effects

C. Both using genetically modified mosquitoes and chemicals have undesired consequences and can be dangerous

D. Using chemicals to kill mosquitoes may have undesired consequences but it is not clear if using genetically modified mosquitoes has any negative consequence

Q22.3.7 MCQ 2022 2M

Consider the following inequalities:

(i) $2x - 1 > 7$

(ii) $2x - 9 < 1$

Which one of the following expressions below satisfies the above two inequalities?

A. $x \leq -4$ **B.** $-4 < x \leq 4$ **C.** $4 < x < 5$ **D.** $x \geq 5$

Q22.3.8 MCQ 2022 2M

Four points $P(0, 1)$, $Q(0, -3)$, $R(-2, -1)$, and $S(2, -1)$ represent the vertices of a quadrilateral.

What is the area enclosed by the quadrilateral?

A. 4 **B.** $4\sqrt{2}$ **C.** 8 **D.** $8\sqrt{2}$

Q22.3.9 MCQ 2022 2M

In a class of five students P, Q, R, S and T, only one student is known to have copied in the exam. The disciplinary committee has investigated the situation and recorded the statements from the students as given below.

Statement of P: R has copied in the exam.
Statement of Q: S has copied in the exam.
Statement of R: P did not copy in the exam.
Statement of S: Only one of us is telling the truth.
Statement of T: R is telling the truth.

The investigating team had authentic information that S never lies.

Based on the information given above, the person who has copied in the exam is

A. R **B.** P **C.** Q **D.** T

Q22.3.10 MCQ 2022 2M

Consider the following square with the four corners and the center marked as P, Q, R, S and T respectively.

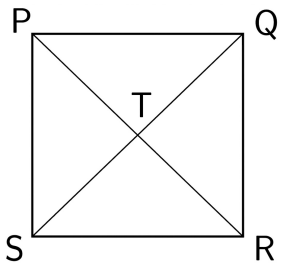


Fig. Q22.3.10

Let X, Y and Z represent the following operations:

X: rotation of the square by 180 degree with respect to the S-Q axis.

Y: rotation of the square by 180 degree with respect to the P-R axis.
Z: rotation of the square by 90 degree clockwise with respect to the axis perpendicular, going into the screen and passing through the point T.

Consider the following three distinct sequences of operation (which are applied in the left to right order):

(1) XYZZ
(2) XY
(3) ZZZZ

Which one of the following statements is correct as per the information provided above?

A. The sequence of operations (1) and (2) are equivalent

B. The sequence of operations (1) and (3) are equivalent

C. The sequence of operations (2) and (3) are equivalent

D. The sequence of operations (1), (2) and (3) are equivalent

Session : 4 | Afternoon | 06/02/22

Q22.4.1 MCQ 2022 1M

Inhaling the smoke from a burning _____ could _____ you quickly

A. tire / tier **B.** tire / tyre

C. tyre / tire **D.** tyre / tier

Q22.4.2 MCQ 2022 1M

A sphere of radius r cm is packed in a box of cubical shape. What should be the minimum volume (in cm^3) of the box that can enclose the sphere?

A. $\frac{r^3}{8}$ **B.** r^3 **C.** $2r^3$ **D.** $8r^3$

Q22.4.3 MCQ 2022 1M

Pipes P and Q can fill a storage tank in full with water in 10 and 6 minutes, respectively. Pipe R draws the water out from the storage tank at a rate of 34 litres per minute. P, Q and R operate at a constant rate.
If it takes one hour to completely empty a full storage tank with all the pipes operating simultaneously, what is the capacity of the storage tank (in litres)?

A. 26.8 **B.** 60.0 **C.** 120.0 **D.** 127.5

Q22.4.4 MCQ 2022 1M

Six persons P, Q, R, S, T and U are sitting around a circular table facing the center not necessarily in the same order. Consider the following statements:

- P sits next to S and T.
- Q sits diametrically opposite to P.
- The shortest distance between S and R is equal to the shortest distance between T and U.

Based on the above statements, Q is a neighbor of:

A. U and S **B.** R and T **C.** R and U **D.** P and S

Q22.4.5 MCQ 2022 1M

A building has several rooms and doors as shown in the top view of the building. The doors are closed initially.
What is the minimum number of doors that need to be opened in order to go from the point P to the point Q?

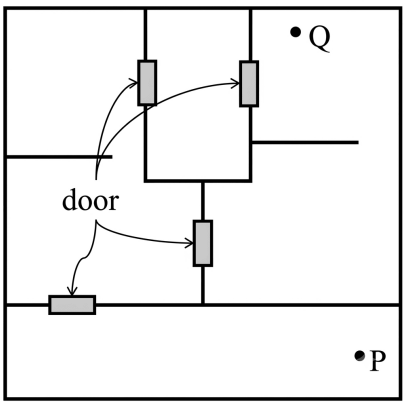


Fig. Q22.4.5

A. 4 **B.** 3 **C.** 2 **D.** 1

Q22.4.6 MCQ 2022 2M

Rice, a versatile and inexpensive source of carbohydrate, is a critical component of diet worldwide. Climate change, causing extreme weather, poses a threat to sustained availability of rice. Scientists are working on developing Green Super Rice (GSR), which is resilient under extreme weather conditions yet gives higher yields sustainably.
Which one of the following is the CORRECT logical inference based on the information given in the above passage?

A. GSR is an alternative to regular rice, but it grows only in an extreme weather

B. GSR may be used in future in response to adverse effects of climate change

C. GSR grows in an extreme weather, but the quantity of produce is lesser than regular rice

D. Regular rice will continue to provide good yields even in extreme weather

Q22.4.7 MCQ 2022 2M

A game consists of spinning an arrow around a stationary disk. When the arrow comes to rest, there are eight equally likely outcomes. It could come to rest in any one of the sectors numbered 1, 2, 3, 4, 5, 6, 7 or 8.
Two such disks are used in a game where their arrows are independently spun.
What is the probability that the sum of the numbers on the resulting sectors upon spinning the two disks is equal to 8 after the arrows come to rest?

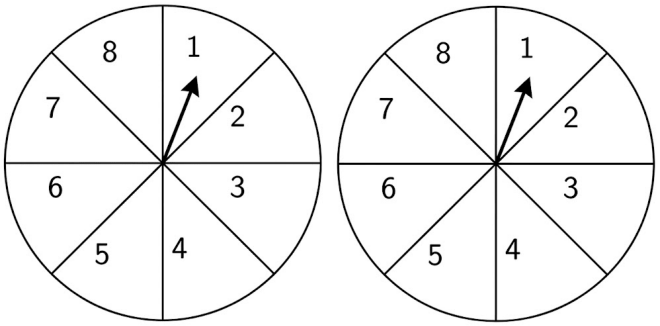


Fig. Q22.4.7

A. $\frac{1}{16}$ **B.** $\frac{5}{64}$ **C.** $\frac{3}{32}$ **D.** $\frac{7}{64}$

Q22.4.8 MCQ 2022 2M

Consider the following inequalities:

(i) $3p - q < 4$
(ii) $3q - p < 12$

Which one of the following expressions below satisfies the above two inequalities?

A. $p + q < 8$ **B.** $p + q = 8$

C. $8 \leq p + q < 16$ **D.** $p + q \geq 16$

Q22.4.9 MCQ 2022 2M

Given below are three statements and four conclusions drawn based on the statements.

Statement 1: Some engineers are writers.
Statement 2: No writer is an actor.
Statement 3: All actors are engineers.

Conclusion I: Some writers are engineers.
Conclusion II: All engineers are actors.
Conclusion III: No actor is a writer.
Conclusion IV: Some actors are writers.

Which one of the following options can be logically inferred?

A. Only conclusion I is correct

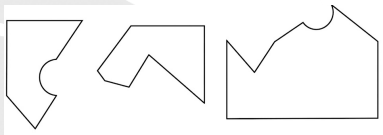
B. Only conclusion II and conclusion III are correct

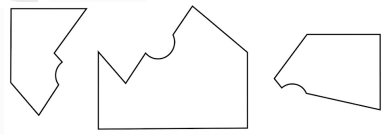
C. Only conclusion I and conclusion III are correct

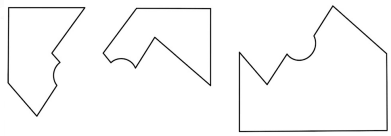
D. Either conclusion III or conclusion IV is correct

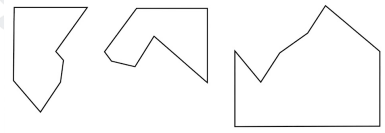
Q22.4.10 MCQ 2022 2M

Which one of the following sets of pieces can be assembled to form a square with a single round hole near the center? Pieces cannot overlap.

A.


B.


C.


D.


Session : 5 | Forenoon | 12/02/22

Q22.5.1 MCQ 2022 1M

You should _____ when to say _____.

A. no / no **B.** no / know **C.** know / know **D.** know / no

Q22.5.2 MCQ 2022 1M

Two straight lines pass through the origin $(x_0, y_0) = (0, 0)$. One of them passes through the point $(x_1, y_1) = (1, 3)$ and the other passes through the point $(x_2, y_2) = (1, 2)$.
What is the area enclosed between the straight lines in the interval $[0, 1]$ on the x-axis?

A. 0.5 **B.** 1.0 **C.** 1.5 **D.** 2.0

Q22.5.3 MCQ 2022 1M

If
$p : q = 1 : 2$
$q : r = 4 : 3$
$r : s = 4 : 5$
and u is 50% more than s, what is the ratio $p : u$?

A. 2:15 **B.** 16:15 **C.** 1:5 **D.** 16:45

Q22.5.4 MCQ 2022 1M

Given the statements:

- P is the sister of Q.
- Q is the husband of R.
- R is the mother of S.
- T is the husband of P.

Based on the above information, T is _____ of S.

A. the grandfather **B.** an uncle

C. the father **D.** a brother

Q22.5.5 MCQ 2022 1M

In the following diagram, the point R is the center of the circle. The lines PQ and ZV are tangential to the circle. The relation among the areas of the squares, PXWR, RUVZ and SPQT is

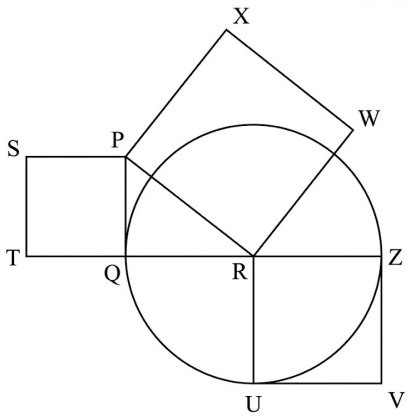


Fig. Q22.5.5

A. Area of SPQT = Area of RUVZ = Area of PXWR

B. Area of SPQT = Area of PXWR − Area of RUVZ

C. Area of PXWR = Area of SPQT − Area of RUVZ

D. Area of PXWR = Area of RUVZ − Area of SPQT

Q22.5.6 MCQ 2022 2M

Healthy eating is a critical component of healthy aging. When should one start eating healthy? It turns out that it is never too early. For example, babies who start eating healthy in the first year are more likely to have better overall health as they get older.
Which one of the following is the CORRECT logical inference based on the information in the above passage?

A. Healthy eating is important for those with good health conditions, but not for others

B. Eating healthy can be started at any age, earlier the better

C. Eating healthy and better overall health are more correlated at a young age, but not older age

D. Healthy eating is more important for adults than kids

Q22.5.7 MCQ 2022 2M

P invested ₹5000 per month for 6 months of a year and Q invested ₹x per month for 8 months of the year in a partnership business. The profit is shared in proportion to the total investment made in that year.
If at the end of that investment year, Q receives $\frac{4}{9}$ of the total profit, what is the value of x (in ₹)?

A. 2500 **B.** 3000 **C.** 4687 **D.** 8437

Q22.5.8 MCQ 2022 2M

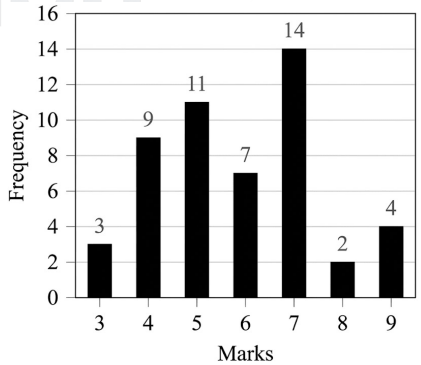


Fig. Q22.5.8

The above frequency chart shows the frequency distribution of marks obtained by a set of students in an exam.
From the data presented above, which one of the following is CORRECT?

A. mean > mode > median **B.** mode > median > mean

C. mode > mean > median **D.** median > mode > mean

Q22.5.9 MCQ | 2022 | 2M

In the square grid shown below, a person standing at P2 position is required to move to P5 position.

The only movement allowed for a step involves, "two moves along one direction followed by one move in a perpendicular direction". The permissible directions for movement are shown as dotted arrows in the right.

For example, a person at a given position **Y** can move only to the positions marked X on the right.

Without occupying any of the shaded squares at the end of each step, the minimum number of steps required to go from P2 to P5 is

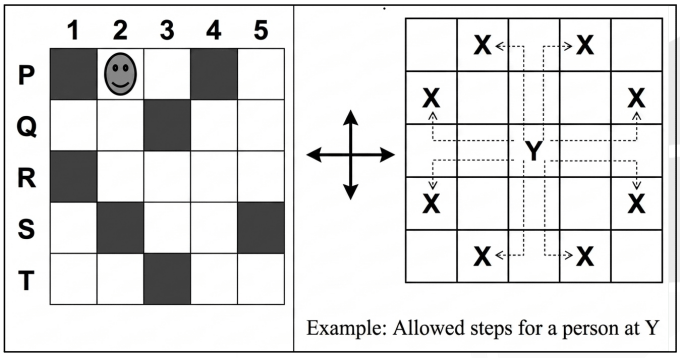


Fig. Q22.5.9

A. 4 **B.** 5 **C.** 6 **D.** 7

Q22.5.10 MCQ | 2022 | 2M

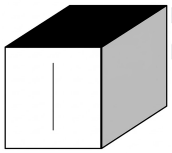
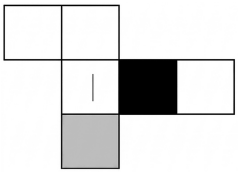
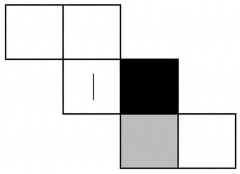
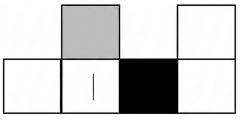


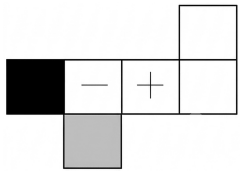
Fig. Q22.5.10

Consider a cube made by folding a single sheet of paper of appropriate shape. The interior faces of the cube are all blank. However, the exterior faces that are not visible in the given view may not be blank. Which one of the following represents a possible unfolding of the cube?

A.


B.


C.


D.


Session : 6 | Afternoon | 12/02/22

Q22.6.1 MCQ | 2022 | 1M

The movie was funny and I _____.

A. could help laughing **B.** couldn't help laughed

C. couldn't help laughing **D.** could helped laughed

Q22.6.2 MCQ | 2022 | 1M

$x : y : z = \frac{1}{2} : \frac{1}{3} : \frac{1}{4}$.

What is the value of $\frac{x + z - y}{y}$?

A. 0.75 **B.** 1.25 **C.** 2.25 **D.** 3.25

Q22.6.3 MCQ | 2022 | 1M

Both the numerator and the denominator of $\frac{3}{4}$ are increased by a positive integer, x, and those of $\frac{15}{17}$ are decreased by the same integer. This operation results in the same value for both the fractions.

What is the value of x?

A. 1 **B.** 2 **C.** 3 **D.** 4

Q22.6.4 MCQ | 2022 | 1M

A survey of 450 students about their subjects of interest resulted in the following outcome:

- 150 students are interested in Mathematics.
- 200 students are interested in Physics.
- 175 students are interested in Chemistry.
- 50 students are interested in Mathematics and Physics.
- 60 students are interested in Physics and Chemistry.
- 40 students are interested in Mathematics and Chemistry.
- 30 students are interested in Mathematics, Physics and Chemistry.
- Remaining students are interested in Humanities.

Based on the above information, the number of students interested in Humanities is

A. 10 **B.** 30 **C.** 40 **D.** 45

Q22.6.5 MCQ 2022 1M

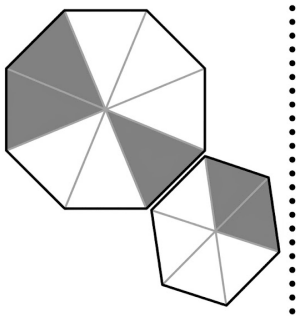
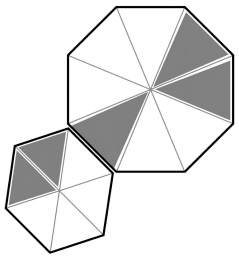
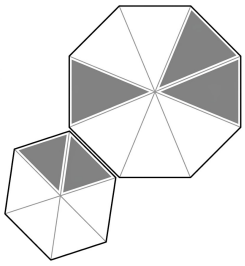
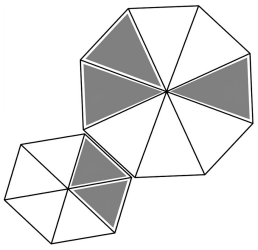


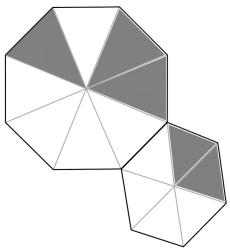
Fig. Q22.6.5

For the picture shown above, which one of the following is the correct picture representing reflection with respect to the mirror shown as the dotted line?

A. 

B. 

C. 

D. 

Q22.6.6 MCQ 2022 2M

In the last few years, several new shopping malls were opened in the city. The total number of visitors in the malls is impressive. However, the total revenue generated through sales in the shops in these malls is generally low. Which one of the following is the CORRECT logical inference based on the information in the above passage?

A. Fewer people are visiting the malls but spending more

B. More people are visiting the malls but not spending enough

C. More people are visiting the malls and spending more

D. Fewer people are visiting the malls and not spending enough

Q22.6.7 MCQ 2022 2M

In a partnership business, the monthly investment by three friends for the first six months is in the ratio 3:4:5. After six months, they had to increase their monthly investments by 10%, 15% and 20%, respectively, of their initial monthly investment. The new investment ratio was kept constant for the next six months.
What is the ratio of their shares in the total profit (in the same order) at the end of the year such that the share is proportional to their individual total investment over the year?

A. 22 : 23 : 24

B. 22 : 33 : 50

C. 33 : 46 : 60

D. 63 : 86 : 110

Q22.6.8 MCQ 2022 2M

Consider the following equations of straight lines:
Line L_1: $2x - 3y = 5$;
Line L_2: $3x + 2y = 8$;
Line L_3: $4x - 6y = 5$;
Line L_4: $6x - 9y = 6$.
Which one among the following is the correct statement?

A. L_1 is parallel to L_2 and L_1 is perpendicular to L_3

B. L_2 is parallel to L_4 and L_2 is perpendicular to L_1

C. L_3 is perpendicular to L_4 and L_3 is parallel to L_2

D. L_4 is perpendicular to L_2 and L_4 is parallel to L_3

Q22.6.9 MCQ 2022 2M

Given below are two statements and four conclusions drawn based on the statements.

Statement 1: Some soaps are clean.
Statement 2: All clean objects are wet.

Conclusion I: Some clean objects are soaps.
Conclusion II: No clean object is a soap.
Conclusion III: Some wet objects are soaps.
Conclusion IV: All wet objects are soaps.

A. Only conclusion I is correct

B. Either conclusion I or conclusion II is correct

C. Either conclusion III or conclusion IV is correct

D. Only conclusion I and conclusion III are correct

Q22.6.10 MCQ 2022 2M

An ant walks in a straight line on a plane leaving behind a trace of its movement. The initial position of the ant is at point P facing east.
The ant first turns 72° anticlockwise at P, and then does the following two steps in sequence exactly FIVE times before halting:

1. moves forward for 10 cm.
2. turns 144° clockwise.

The pattern made by the trace left behind by the ant is .

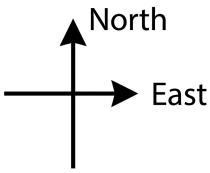
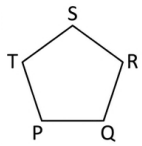
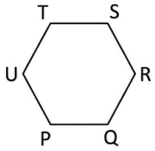


Fig. Q22.6.10

A. 

PQ = QR = RS = ST = TP = 10 cm

B. 

PQ = QR = RS = ST = TU = UP = 10 cm

C.

SQ = QT = TR = RP = PS = 10 cm

D.

SW = WR = RP = PT = TQ = QU = US = 10 cm

Session : 7 | Forenoon | 13/02/22

Q22.7.1 MCQ 2022 1M

After playing _____ hours of tennis, I am feeling _____ tired to walk back.

A. too / too **B.** too / two **C.** two / two **D.** two / too

Q22.7.2 MCQ 2022 1M

The average of the monthly salaries of M, N and S is ₹4000.
The average of the monthly salaries of N, S and P is ₹5000.
The monthly salary of P is ₹6000.
What is the monthly salary of M as a percentage of the monthly salary of P?

A. 50% **B.** 75% **C.** 100% **D.** 125%

Q22.7.3 MCQ 2022 1M

A person travelled 80 km in 6 hours. If the person travelled the first part with a uniform speed of 10 kmph and the remaining part with a uniform speed of 18 kmph.
What percentage of the total distance is travelled at a uniform speed of 10 kmph?

A. 28.25 **B.** 37.25 **C.** 43.75 **D.** 50.00

Q22.7.4 MCQ 2022 1M

Four girls P, Q, R and S are studying languages in a University. P is learning French and Dutch. Q is learning Chinese and Japanese. R is learning Spanish and French. S is learning Dutch and Japanese.
Given that: French is easier than Dutch; Chinese is harder than Japanese; Dutch is easier than Japanese, and Spanish is easier than French.
Based on the above information, which girl is learning the most difficult pair of languages?

A. P **B.** Q **C.** R **D.** S

Q22.7.5 MCQ 2022 1M

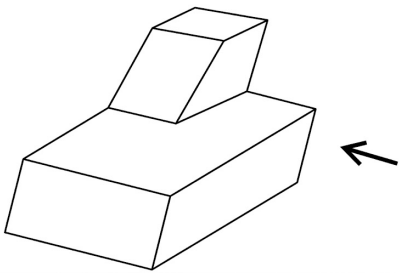
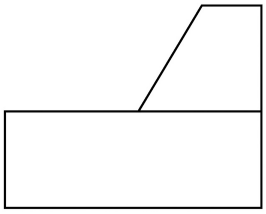
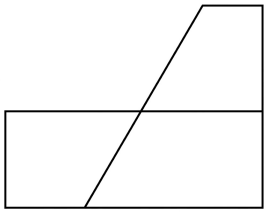
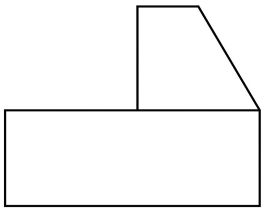


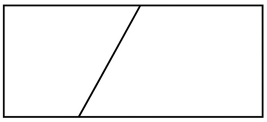
Fig. Q22.7.5

A block with a trapezoidal cross-section is placed over a block with rectangular cross section as shown above. Which one of the following is the correct drawing of the view of the 3D object as viewed in the direction indicated by an arrow in the above figure?

A. 

B. 

C. 

D. 

Q22.7.6 MCQ 2022 2M

Humans are naturally compassionate and honest. In a study using strategically placed wallets that appear "lost", it was found that wallets with money are more likely to be returned than wallets without money. Similarly, wallets that had a key and money are more likely to be returned than wallets with the same amount of money alone. This suggests that the primary reason for this behavior is compassion and empathy.

Which one of the following is the CORRECT logical inference based on the information in the above passage?

A. Wallets with a key are more likely to be returned because people do not care about money

B. Wallets with a key are more likely to be returned because people relate to suffering of others

C. Wallets used in experiments are more likely to be returned than wallets that are really lost

D. Money is always more important than keys

Q22.7.7 MCQ 2022 2M

A rhombus is formed by joining the midpoints of the sides of a unit square.

What is the diameter of the largest circle that can be inscribed within the rhombus?

A. $\frac{1}{\sqrt{2}}$ **B.** $\frac{1}{2\sqrt{2}}$ **C.** $\sqrt{2}$ **D.** $2\sqrt{2}$

Q22.7.8 MCQ 2022 2M

An equilateral triangle, a square and a circle have equal areas.

What is the ratio of the perimeters of the equilateral triangle to square to circle?

A. $3\sqrt{3} : 2 : \sqrt{\pi}$ **B.** $\sqrt{(3\sqrt{3})} : 2 : \sqrt{\pi}$

C. $\sqrt{(3\sqrt{3})} : 4 : 2\sqrt{\pi}$ **D.** $\sqrt{(3\sqrt{3})} : 2 : 2\sqrt{\pi}$

Q22.7.9 MCQ 2022 2M

Given below are three conclusions drawn based on the following three statements.

Statement 1: All teachers are professors.
Statement 2: No professor is a male.
Statement 3: Some males are engineers.

Conclusion I: No engineer is a professor.
Conclusion II: Some engineers are professors.
Conclusion III: No male is a teacher.

Which one of the following options can be logically inferred?

A. Only conclusion III is correct

B. Only conclusion I and conclusion II are correct

C. Only conclusion II and conclusion III are correct

D. Only conclusion I and conclusion III are correct

Q22.7.10 MCQ 2022 2M

In a 12-hour clock that runs correctly, how many times do the second, minute, and hour hands of the clock coincide, in a 12-hour duration from 3 PM in a day to 3 AM the next day?

A. 11 **B.** 12 **C.** 144 **D.** 2

Session : 8 | Afternoon | 13/02/22

Q22.8.1 MCQ 2022 1M

Writing too many things on the _____ while teaching could make the students get _____.

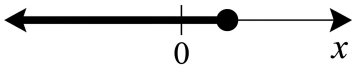
A. bored / board **B.** board / bored

C. board / board **D.** bored / bored

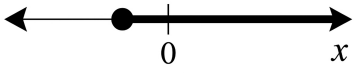
Q22.8.2 MCQ 2022 1M

Which one of the following is a representation (not to scale and in bold) of all values of x satisfying the inequality $2 - 5x \leq -\frac{6x-5}{3}$ on the real number line?

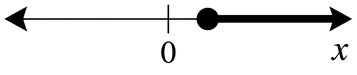
A.



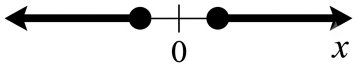
B.



C.



D.



Q22.8.3 MCQ 2022 1M

If $f(x) = 2\ln(\sqrt{e^x})$, what is the area bounded by $f(x)$ for the interval [0, 2] on the x-axis?

A. $\frac{1}{2}$ **B.** 1 **C.** 2 **D.** 4

Q22.8.4 MCQ 2022 1M

A person was born on the fifth Monday of February in a particular year.

Which one of the following statements is correct based on the above information?

A. The 2nd February of that year is a Tuesday

B. There will be five Sundays in the month of February in that year

C. The 1st February of that year is a Sunday

D. All Mondays of February in that year have even dates

Q22.8.5 MCQ 2022 1M

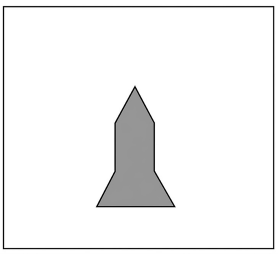
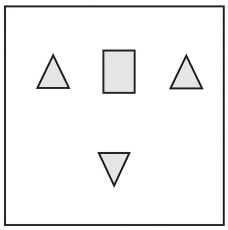


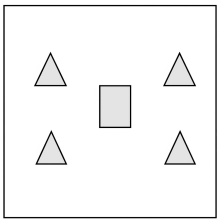
Fig. Q22.8.5

Which one of the groups given below can be assembled to get the shape that is shown above using each piece only once without overlapping with each other? (rotation and translation operations may be used).

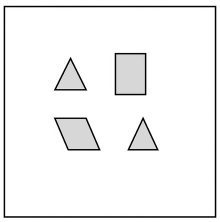
A.



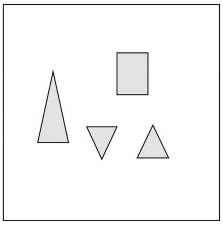
B.



C.



D.



Q22.8.6 MCQ 2022 2M

Fish belonging to species S in the deep sea have skins that are extremely black (ultra-black skin). This helps them not only to avoid predators but also sneakily attack their prey. However, having this extra layer of black pigment results in lower collagen on their skin, making their skin more fragile.

Which one of the following is the CORRECT logical inference based on the information in the above passage?

A. Having ultra-black skin is only advantageous to species S

B. Species S with lower collagen in their skin are at an advantage because it helps them avoid predators

C. Having ultra-black skin has both advantages and disadvantages to species S

D. Having ultra-black skin is only disadvantageous to species S but advantageous only to their predators

Q22.8.7 MCQ 2022 2M

For the past m days, the average daily production at a company was 100 units per day.
If today's production of 180 units changes the average to 110 units per day, what is the value of m?

A. 18 **B.** 10 **C.** 7 **D.** 5

Q22.8.8 MCQ 2022 2M

Consider the following functions for non-zero positive integers, p and q.

$$f(p,q) = \underbrace{p \times p \times p \times \ldots\ldots \cdots \times p}_{q \text{ terms}} = p^q; \quad f(p,1) = p$$

$$g(p,q) = p^{p^{p^{p^{\cdots}}}} \text{ (up to } q \text{ terms)} \quad ; \quad g(p,1) = p$$

Which one of the following options is correct based on the above?

A. $f(2,2) = g(2,2)$ **B.** $f(g(2,2),2) < f(2,g(2,2))$

C. $g(2,1) \neq f(2,1)$ **D.** $f(3,2) > g(3,2)$

Q22.8.9 MCQ 2022 2M

Four cities P, Q, R and S are connected through one-way routes as shown in the figure. The travel time between any two connected cities is one hour. The boxes beside each city name describe the starting time of first train of the day and their frequency of operation. For example, from city P, the first trains of the day start at 8 AM with a frequency of 90 minutes to each of R and S. A person does not spend additional time at any city other than the waiting time for the next connecting train.
If the person starts from R at 7 AM and is required to visit S and return to R, what is the minimum time required?

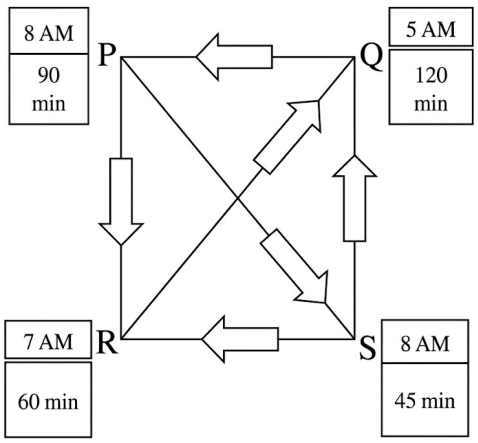


Fig. Q22.8.9

A. 6 hours 30 minutes **B.** 3 hours 45 minutes

C. 4 hours 30 minutes **D.** 5 hours 15 minutes

Q22.8.10 MCQ 2022 2M

Equal sized circular regions are shaded in a square sheet of paper of 1 cm side length. Two cases, case M and case N, are considered as shown in the figures below. In the case M, four circles are shaded in the square sheet and in the case N, nine circles are shaded in the square sheet as shown.
What is the ratio of the areas of unshaded regions of case M to that of case N?

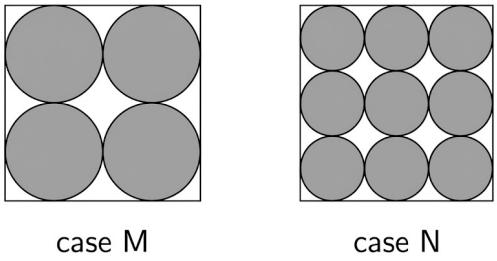


Fig. Q22.8.10

A. 2:3 **B.** 1:1 **C.** 3:2 **D.** 2:1

Answer Key

Note: MTA Marks To All awarded — see detailed solutions.

Session : 1					
Q22.1.1 D	Q22.1.2 C	Q22.1.3 D	Q22.1.4 A	Q22.1.5 B	Q22.1.6 B
Q22.1.7 D	Q22.1.8 D	Q22.1.9 B	Q22.1.10 D		
Session : 2					
Q22.2.1 D	Q22.2.2 A	Q22.2.3 A	Q22.2.4 A	Q22.2.5 C	Q22.2.6 C
Q22.2.7 A	Q22.2.8 B	Q22.2.9 A	Q22.2.10 C		
Session : 3					
Q22.3.1 C	Q22.3.2 D	Q22.3.3 B	Q22.3.4 B	Q22.3.5 C	Q22.3.6 D
Q22.3.7 C	Q22.3.8 C	Q22.3.9 B	Q22.3.10 B		
Session : 4					
Q22.4.1 C	Q22.4.2 D	Q22.4.3 C	Q22.4.4 C	Q22.4.5 C	Q22.4.6 B
Q22.4.7 D	Q22.4.8 A	Q22.4.9 C	Q22.4.10 C		
Session : 5					
Q22.5.1 D	Q22.5.2 A	Q22.5.3 D	Q22.5.4 B	Q22.5.5 B	Q22.5.6 B
Q22.5.7 B	Q22.5.8 B	Q22.5.9 B	Q22.5.10 MTA		
Session : 6					
Q22.6.1 C	Q22.6.2 B	Q22.6.3 C	Q22.6.4 D	Q22.6.5 A	Q22.6.6 B
Q22.6.7 D	Q22.6.8 D	Q22.6.9 D	Q22.6.10 C		
Session : 7					
Q22.7.1 D	Q22.7.2 A	Q22.7.3 C	Q22.7.4 B	Q22.7.5 A	Q22.7.6 B
Q22.7.7 A	Q22.7.8 B	Q22.7.9 A	Q22.7.10 MTA		
Session : 8					
Q22.8.1 B	Q22.8.2 C	Q22.8.3 C	Q22.8.4 A	Q22.8.5 B OR C	Q22.8.6 C
Q22.8.7 C	Q22.8.8 A	Q22.8.9 A	Q22.8.10 B		

YEAR

2021

8 Sessions · 80 Questions

Sessions Summary

Session	Participating Branches	Total Qs
Session 1	CE1, IN, CY	10
Session 2	CE2, PH, AR, BM, AE, MN	10
Session 3	EE, GG, AG, EY	10
Session 4	EC, PE, ES, ST, PI	10
Session 5	CS1, MA, BT, CH, TF	10
Session 6	CS2, XE, XL	10
Session 7	ME1, XH	10
Session 8	ME2, MT	10

PrepFusion

Year Snapshot

Total Questions: 80 **MCQ:** 80 **MSQ:** 0 **NAT:** 0

Session : 1 | Forenoon | 06/02/21

Q21.1.1 MCQ 2021 1M

Getting to the top is _____ than staying on top.

A. more easy **B.** much easy **C.** easiest **D.** easier

Q21.1.2 MCQ 2021 1M

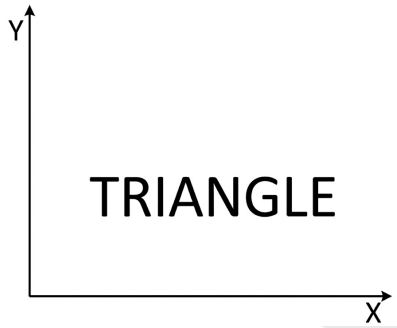


Fig. Q21.1.2

The mirror image of the text about the x-axis is:

A. ⊥RIAИGLE **B.** ⊥RIAИƆLE

C. ⊥RIAИƆLE **D.** ⊥RIAИƆLƎ

Q21.1.3 MCQ 2021 1M

In a company, 35% of the employees drink coffee, 40% of the employees drink tea and 10% of the employees drink both tea and coffee. What % of employees drink neither tea nor coffee?

A. 15 **B.** 25 **C.** 35 **D.** 40

Q21.1.4 MCQ 2021 1M

$\oplus$ and $\odot$ are two operators on numbers p and q such that

$p \oplus q = \dfrac{p^2 + q^2}{pq}$ and $p \odot q = \dfrac{p^2}{q}$;

If $x \oplus y = 2 \odot 2$, then $x =$

A. $\dfrac{y}{2}$ **B.** y **C.** $\dfrac{3y}{2}$ **D.** $2y$

Q21.1.5 MCQ 2021 1M

Four persons P, Q, R and S are to be seated in a row, all facing the same direction, but not necessarily in the same order. P and R cannot sit adjacent to each other. S should be seated to the right of Q. The number of distinct seating arrangements possible is:

A. 2 **B.** 4 **C.** 6 **D.** 8

Q21.1.6 MCQ 2021 2M

Statement: Either P marries Q or X marries Y.
Among the options below, the logical NEGATION of the above statement is:

A. P does not marry Q and X marries Y.

B. Neither P marries Q nor X marries Y.

C. X does not marry Y and P marries Q.

D. P marries Q and X marries Y.

Q21.1.7 MCQ 2021 2M

Consider two rectangular sheets, Sheet M and Sheet N of dimensions 6 cm × 4 cm each.
Folding operation 1: The sheet is folded into half by joining the short edges of the current shape.
Folding operation 2: The sheet is folded into half by joining the long edges of the current shape.
Folding operation 1 is carried out on Sheet M three times.
Folding operation 2 is carried out on Sheet N three times.
The ratio of perimeters of the final folded shape of Sheet N to the final folded shape of Sheet M is _____.

A. 13:7 **B.** 3:2 **C.** 7:5 **D.** 5:13

Q21.1.8 MCQ 2021 2M

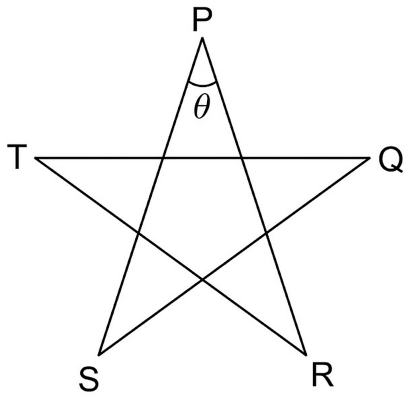


Fig. Q21.1.8

Five line segments of equal lengths, PR, PS, QS, QT and RT are used to form a star as shown in the figure above. The value of θ, in degrees, is _____

A. 36 **B.** 45 **C.** 72 **D.** 108

Q21.1.9 MCQ 2021 2M

A function, λ, is defined by

$$\lambda(p,q) = \begin{cases} (p-q)^2, & \text{if } p \geq q, \\ p+q, & \text{if } p < q. \end{cases}$$

The value of the expression $\dfrac{\lambda(-(-3+2),(-2+3))}{(-(-2+1))}$ is:

A. -1 **B.** 0 **C.** $\frac{16}{3}$ **D.** 16

Q21.1.10 MCQ 2021 2M

Humans have the ability to construct worlds entirely in their minds, which don't exist in the physical world. So far as we know, no other species possesses this ability. This skill is so important that we have different words to refer to its different flavors, such as imagination, invention and innovation.

Based on the above passage, which one of the following is TRUE?

A. No species possess the ability to construct worlds in their minds.

B. The terms imagination, invention and innovation refer to unrelated skills.

C. We do not know of any species other than humans who possess the ability to construct mental worlds.

D. Imagination, invention and innovation are unrelated to the ability to construct mental worlds.

Session : 2 | Afternoon | 06/02/21

Q21.2.1 MCQ 2021 1M

(i) Arun and Aparna are here.
(ii) Arun and Aparna is here.
(iii) Arun's families is here.
(iv) Arun's family is here.

Which of the above sentences are grammatically CORRECT?

A. (i) and (ii) **B.** (i) and (iv)

C. (ii) and (iv) **D.** (iii) and (iv)

Q21.2.2 MCQ 2021 1M

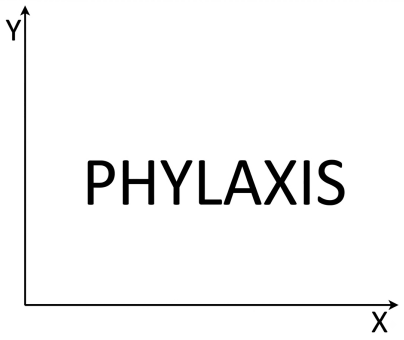


Fig. Q21.2.2

The mirror image of the text PHYLAXIS about the x-axis is

A. PHʎΓ∀XIS **B.** ⅁HʎΓ∀XIS

C. ⅁HʎΓ∀XIS **D.** ⅁HʎΓ∀XIS

Q21.2.3 MCQ 2021 1M

Two identical cube shaped dice each with faces numbered 1 to 6 are rolled simultaneously. The probability that an even number is rolled out on each dice is

A. $\frac{1}{36}$ **B.** $\frac{1}{12}$ **C.** $\frac{1}{8}$ **D.** $\frac{1}{4}$

Q21.2.4 MCQ 2021 1M

$\oplus$ and $\odot$ are two operators on numbers p and q such that

$$p \odot q = p - q, \quad \text{and} \quad p \oplus q = p \times q$$

Then, $(9 \odot (6 \oplus 7)) \odot (7 \oplus (6 \odot 5)) =$

A. 40 **B.** -26 **C.** -33 **D.** -40

Q21.2.5 MCQ 2021 1M

Four persons P, Q, R and S are to be seated in a row. R should not be seated at the second position from the left end of the row. The number of distinct seating arrangements possible is:

A. 6 **B.** 9 **C.** 18 **D.** 24

Q21.2.6 MCQ 2021 2M

On a planar field, you travelled 3 units East from a point O. Next you travelled 4 units South to arrive at point P. Then you travelled from P in the North-East direction such that you arrive at a point that is 6 units East of point O. Next, you travelled in the North-West direction, so that you arrive at point Q that is 8 units North of point P.

The distance of point Q to point O, in the same units, should be _____

A. 3 **B.** 4 **C.** 5 **D.** 6

Q21.2.7 MCQ | 2021 | 2M

The author said, "Musicians rehearse before their concerts. Actors rehearse their roles before the opening of a new play. On the other hand, I find it strange that many public speakers think they can just walk on to the stage and start speaking. In my opinion, it is no less important for public speakers to rehearse their talks."
Based on the above passage, which one of the following is TRUE?

A. The author is of the opinion that rehearsing is important for musicians, actors and public speakers.

B. The author is of the opinion that rehearsing is less important for public speakers than for musicians and actors.

C. The author is of the opinion that rehearsing is more important only for musicians than public speakers.

D. The author is of the opinion that rehearsal is more important for actors than musicians.

Q21.2.8 MCQ | 2021 | 2M

1. Some football players play cricket.
2. All cricket players play hockey.

Among the options given below, the statement that logically follows from the two statements 1 and 2 above, is:

A. No football player plays hockey.

B. Some football players play hockey.

C. All football players play hockey.

D. All hockey players play football.

Q21.2.9 MCQ | 2021 | 2M

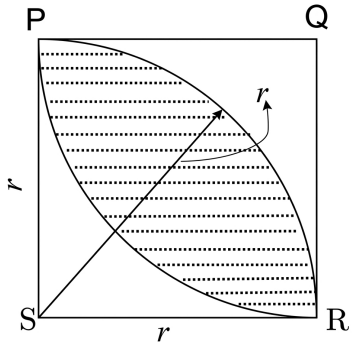


Fig. Q21.2.9

In the figure shown above, PQRS is a square. The shaded portion is formed by the intersection of sectors of circles with radius equal to the side of the square and centers at S and Q.
The probability that any point picked randomly within the square falls in the shaded area is _____

A. $4-\frac{\pi}{2}$ **B.** $\frac{1}{2}$ **C.** $\frac{\pi}{2}-1$ **D.** $\frac{\pi}{4}$

Q21.2.10 MCQ | 2021 | 2M

In an equilateral triangle PQR, side PQ is divided into four equal parts, side QR is divided into six equal parts and side PR is divided into eight equal parts. The length of each subdivided part in cm is an integer. The minimum area of the triangle PQR possible, in cm^2, is

A. 18 **B.** 24 **C.** $48\sqrt{3}$ **D.** $144\sqrt{3}$

Session : 3 | Forenoon | 07/02/21

Q21.3.1 MCQ | 2021 | 1M

The people _____ were at the demonstration were from all sections of society.

A. whose **B.** which **C.** who **D.** whom

Q21.3.2 MCQ | 2021 | 1M

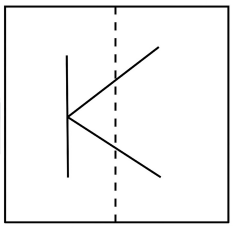
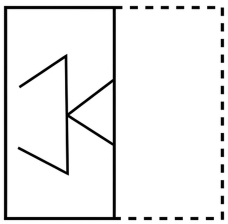
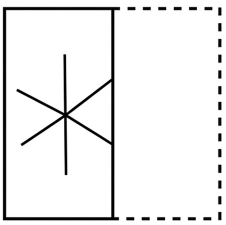
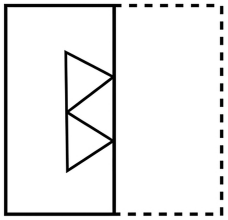


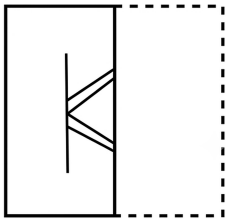
Fig. Q21.3.2

A transparent square sheet shown above is folded along the dotted line. The folded sheet will look like _____.

A.


B.


C.


D.


Q21.3.3 MCQ 2021 1M

For a regular polygon having 10 sides, the interior angle between the sides of the polygon, in degrees, is:

A. 396 **B.** 324 **C.** 216 **D.** 144

Q21.3.4 MCQ 2021 1M

Which one of the following numbers is exactly divisible by $(11^{13} + 1)$?

A. $11^{26} + 1$ **B.** $11^{33} + 1$

C. $11^{39} - 1$ **D.** $11^{52} - 1$

Q21.3.5 MCQ 2021 1M

Oasis is to *sand* as *island* is to _____

Which one of the following options maintains a similar logical relation in the above sentence?

A. Stone **B.** Land **C.** Water **D.** Mountain

Q21.3.6 MCQ 2021 2M

The importance of sleep is often overlooked by students when they are preparing for exams. Research has consistently shown that sleep deprivation greatly reduces the ability to recall the material learnt. Hence, cutting down on sleep to study longer hours can be counterproductive.

Which one of the following statements is the CORRECT inference from the above passage?

A. Sleeping well alone is enough to prepare for an exam. Studying has lesser benefit.

B. Students are efficient and are not wrong in thinking that sleep is a waste of time.

C. If a student is extremely well prepared for an exam, he needs little or no sleep.

D. To do well in an exam, adequate sleep must be part of the preparation.

Q21.3.7 MCQ 2021 2M

In the figure shown above, each inside square is formed by joining the midpoints of the sides of the next larger square. The area of the smallest square (shaded) as shown, in cm^2 is:

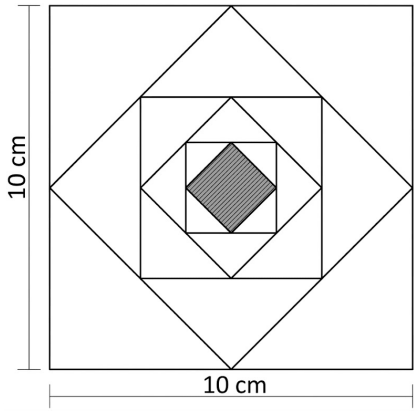


Fig. Q21.3.7

A. 12.50 **B.** 6.25 **C.** 3.125 **D.** 1.5625

Q21.3.8 MCQ 2021 2M

Let X be a continuous random variable denoting the temperature measured. The range of temperature is $[0, 100]$ degree Celsius and let the probability density function of X be $f(x) = 0.01$ for $0 \leq X \leq 100$.

The mean of X is _____

A. 2.5 **B.** 5.0 **C.** 25.0 **D.** 50.0

Q21.3.9 MCQ 2021 2M

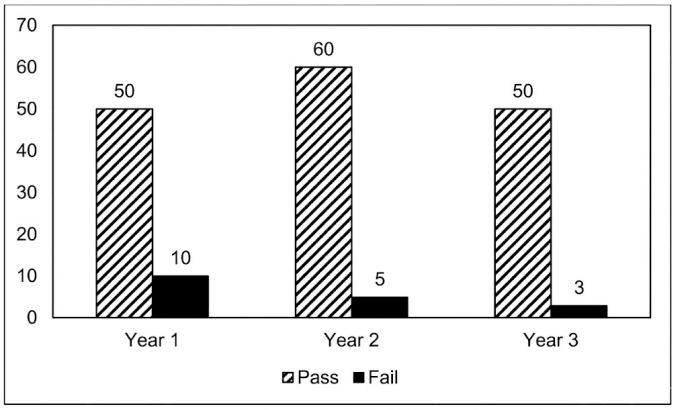


Fig. Q21.3.9

The number of students passing or failing in an exam for a particular subject are presented in the bar chart above. Students who pass the exam cannot appear for the exam again. Students who fail the exam in the first attempt must appear for the exam in the following year. Students always pass the exam in their second attempt.

The number of students who took the exam for the first time in the year 2 and the year 3 respectively, are _____.

A. 65 and 53 **B.** 60 and 50 **C.** 55 and 53 **D.** 55 and 48

Q21.3.10 MCQ 2021 2M

Seven cars P, Q, R, S, T, U and V are parked in a row not necessarily in that order. The cars T and U should be parked next to each other. The cars S and V also should be parked next to each other, whereas P and Q cannot be parked next to each other. Q and S must be parked next to each other. R is parked to the immediate right of V. T is parked to the left of U.
Based on the above statements, the only INCORRECT option given below is:

A. There are two cars parked in between Q and V.

B. Q and R are not parked together.

C. V is the only car parked in between S and R.

D. Car P is parked at the extreme end.

Session : 4 | Afternoon | 07/02/21

Q21.4.1 MCQ 2021 1M

The current population of a city is 11,02,500. If it has been increasing at the rate of 5% per annum, what was its population 2 years ago?

A. 9,92,500 **B.** 9,95,006 **C.** 10,00,000 **D.** 12,51,506

Q21.4.2 MCQ 2021 1M

p and q are positive integers and $\frac{p}{q} + \frac{q}{p} = 3$.

then, $\frac{p^2}{q^2} + \frac{q^2}{p^2} =$

A. 3 **B.** 7 **C.** 9 **D.** 11

Q21.4.3 MCQ 2021 1M

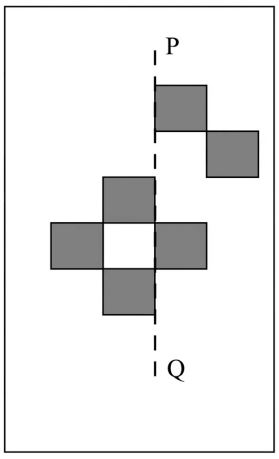


Fig. Q21.4.3

The least number of squares that must be added so that the line P-Q becomes the line of symmetry is ____

A. 4 **B.** 3 **C.** 6 **D.** 7

Q21.4.4 MCQ 2021 1M

Nostalgia is to *anticipation* as _____ is to _____.
Which one of the following options maintains a similar logical relation in the above sentence?

A. Present, past **B.** Future, past

C. Past, future **D.** Future, present

Q21.4.5 MCQ 2021 1M

Consider the following sentences:

(i) I woke up from sleep
(ii) I woked up from sleep.
(iii) I was woken up from sleep.
(iv) I was wokened up from sleep.

Which of the above sentences are grammatically CORRECT?

A. (i) and (ii) **B.** (i) and (iii)

C. (ii) and (iii) **D.** (i) and (iv)

Q21.4.6 MCQ 2021 2M

Given below are two statements and two conclusions.

Statement 1: All purple are green.
Statement 2: All black are green.

Conclusion I: Some black are purple.
Conclusion II: No black is purple.

Based on the above statements and conclusions, which one of the following options is logically CORRECT?

A. Only conclusion I is correct.

B. Only conclusion II is correct.

C. Either conclusion I or II is correct.

D. Both conclusion I and II are correct.

Q21.4.7 MCQ 2021 2M

Computers are ubiquitous. They are used to improve efficiency in almost all fields from agriculture to space exploration. Artificial intelligence (AI) is currently a hot topic. AI enables computers to learn, given enough training data. For humans, sitting in front of a computer for long hours can lead to health issues.

Which of the following can be deduced from the above passage?

(i) Nowadays, computers are present in almost all places.
(ii) Computers cannot be used for solving problems in engineering.
(iii) For humans, there are both positive and negative effects of using computers.
(iv) Artificial intelligence can be done without data.

A. (ii) and (iii) **B.** (ii) and (iv)

C. (i), (iii) and (iv) **D.** (i) and (iii)

Q21.4.8 MCQ 2021 2M

Consider a square sheet of side 1 unit. In the first step, it is cut along the main diagonal to get two triangles. In the next step, one of the cut triangles is revolved about its short edge to form a solid cone. The volume of the resulting cone, in cubic units, is _____

A. $\frac{\pi}{3}$ **B.** $\frac{2\pi}{3}$ **C.** $\frac{3\pi}{2}$ **D.** 3π

Q21.4.9 MCQ 2021 2M

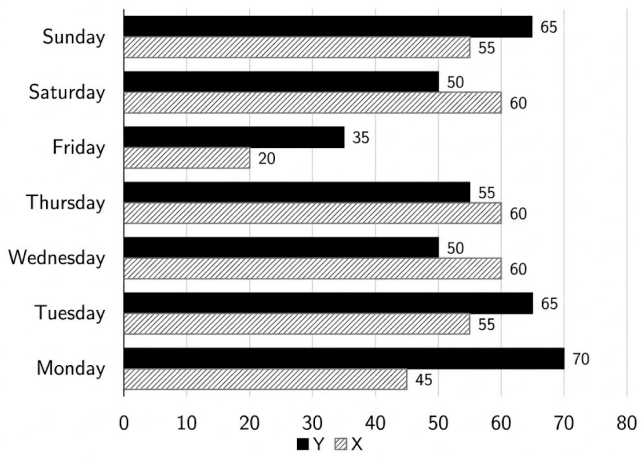


Fig. Q21.4.9

The number of minutes spent by two students, X and Y, exercising every day in a given week are shown in the bar chart above.
The number of days in the given week in which one of the students spent a minimum of 10% more than the other student, on a given day, is

A. 4 **B.** 5 **C.** 6 **D.** 7

Q21.4.10 MCQ 2021 2M

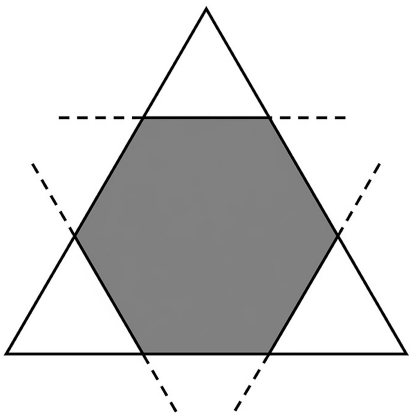


Fig. Q21.4.10

Corners are cut from an equilateral triangle to produce a regular convex hexagon as shown in the figure above.
The ratio of the area of the regular convex hexagon to the area of the original equilateral triangle is

A. 2:3 **B.** 3:4 **C.** 4:5 **D.** 5:6

Session : 5 | Forenoon | 13/02/21

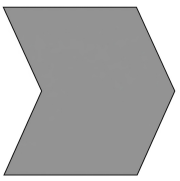
Q21.5.1 MCQ 2021 1M

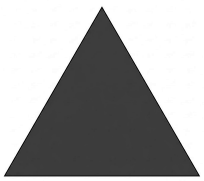
The ratio of boys to girls in a class is 7 to 3.
Among the options below, an acceptable value for the total number of students in the class is:

A. 21 **B.** 37 **C.** 50 **D.** 73

Q21.5.2 MCQ 2021 1M

A polygon is convex if, for every pair of points, P and Q belonging to the polygon, the line segment PQ lies completely inside or on the polygon.
Which one of the following is NOT a convex polygon?

A. 

B. 

C.

D.

Q21.5.3 MCQ 2021 1M

Consider the following sentences:

(i) Everybody in the class is prepared for the exam.
(ii) Babu invited Danish to his home because he enjoys playing chess.

Which of the following is the CORRECT observation about the above two sentences?

A. (i) is grammatically correct and (ii) is unambiguous

B. (i) is grammatically incorrect and (ii) is unambiguous

C. (i) is grammatically correct and (ii) is ambiguous

D. (i) is grammatically incorrect and (ii) is ambiguous

Q21.5.4 MCQ 2021 1M

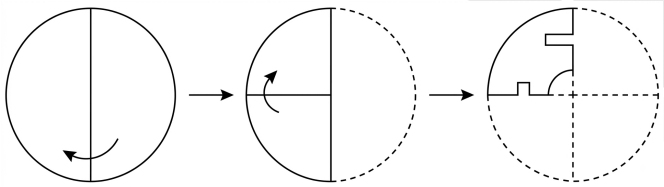
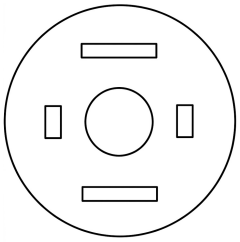
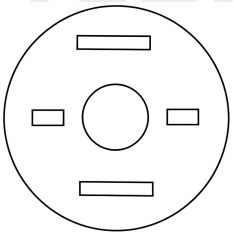
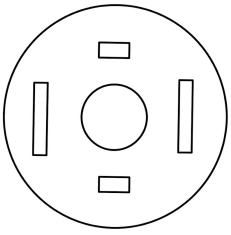


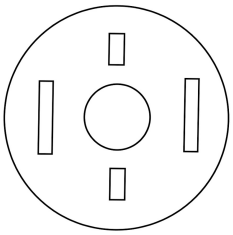
Fig. Q21.5.4

A circular sheet of paper is folded along the lines in the directions shown. The paper, after being punched in the final folded state as shown and unfolded in the reverse order of folding, will look like _____.

A.


B.


C.


D.


Q21.5.5 MCQ 2021 1M

_____ is to *surgery* as *writer* is to _____.

Which one of the following options maintains a similar logical relation in the above sentence?

A. Plan, outline **B.** Hospital, library

C. Doctor, book **D.** Medicine, grammar

Q21.5.6 MCQ 2021 2M

We have 2 rectangular sheets of paper, M and N, of dimensions 6 cm × 1 cm each. Sheet M is rolled to form an open cylinder by bringing the short edges of the sheet together. Sheet N is cut into equal square patches and assembled to form the largest possible closed cube. Assuming the ends of the cylinder are closed, the ratio of the volume of the cylinder to that of the cube is _____

A. $\frac{\pi}{2}$ **B.** $\frac{3}{\pi}$ **C.** $\frac{9}{\pi}$ **D.** 3π

Q21.5.7 MCQ 2021 2M

Items	Cost (₹)	Profit %	Marked Price (₹)
P	5,400	- - -	5,860
Q	- - -	25	10,000

Details of prices of two items P and Q are presented in the above table. The ratio of cost of item P to cost of item Q is 3:4. Discount is calculated as the difference between the marked price and the selling price. The profit percentage is calculated as the ratio of the difference between selling price and cost, to the cost

$$\left(\text{Profit \%} = \frac{\text{Selling price} - \text{Cost}}{\text{Cost}} \times 100\right)$$

The discount on item Q, as a percentage of its marked price, is _____

A. 25 **B.** 12.5 **C.** 10 **D.** 5

Q21.5.8 MCQ 2021 2M

There are five bags each containing identical sets of ten distinct chocolates. One chocolate is picked from each bag.

The probability that at least two chocolates are identical is

A. 0.3024 **B.** 0.4235 **C.** 0.6976 **D.** 0.8125

Q21.5.9 MCQ 2021 2M

Given below are two statements 1 and 2, and two conclusions I and II

Statement 1: All bacteria are microorganisms.
Statement 2: All pathogens are microorganisms.

Conclusion I: Some pathogens are bacteria.
Conclusion II: All pathogens are not bacteria.

Based on the above statements and conclusions, which one of the following options is logically CORRECT?

A. Only conclusion I is correct

B. Only conclusion II is correct

C. Either conclusion I or II is correct

D. Neither conclusion I nor II is correct

Q21.5.10 MCQ | 2021 | 2M

Some people suggest anti-obesity measures (AOM) such as displaying calorie information in restaurant menus. Such measures sidestep addressing the core problems that cause obesity: poverty and income inequality.
Which one of the following statements summarizes the passage?

A. The proposed AOM addresses the core problems that cause obesity.

B. If obesity reduces, poverty will naturally reduce, since obesity causes poverty.

C. AOM are addressing the core problems and are likely to succeed.

D. AOM are addressing the problem superficially.

Session : 6 | Afternoon | 13/02/21

Q21.6.1 MCQ | 2021 | 1M

Gauri said that she can play the keyboard _____ her sister.

A. as well as **B.** as better as

C. as nicest as **D.** as worse as

Q21.6.2 MCQ | 2021 | 1M

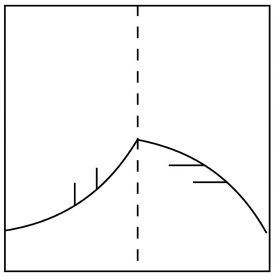
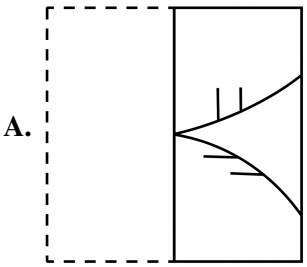
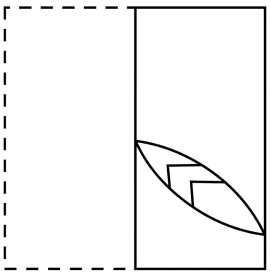
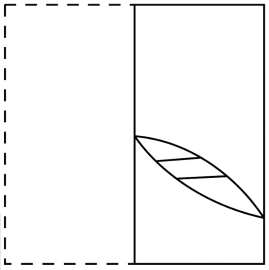


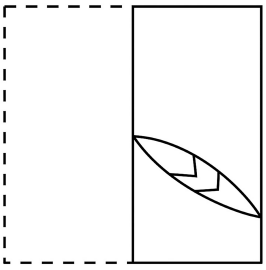
Fig. Q21.6.2

A transparent square sheet shown above is folded along the dotted line. The folded sheet will look like _____.

A. 

B. 

C. 

D. 

Q21.6.3 MCQ | 2021 | 1M

If θ is the angle, in degrees, between the longest diagonal of the cube and any one of the edges of the cube, then, $\cos\theta =$

A. $\frac{1}{2}$ **B.** $\frac{1}{\sqrt{3}}$ **C.** $\frac{1}{\sqrt{2}}$ **D.** $\frac{\sqrt{3}}{2}$

Q21.6.4 MCQ | 2021 | 1M

If $(x-\frac{1}{2})^2-(x-\frac{3}{2})^2 = x+2$, then the value of x is:

A. 2 **B.** 4 **C.** 6 **D.** 8

Q21.6.5 MCQ | 2021 | 1M

Pen : Write :: Knife : _____
Which of the the following options maintains a similar logical relation to the above?

A. Vegetables **B.** Sharp **C.** Cut **D.** Blunt

Q21.6.6 MCQ | 2021 | 2M

Listening to music during exercise improves exercise performance and reduces discomfort. Scientists researched whether listening to music while studying can help students learn better and the results were inconclusive. Students who needed external stimulation for studying fared worse while students who did not need any external stimulation benefited from music.
Which one of the following statements is the CORRECT inference of the above passage?

A. Listening to music has no effect on learning and a positive effect on physical exercise.

B. Listening to music has a clear positive effect both on physical exercise and on learning.

C. Listening to music has a clear positive effect on physical exercise. Music has a positive effect on learning only in some students.

D. Listening to music has a clear positive effect on learning in all students. Music has a positive effect only in some students who exercise.

Q21.6.7 MCQ 2021 2M

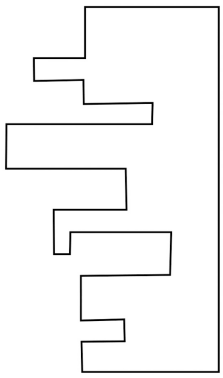
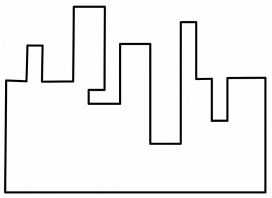
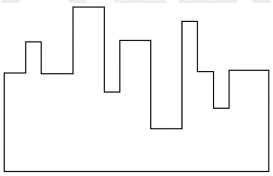
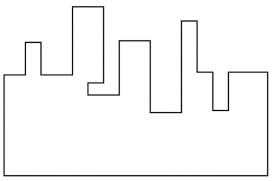


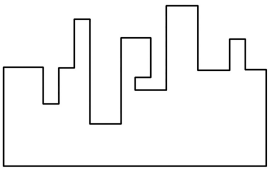
Fig. Q21.6.7

A jigsaw puzzle has 2 pieces. One of the pieces is shown above. Which one of the given options for the missing piece when assembled will form a rectangle? The piece can be moved, rotated or flipped to assemble with the above piece.

A.


B.


C.


D.


Q21.6.8 MCQ 2021 2M

The number of students in three classes is in the ratio 3:13:6. If 18 students are added to each class, the ratio changes to 15:35:21.
The total number of students in all the three classes in the beginning was:

A. 22 **B.** 66 **C.** 88 **D.** 110

Q21.6.9 MCQ 2021 2M

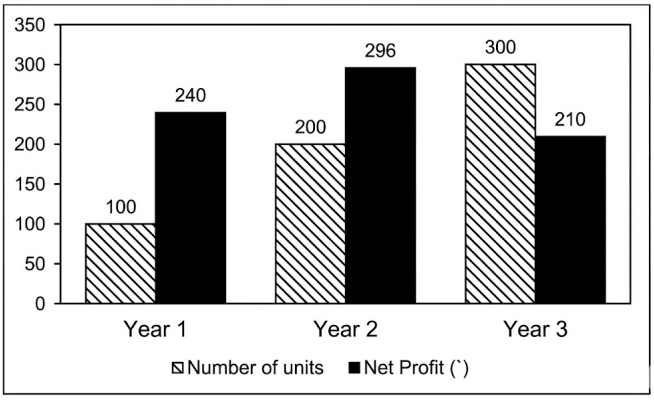


Fig. Q21.6.9

The number of units of a product sold in three different years and the respective net profits are presented in the figure above. The cost/unit in Year 3 was 1, which was half the cost/unit in Year 2. The cost/unit in Year 3 was one-third of the cost/unit in Year 1. Taxes were paid on the selling price at 10%, 13% and 15% respectively for the three years. Net profit is calculated as the difference between the selling price and the sum of cost and taxes paid in that year.
The ratio of the selling price in Year 2 to the selling price in Year 3 is _____.

A. 4:3 **B.** 1:1 **C.** 3:4 **D.** 1:2

Q21.6.10 MCQ 2021 2M

Six students P, Q, R, S, T and U, with distinct heights, compare their heights and make the following observations.
Observation I: S is taller than R.
Observation II: Q is the shortest of all.
Observation III: U is taller than only one student.
Observation IV: T is taller than S but is not the tallest.
The number of students that are taller than R is the same as the number of students shorter than _____.

A. T **B.** R **C.** S **D.** P

Session : 7 | Forenoon | 14/02/21

Q21.7.1 MCQ 2021 1M

Consider the following sentences:

(i) After his surgery, Raja hardly could walk.
(ii) After his surgery, Raja could barely walk.
(iii) After his surgery, Raja barely could walk.
(iv) After his surgery, Raja could hardly walk.

Which of the above sentences are grammatically CORRECT?

A. (i) and (ii) **B.** (i) and (iii)

C. (iii) and (iv) **D.** (ii) and (iv)

Q21.7.2 MCQ 2021 1M

Ms. X came out of a building through its front door to find her shadow due to the morning sun falling to her right side with the building to her back. From this, it can be inferred that building is facing _____

A. North **B.** East **C.** West **D.** South

Q21.7.3 MCQ 2021 1M

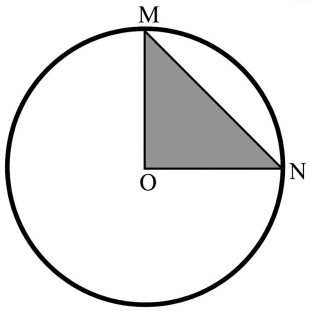


Fig. Q21.7.3

In the above figure, O is the center of the circle and, M and N lie on the circle.
The area of the right triangle MON is 50 cm^2.
What is the area of the circle in cm^2?

A. 2π **B.** 50π **C.** 75π **D.** 100π

Q21.7.4 MCQ 2021 1M

If $\begin{cases} " \oplus " \text{ means } " - ", \\ " \otimes " \text{ means } " \div ", \\ "\Delta" \text{ means } " + ", \\ "\nabla" \text{ means } " \times ", \end{cases}$

then, the value of the expression $\Delta 2 \oplus 3\Delta((4 \otimes 2)\nabla 4) =$

A. -1 **B.** -0.5 **C.** 6 **D.** 7

Q21.7.5 MCQ 2021 1M

"The increased consumption of leafy vegetables in the recent months is a clear indication that the people in the state have begun to lead a healthy lifestyle"
Which of the following can be logically inferred from the information presented in the above statement?

A. The people in the state did not consume leafy vegetables earlier.

B. Consumption of leafy vegetables may not be the only indicator of healthy lifestyle.

C. Leading a healthy lifestyle is related to a diet with leafy vegetables.

D. The people in the state have increased awareness of health hazards causing by consumption of junk foods.

Q21.7.6 MCQ 2021 2M

Oxpeckers and rhinos manifest a symbiotic relationship in the wild. The oxpeckers warn the rhinos about approaching poachers, thus possibly saving the lives of the rhinos. Oxpeckers also feed on the parasitic ticks found on rhinos. In the symbiotic relationship described above, the primary benefits for oxpeckers and rhinos respectively are,

A. Oxpeckers get a food source, rhinos have no benefit.

B. Oxpeckers save their habitat from poachers while the rhinos have no benefit.

C. Oxpeckers get a food source, rhinos may be saved from the poachers.

D. Oxpeckers save the lives of poachers, rhinos save their own lives.

Q21.7.7 MCQ 2021 2M

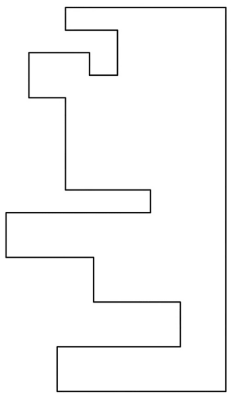
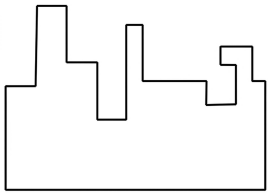
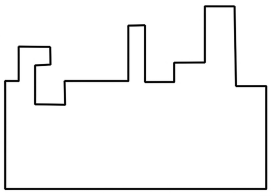
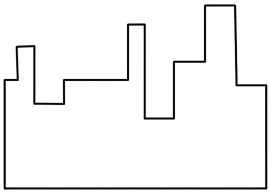


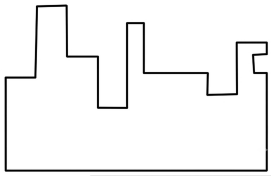
Fig. Q21.7.7

A jigsaw puzzle has 2 pieces. One of the pieces is shown above. Which one of the given options for the missing piece when assembled will form a rectangle? The piece can be moved, rotated or flipped to assemble with the above piece.

A. 

B. 

C. 

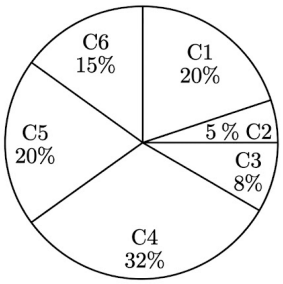
D. 

Q21.7.8 MCQ 2021 2M

The number of hens, ducks and goats in farm P are 65, 91 and 169, respectively. The total number of hens, ducks and goats in a nearby farm Q is 416. The ratio of hens:ducks:goats in farm Q is 5:14:13. All the hens, ducks and goats are sent from farm Q to farm P.
The new ratio of hens:ducks:goats in farm P is _____

A. 5:7:13 **B.** 5:14:13 **C.** 10:21:26 **D.** 21:10:26

Q21.7.9 MCQ 2021 2M



Company	Ratio
C1	3:2
C2	1:4
C3	5:3
C4	2:3
C5	9:1
C6	3:4

Fig. Q21.7.9

The distribution of employees at the rank of executives, across different companies C1, C2, ..., C6 is presented in the chart given above. The ratio of executives with a management degree to those without a management degree in each of these companies is provided in the table above. The total number of executives across all companies is 10,000.
The total number of management degree holders among the executives in companies C2 and C5 together is _____.

A. 225 **B.** 600 **C.** 1900 **D.** 2500

Q21.7.10 MCQ 2021 2M

Five persons P, Q, R, S and T are sitting in a row not necessarily in the same order. Q and R are separated by one person, and S should not be seated adjacent to Q.
The number of distinct seating arrangements possible is:

A. 4 **B.** 8 **C.** 10 **D.** 16

Session : 8 | Afternoon | 14/02/21

Q21.8.1 MCQ 2021 1M

Five persons P, Q, R, S and T are to be seated in a row, all facing the same direction, but not necessarily in the same order. P and T cannot be seated at either end of the row. P should not be seated adjacent to S. R is to be seated at the second position from the left end of the row. The number of distinct seating arrangements possible is:

A. 2 **B.** 3 **C.** 4 **D.** 5

Q21.8.2 MCQ 2021 1M

Consider the following sentences:

(i) The number of candidates who appear for the GATE examination is staggering.
(ii) A number of candidates from my class are appearing for the GATE examination.
(iii) The number of candidates who appear for the GATE examination are staggering.
(iv) A number of candidates from my class is appearing for the GATE examination.

Which of the above sentences are grammatically CORRECT?

A. (i) and (ii) **B.** (i) and (iii)

C. (ii) and (iii) **D.** (ii) and (iv)

Q21.8.3 MCQ 2021 1M

A digital watch X beeps every 30 seconds while watch Y beeps every 32 seconds. They beeped together at 10 AM. The immediate next time that they will beep together is ___

A. 10.08 AM **B.** 10.42 AM

C. 11.00 AM **D.** 10.00 PM

Q21.8.4 MCQ 2021 1M

If $\oplus \div \odot = 2$; $\oplus \div \Delta = 3$; $\odot + \Delta = 5$; $\Delta \times \otimes = 10$, Then, the value of $(\otimes - \oplus)^2$, is:

A. 0 **B.** 1 **C.** 4 **D.** 16

Q21.8.5 MCQ 2021 1M

The front door of Mr. X's house faces East. Mr. X leaves the house, walking 50 m straight from the back door that is situated directly opposite to the front door. He then turns to his right, walks for another 50 m and stops. The direction of the point Mr. X is now located at with respect to the starting point is _____

A. South-East **B.** North-East **C.** West **D.** North-West

Q21.8.6 MCQ 2021 2M

Given below are two statements 1 and 2, and two conclusions I and II.

Statement 1: All entrepreneurs are wealthy.
Statement 2: All wealthy are risk seekers.

Conclusion I: All risk seekers are wealthy.
Conclusion II: Only some entrepreneurs are risk seekers.

Based on the above statements and conclusions, which one of the following options is CORRECT?

A. Only conclusion I is correct

B. Only conclusion II is correct

C. Neither conclusion I nor II is correct

D. Both conclusions I and II are correct

Q21.8.7 MCQ 2021 2M

A box contains 15 blue balls and 45 black balls. If 2 balls are selected randomly, without replacement, the probability of an outcome in which the first selected is a blue ball and the second selected is a black ball, is ____

A. 3/16 **B.** 45/236 **C.** 1/4 **D.** 3/4

Q21.8.8 MCQ 2021 2M

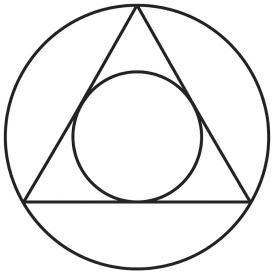


Fig. Q21.8.8

The ratio of the area of the inscribed circle to the area of the circumscribed circle of an equilateral triangle is _____

A. 1/8 **B.** 1/6 **C.** 1/4 **D.** 1/2

Q21.8.9 MCQ 2021 2M

Consider a square sheet of side 1 unit. The sheet is first folded along the main diagonal. This is followed by a fold along its line of symmetry. The resulting folded shape is again folded along its line of symmetry. The area of each face of the final folded shape, in square units, equal to

A. 1/4 **B.** 1/8 **C.** 1/16 **D.** 1/32

Q21.8.10 MCQ 2021 2M

The world is going through the worst pandemic in the past hundred years. The air travel industry is facing a crisis, as the resulting quarantine requirement for travelers led to weak demand.
In relation to the first sentence above, what does the second sentence do?

A. Restates an idea from the first sentence.

B. Second sentence entirely contradicts the first sentence.

C. The two statements are unrelated.

D. States an effect of the first sentence.

Answer Key

Session : 1					
Q21.1.1 D	Q21.1.2 B	Q21.1.3 C	Q21.1.4 B	Q21.1.5 C	Q21.1.6 B
Q21.1.7 A	Q21.1.8 A	Q21.1.9 B	Q21.1.10 C		
Session : 2					
Q21.2.1 B	Q21.2.2 B	Q21.2.3 D	Q21.2.4 D	Q21.2.5 C	Q21.2.6 C
Q21.2.7 A	Q21.2.8 B	Q21.2.9 C	Q21.2.10 D		
Session : 3					
Q21.3.1 C	Q21.3.2 C	Q21.3.3 D	Q21.3.4 D	Q21.3.5 C	Q21.3.6 D
Q21.3.7 C	Q21.3.8 D	Q21.3.9 D	Q21.3.10 A		
Session : 4					
Q21.4.1 C	Q21.4.2 B	Q21.4.3 C	Q21.4.4 C	Q21.4.5 B	Q21.4.6 C
Q21.4.7 D	Q21.4.8 A	Q21.4.9 C	Q21.4.10 A		
Session : 5					
Q21.5.1 C	Q21.5.2 A	Q21.5.3 C	Q21.5.4 A	Q21.5.5 C	Q21.5.6 C
Q21.5.7 C	Q21.5.8 C	Q21.5.9 C OR D	Q21.5.10 D		
Session : 6					
Q21.6.1 A	Q21.6.2 B	Q21.6.3 B	Q21.6.4 B	Q21.6.5 C	Q21.6.6 C
Q21.6.7 A	Q21.6.8 C	Q21.6.9 A	Q21.6.10 C		
Session : 7					
Q21.7.1 D	Q21.7.2 D	Q21.7.3 D	Q21.7.4 D	Q21.7.5 C	Q21.7.6 C
Q21.7.7 A	Q21.7.8 C	Q21.7.9 C	Q21.7.10 D		
Session : 8					
Q21.8.1 B	Q21.8.2 A	Q21.8.3 A	Q21.8.4 B	Q21.8.5 D	Q21.8.6 C
Q21.8.7 B	Q21.8.8 C	Q21.8.9 B	Q21.8.10 D		

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